

Analyses of detector richness variance and between-detector beta diversity

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Setup

Set up libraries, options, import and process data

```
# setup -----
library(tidyverse)
library(readxl)
library(patchwork)
library(brms)
library(bayesplot)
library(distributional)
library(tidybayes)
library(ggdist)
library(vegan)
library(patchwork)

options(brms.file_refit = "on_change")
options(mc.cores = parallel::detectCores())
theme_set(theme_minimal())

if (!dir.exists("fits")) dir.create("fits")

set.seed(932)

## Plotting functions -----

post_prior_beta <- function(x){
  x |>
    as_draws_df("b_exercise2") |>
    ggplot(aes(x = b_exercise2)) +
    stat_slab(aes(xdist = dist_normal(0, 1)),
              fill = "#00BA38", alpha = 0.5, inherit.aes = F,
              data = tibble()) +
    geom_density(fill = "#F8766D", color = NA, alpha = 0.5)
}

post_slab <- function(data, x, line){
  data |>
```

```

ggplot(aes(x = {{x}})) +
  stat_slab(aes(fill_ramp = after_stat(level)),
    .width = c(.50, .80, .95, 1),
    slab_color = "navy",
    fill = "skyblue") +
  scale_fill_ramp_discrete() +
  guides(fill_ramp = "none") +
  geom_vline(xintercept = line, linetype = 2) +
  stat_pointinterval(aes(interval_color = after_stat(level)),
    point_interval = "median_qi",
    side = "top",
    .width = c(.50, .80, .95),
    interval_size_domain = c(1.5, 5)) +
  scale_color_discrete(palette = c("grey60", "grey40", "black"),
    aesthetics = "interval_color",
    guide = NULL) +
  theme(axis.text.y = element_blank(),
    legend.position = "none",
    panel.grid.major.y = element_blank(),
    panel.grid.minor.y = element_blank())
}

plot_post <- function(data, line = 0, fit = F){
  if(fit){
    data <- data |>
      as_draws_df() |>
      select(value = b_exercise2)
  }
  post_slab(data, value, line)
}

plot_comp <- function(x, comparison){
  x |>
    epred_draws(newdata = tibble(exercise = as.character(1:2)),
      re_formula = NA) |>
    ungroup() |>
    select(.draw, exercise, .epred) |>
    pivot_wider(names_from = exercise, values_from = .epred,
      names_prefix = "ex") |>
    mutate(contrast = ex2 - ex1,
      lift = ex2 / ex1 - 1) |>
    post_slab({{comparison}}, line = 0)
}

plot_pred <- function(x){
  tibble(exercise = as.character(1:2),
    SU = NA) |>
  add_epred_draws(x, re_formula = NA) |>
  ggplot(aes(x = .epred, y = exercise)) +
  stat_dotsinterval(aes(interval_color = after_stat(level)),
    quantiles = 100,
    .width = c(0.5, 0.8, 0.95),
    interval_size_domain = c(1, 6),

```

```

    slab_fill = "skyblue",
    slab_color = NA) +
  scale_color_discrete(palette = c("grey60", "grey40", "black"),
    aesthetics = "interval_color",
    guide = NULL) +
  scale_y_discrete(limits = rev,
    expand = expansion(mult = c(.15, 0), add = c(0, -1)))
}

```

```

# data import -----

## 2023 -----

species_2023 <- read_xls("data/2023/23pre_post_training_sp.xls") |>
  select(event = 'Numero Plot', SU = 'Subplot', Layer = 'Codice Strato',
    species = 'Genere e Specie Pignatti', cover = 'Copertura %')

#recode event into detector and exercise
lookup <- tibble(event = 32:39,
  exercise = rep(1:2, each = 4),
  detector = rep(LETTERS[1:4], times = 2))

species_2023 <- species_2023 |>
  left_join(lookup) |>
  select(!event) |>
  rename(Detector = detector)

#convert 2023 species layers

```

layers coded as:

1. tree (woody + pertinent lianas and climbers) $h > 5$ m
2. shrub (woody + pertinent lianas and climbers) $0.5 < h \leq 5$
3. herb(however herbaceous and ferns; woody only if $h \leq 0.5$ m) $h \leq 0.5$ m
4. moss (terricolous bryophytes and t-lichens) on mineral, organic soil

```

species_2023 <- species_2023 |>
  filter(Layer !=4) |> # 2025 doesn't have moss
  mutate(Layer = factor(Layer, labels = c("T", "S", "H")))

## 2025 -----

### 1° exercise -----

species_2025_bf <- list.files("data/2025/25_02", full.names = T) |>
  map(\(x) read_excel(x, sheet = "Species")) |>
  list_rbind() |>
  select(!Notes) |>
  mutate(SU = as_factor(SU)) |>
  rename(species = "species (Genus + species)",
    cover = "Cover(perc.)") |>
  mutate(cover = as.numeric(gsub(",", ".", cover))) |>
  filter(Layer %in% c("T", "S", "H")) |> #layer codes

```

```

separate_wider_delim(Plot.code, delim = ".", names = c("Detector", "Anno")) |>
select(!Anno)

### 2' exercise -----

second_2025 <- list.files("data/2025/27_02", full.names = T)
species_sheets <- second_2025[str_detect(second_2025, regex("species", ignore_case = T))]
```

```

species_2025_af <- lapply(species_sheets, function(x) read_csv(x, col_names = F, skip = 1)) |>
  bind_rows() |>
  select(id = X1, SU = X2, Layer=X3, species=X4, cover=X5) |>
  separate_wider_delim(id, delim = ".", names = c("Detector", "Anno")) |>
  select(!Anno)|>
  mutate(SU = as_factor(SU))

### join the exercises -----

species_2025 <- bind_rows(species_2025_bf, species_2025_af, .id = "exercise") |>
  filter(cover>0) #f and c inserted "0" as cover obtaining all the species...

## uniform column types -----

species_2023 <- species_2023 |>
  mutate(SU = factor(SU),
         exercise = as.character(exercise),
         Layer = as.character(Layer))

## check species names -----

# remove leading and trailing whitespaces
species_2023$species <- trimws(species_2023$species)

species_2025$species <- trimws(species_2025$species)

# 2023
unique(species_2023$species) |> sort() #sort and check all the species surveyed

```

```

## [1] "Acer campestre"      "Ajuga reptans"
## [3] "Ajuga sp."          "Allium sp."
## [5] "Anemone apennina"   "Anemone nemorosa"
## [7] "Arisarum cfr. vulgare" "Arisarum vulgare"
## [9] "Aristolochia clematitis" "Aristolochia rotunda"
## [11] "Arum maculatum"     "Arum sp."
## [13] "Brachypodium sp."   "Brachypodium sylvaticum"
## [15] "Carex hallerana"    "Carpinus betulus"
## [17] "Colchicum lusitanum" "Colchicum sp."
## [19] "Cornus mas"         "Crataegus monogyna"
## [21] "Crataegus oxyacantha" "Crocus sp."
## [23] "Cruciata glabra"    "Cruciata laevipes"
## [25] "Cyclamen repandum"  "Daphne sp."
## [27] "Euonymus europaeus" "Fraxinus ornus"
## [29] "Fraxinus oxycarpa"  "Gagea sp."
## [31] "Galium sp."         "Geranium sp."

```

```
## [33] "Hedera helix"           "Helleborus foetidus"
## [35] "Helleborus sp."        "Helleborus viridis"
## [37] "Juniperus communis"    "Lathyrus sp."
## [39] "Lathyrus venetus"      "Ligustrum vulgare"
## [41] "Lonicera caprifolium"  "Lonicera sp."
## [43] "Lonicera xylosteum"    "Luzula forsteri"
## [45] "Malus sp."             "Malus sylvestris"
## [47] "Melica uniflora"       "Oenanthe pimpinelloides"
## [49] "Orobancha hederaceae"  "Orobancha sp."
## [51] "Phillyrea latifolia"   "Pirus pyraeaster"
## [53] "Platanthera sp."       "Poa nemoralis"
## [55] "Prunella vulgaris"     "Prunus sp."
## [57] "Prunus spinosa"        "Pyrus pyraeaster"
## [59] "Quercus cerris"        "Quercus ilex"
## [61] "Quercus pubescens"     "Quercus robur"
## [63] "Quercus sp."           "Ranunculus ficaria"
## [65] "Ranunculus lanuginosus" "Ranunculus sp."
## [67] "Rhamnus alaternus"     "Rhamnus catharticus"
## [69] "Rhamnus sp."           "Rosa arvensis"
## [71] "Rosa cfr. arvensis"    "Rosa cfr. sempervirens"
## [73] "Rosa sp."              "Rubia peregrina"
## [75] "Rubus sp."             "Rubus ulmifolius"
## [77] "Ruscus aculeatus"      "Smilax aspera"
## [79] "Smilax sp."            "Sorbus domestica"
## [81] "Sorbus torminalis"     "Staphylea pinnata"
## [83] "Symphytum bulbosum"    "Symphytum sp."
## [85] "Symphytum tuberosum"   "Tamus communis"
## [87] "Ulmus minor"           "Viburnum tinus"
## [89] "Viola alba"            "Viola sp."
## [91] "Vitis sp."             "Vitis vinifera"
```

```
# unify and correct differences among species names
```

```
species_2023[species_2023$species == "Arisarum cfr. vulgare", "species"] = "Arisarum vulgare"
species_2023[species_2023$species == "Rosa cfr. sempervirens", "species"] = "Rosa sempervirens"
```

```
# 2025
```

```
unique(species_2025$species) |> sort()
```

```
## [1] "Acer monspessulanum"
## [2] "Alium sp."
## [3] "Anemone apennina"
## [4] "Anemonoides apennina"
## [5] "Arbutus unedo"
## [6] "Arcticum nemorosum"
## [7] "Arctium cfr. minus"
## [8] "Arctium cfr. nemorosus"
## [9] "Arctium minus"
## [10] "Arctium nemorosum"
## [11] "Arctium nemorosum cfr"
## [12] "Asparagus acutifolius"
## [13] "Asplenium onopteris"
## [14] "Betonica officinalis"
```

[15] "Brachypodium sylvaticum"
 ## [16] "Brachypodium rupestre"
 ## [17] "Brachypodium sp."
 ## [18] "Brachypodium sylvaticum"
 ## [19] "Carex distachya"
 ## [20] "Carex flacca"
 ## [21] "Carex olbiensis"
 ## [22] "Carex sp."
 ## [23] "Cornus mas"
 ## [24] "Crataegus monogyna"
 ## [25] "Crataegus monogyna"
 ## [26] "Crocus etruscus"
 ## [27] "Cyclamen repandum"
 ## [28] "Daphne laureola"
 ## [29] "Erica arborea"
 ## [30] "Erica scoparia"
 ## [31] "Erica sp."
 ## [32] "Euonymus europaeus"
 ## [33] "Euonymus latifolius"
 ## [34] "Festuca heterophylla"
 ## [35] "foglia morbida"
 ## [36] "Fraxinus ornus"
 ## [37] "Galium aparine"
 ## [38] "Galium hirsutum"
 ## [39] "Gallium aparine"
 ## [40] "Genista sp."
 ## [41] "Gladiolus sp."
 ## [42] "Hedera helix"
 ## [43] "Helleborus cfr. viridis"
 ## [44] "Helleborus cfr. viridis subsp. bocconeii"
 ## [45] "Helleborus foetidus"
 ## [46] "Helleborus sp.?"
 ## [47] "Helleborus viridis"
 ## [48] "Juniperus communis"
 ## [49] "lathyrus sp."
 ## [50] "Lathyrus sp."
 ## [51] "Lathyrus venetus"
 ## [52] "Ligustrum lucidum"
 ## [53] "Ligustrum vulgare"
 ## [54] "Lonicera etrusca"
 ## [55] "Lunaria annua"
 ## [56] "Luzula forsteri"
 ## [57] "Lychnis flos-coculi"
 ## [58] "Melica uniflora"
 ## [59] "Oenanthe pimpinelloides"
 ## [60] "Olea europaea"
 ## [61] "Olea europaea var. sylvestris"
 ## [62] "Osyris alba"
 ## [63] "Paliurus spina-christi"
 ## [64] "Phillyrea angustifolia"
 ## [65] "Phillyrea latifolia"
 ## [66] "Phillyrea sp."
 ## [67] "Phyllirea angustifolia"
 ## [68] "Phyllirea latifolia"

```
## [69] "Pistacia lentiscus"
## [70] "Poa sp."
## [71] "Poa sylvicola"
## [72] "Prunella vulgaris"
## [73] "Prunus spinosa"
## [74] "Pyracantha coccinea"
## [75] "Quercus cerris"
## [76] "Quercus ilex"
## [77] "Quercus pubescens"
## [78] "Ranunculus ficaria"
## [79] "Rhamnus alaternus"
## [80] "Rosa sempervirens"
## [81] "Rubia peregrina"
## [82] "Rubus hirtus"
## [83] "Rubus hulmifolius"
## [84] "Rubus sp"
## [85] "Rubus sp."
## [86] "Rubus ulmifolius"
## [87] "Ruscus aculeatus"
## [88] "Serratula cfr. tinctoria"
## [89] "Serratula tinctoria"
## [90] "Smilas aspera"
## [91] "Smilax aspera"
## [92] "Sorbus domestica"
## [93] "Sorbus domesticus"
## [94] "Sorbus sp."
## [95] "Sorbus torminalis"
## [96] "Stachys officinalis"
## [97] "Stachys sp"
## [98] "Surbus domestica"
## [99] "Symphytum bulbosum"
## [100] "Symphytum cfr. officinale"
## [101] "Symphytum officinale"
## [102] "Symphytum sp"
## [103] "Symphytum sp."
## [104] "Symphytum tuberosum"
## [105] "Viburnum tinus"
## [106] "Vicia sp."
## [107] "Vicia sp. (fg larghe pelose)"
## [108] "Vicia sp. (fg strette)"
## [109] "Vicia sp. (foglie larghe)"
## [110] "Viola alba"
## [111] "Viola alba subsp. dehnhardtii"
## [112] "Viola alba subsp. dennhardtii"
## [113] "Viola alba/odorata"
## [114] "Viola odorata"
```

```
# correct species names
```

```
species_2025 <- species_2025 |>
  # collapse cfr. to most likely species
  mutate(species = str_remove(species, "cfr\\.|cfr")) |>
  # remove subsp. and var.
  mutate(species = str_remove(species, "(var|subsp)\\..*$")) |>
```

```
# change double space to single
mutate(species = str_squish(species))

### fix typos -----
```

here the goal is to fix only those typos that result in the same species being recorded with different names. Typos and notes that don't lead to this are kept

```
species_2025$species |> unique() |> sort()
```

```
## [1] "Acer monspessulanum"      "Alium sp."
## [3] "Anemone apennina"        "Anemonoides apennina"
## [5] "Arbutus unedo"           "Arcticum nemorosum"
## [7] "Arctium minus"           "Arctium nemorosum"
## [9] "Arctium nemorosus"       "Asparagus acutifolius"
## [11] "Asplenium onopteris"     "Betonica officinalis"
## [13] "Brachypodium rupestre"   "Brachypodium sp."
## [15] "Brachypodium sylvaticum" "Carex distachya"
## [17] "Carex flacca"           "Carex olbiensis"
## [19] "Carex sp."              "Cornus mas"
## [21] "Crataegus monogyna"      "Crategus monogyna"
## [23] "Crocus etruscus"         "Cyclamen repandum"
## [25] "Daphne laureola"        "Erica arborea"
## [27] "Erica scoparia"         "Erica sp."
## [29] "Euonymus europaeus"     "Euonymus latifolius"
## [31] "Festuca heterophylla"    "foglia morbida"
## [33] "Fraxinus ornus"         "Galium aparine"
## [35] "Galium hirsutum"        "Gallium aparine"
## [37] "Genista sp."            "Gladiolus sp."
## [39] "Hedera helix"           "Helleborus foetidus"
## [41] "Helleborus sp.?"        "Helleborus viridis"
## [43] "Juniperus communis"     "lathyrus sp."
## [45] "Lathyrus sp."           "Lathyrus venetus"
## [47] "Ligustrum lucidum"      "Ligustrum vulgare"
## [49] "Lonicera etrusca"       "Lunaria annua"
## [51] "Luzula forsteri"        "Lychnis flos-coculi"
## [53] "Melica uniflora"        "Oenanthe pimpinelloides"
## [55] "Olea europaea"          "Osyris alba"
## [57] "Paliurus spina-christi"  "Phillyrea angustifolia"
## [59] "Phillyrea latifolia"    "Phillyrea sp."
## [61] "Phyllirea angustifolia" "Phyllirea latifolia"
## [63] "Pistacia lentiscus"     "Poa sp."
## [65] "Poa sylvicola"          "Prunella vulgaris"
## [67] "Prunus spinosa"         "Pyracantha coccinea"
## [69] "Quercus cerris"         "Quercus ilex"
## [71] "Quercus pubescens"      "Ranunculus ficaria"
## [73] "Rhamnus alaternus"     "Rosa sempervirens"
## [75] "Rubia peregrina"        "Rubus hirtus"
## [77] "Rubus hulmifolius"      "Rubus sp"
## [79] "Rubus sp."              "Rubus ulmifolius"
## [81] "Ruscus aculeatus"       "Serratula tinctoria"
## [83] "Smilas aspera"          "Smilax aspera"
## [85] "Sorbus domestica"       "Sorbus domesticus"
```

```
## [87] "Sorbus sp." "Sorbus torminalis"
## [89] "Stachys officinalis" "Stachys sp"
## [91] "Surbus domestica" "Symphytum bulbosum"
## [93] "Symphytum officinale" "Symphytum sp"
## [95] "Symphytum sp." "Symphytum tuberosum"
## [97] "Viburnum tinus" "Vicia sp."
## [99] "Vicia sp. (fg larghe pelose)" "Vicia sp. (fg strette)"
## [101] "Vicia sp. (foglie larghe)" "Viola alba"
## [103] "Viola alba/odorata" "Viola odorata"
```

```
species_2025 <- species_2025 |>
  mutate(species = str_replace_all(species,
    c("Anemone" = "Anemonoides",
      "Arcticum" = "Arctium",
      "nemorosus" = "nemorosum",
      "Crategus" = "Crataegus",
      "Phyllirea" = "Phillyrea",
      "domesticus" = "domestica",
      "Surbus" = "Sorbus",
      "Symphytum sp$" = "Symphytum sp.",
      "Gallium" = "Galium",
      "lathyrus" = "Lathyrus",
      "Rubus sp$" = "Rubus sp.",
      "hulmifolius" = "ulmifolius",
      "Smilas" = "Smilax")))

species_2025$species |> unique() |> sort()
```

```
## [1] "Acer monspessulanum" "Alium sp."
## [3] "Anemonoides apennina" "Arbutus unedo"
## [5] "Arctium minus" "Arctium nemorosum"
## [7] "Asparagus acutifolius" "Asplenium onopteris"
## [9] "Betonica officinalis" "Brachypodium rupestre"
## [11] "Brachypodium sp." "Brachypodium sylvaticum"
## [13] "Carex distachya" "Carex flacca"
## [15] "Carex olbiensis" "Carex sp."
## [17] "Cornus mas" "Crataegus monogyna"
## [19] "Crocus etruscus" "Cyclamen repandum"
## [21] "Daphne laureola" "Erica arborea"
## [23] "Erica scoparia" "Erica sp."
## [25] "Euonymus europaeus" "Euonymus latifolius"
## [27] "Festuca heterophylla" "foglia morbida"
## [29] "Fraxinus ornus" "Galium aparine"
## [31] "Galium hirsutum" "Genista sp."
## [33] "Gladiolus sp." "Hedera helix"
## [35] "Helleborus foetidus" "Helleborus sp.?"
## [37] "Helleborus viridis" "Juniperus communis"
## [39] "Lathyrus sp." "Lathyrus venetus"
## [41] "Ligustrum lucidum" "Ligustrum vulgare"
## [43] "Lonicera etrusca" "Lunaria annua"
## [45] "Luzula forsteri" "Lychnis flos-coculi"
## [47] "Melica uniflora" "Oenanthe pimpinelloides"
## [49] "Olea europaea" "Osyris alba"
## [51] "Paliurus spina-christi" "Phillyrea angustifolia"
```

```
## [53] "Phillyrea latifolia"      "Phillyrea sp."
## [55] "Pistacia lentiscus"      "Poa sp."
## [57] "Poa sylvicola"          "Prunella vulgaris"
## [59] "Prunus spinosa"          "Pyracantha coccinea"
## [61] "Quercus cerris"          "Quercus ilex"
## [63] "Quercus pubescens"       "Ranunculus ficaria"
## [65] "Rhamnus alaternus"       "Rosa sempervirens"
## [67] "Rubia peregrina"         "Rubus hirtus"
## [69] "Rubus sp."              "Rubus ulmifolius"
## [71] "Ruscus aculeatus"        "Serratula tinctoria"
## [73] "Smilax aspera"           "Sorbus domestica"
## [75] "Sorbus sp."              "Sorbus torminalis"
## [77] "Stachys officinalis"     "Stachys sp"
## [79] "Symphytum bulbosum"      "Symphytum officinale"
## [81] "Symphytum sp."           "Symphytum tuberosum"
## [83] "Viburnum tinus"          "Vicia sp."
## [85] "Vicia sp. (fg larghe pelose)" "Vicia sp. (fg strette)"
## [87] "Vicia sp. (foglie larghe)" "Viola alba"
## [89] "Viola alba/odorata"      "Viola odorata"
```

```
# calculate beta -----

# function from long df to input for vegdist
vegan_cook <- function(x){
  mat <- x |>
    mutate(cover = log1p(cover)) |>
    pivot_wider(names_from = species, values_from = cover,
      #some detectors reported the same species twice, use the first one
      values_fn = ~ .x[[1]], values_fill = 0) |>
    column_to_rownames("Detector") |>
    as.matrix()
}

# function to map vegdist and wrangle output

tidyvegdist <- function(x, method, binary){
  x |>
    map(\(x) vegdist(x, method = method, binary = binary) |>
      as.matrix() |>
      `diag<-`(NA) |>
      as_tibble() |>
      pivot_longer(everything(), names_to = "detector") |>
      filter(!is.na(value))) |>
    bind_rows(.id = "ex.SU") |>
    separate_wider_delim(ex.SU, delim = ".", names = c("exercise", "SU")) |>
    rename(dissimilarity = value)
}

#dataframes for beta diversity
#only herb layer, group for group_split later
vegans_2025 <- species_2025 |>
  filter(Layer == "H") |>
  select(!c(Layer)) |>
  group_by(exercise, SU)
```

```

#detector m didn't sample species before training in SU 7, remove also from after
vegans_2025 <- vegans_2025 |>
  filter(!(Detector == "m" & SU == "7"))

vegans_2023 <- species_2023 |>
  filter(Layer == "H") |>
  select(!c(Layer)) |>
  group_by(exercise, SU)

#list names
keys_2025 <- vegans_2025 |>
  group_keys()

keys_2025 <- paste(keys_2025$exercise, keys_2025$SU, sep = ".")

keys_2023 <- vegans_2023 |>
  group_keys()

keys_2023 <- paste(keys_2023$exercise, keys_2023$SU, sep = ".")

#make list of matrices for each exercise and SU
vegans_2025 <- vegans_2025 |>
  group_split(.keep = F) |>
  map(vegan_cook)

names(vegans_2025) <- keys_2025

vegans_2023 <- vegans_2023 |>
  group_split(.keep = F) |>
  map(vegan_cook)

names(vegans_2023) <- keys_2023

rm(keys_2023, keys_2025)

## Jaccard -----

jaccard_2025 <- vegans_2025 |>
  tidyvegdist(method = "jaccard", binary = T)

jaccard_2023 <- vegans_2023 |>
  tidyvegdist(method = "jaccard", binary = T)

## Bray -----

bray_2025 <- vegans_2025 |>
  tidyvegdist(method = "bray", binary = F)

bray_2023 <- vegans_2023 |>
  tidyvegdist(method = "bray", binary = F)

## Euclidean -----

```

```
euc_2025 <- vegans_2025 |>
  tidyvegdist(method = "euclidean", binary = F) |>
  mutate(dissimilarity = dissimilarity/max(dissimilarity))

euc_2023 <- vegans_2023 |>
  tidyvegdist(method = "euclidean", binary = F) |>
  mutate(dissimilarity = dissimilarity/max(dissimilarity))

## nudge 0 and 1s for beta regression -----

jaccard_2025 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 0 x 4
## # i 4 variables: exercise <chr>, SU <chr>, detector <chr>, dissimilarity <dbl>

jaccard_2023 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 0 x 4
## # i 4 variables: exercise <chr>, SU <chr>, detector <chr>, dissimilarity <dbl>

bray_2025 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 0 x 4
## # i 4 variables: exercise <chr>, SU <chr>, detector <chr>, dissimilarity <dbl>

bray_2023 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 0 x 4
## # i 4 variables: exercise <chr>, SU <chr>, detector <chr>, dissimilarity <dbl>

euc_2025 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 2 x 4
##   exercise SU      detector dissimilarity
##   <chr>    <chr> <chr>          <dbl>
## 1 1      8      m              1
## 2 1      8      1              1

euc_2023 |>
  filter(dissimilarity == 0 | dissimilarity == 1)

## # A tibble: 2 x 4
##   exercise SU      detector dissimilarity
##   <chr>    <chr> <chr>          <dbl>
## 1 1      1      C              1
## 2 1      1      A              1
```

```
euc_2025 <- euc_2025 |>
  mutate(dissimilarity = pmin(1 - 1e-06, pmax(1e-06, dissimilarity)))

euc_2023 <- euc_2023 |>
  mutate(dissimilarity = pmin(1 - 1e-06, pmax(1e-06, dissimilarity)))
```

Models based on richness

```
# richness models -----

richness_2023 <- species_2023 |>
  group_by(Detector, SU, exercise) |>
  distinct(species) |>
  count() |>
  mutate(Detector = as.factor(paste(Detector, exercise, sep = ".")),
         exercise = as.factor(exercise))

richness_2025 <- species_2025 |>
  group_by(Detector, SU, exercise) |>
  distinct(species) |>
  count() |>
  mutate(Detector = as.factor(paste(Detector, exercise, sep = ".")),
         exercise = as.factor(exercise))

# formula

rich_formula <- brmsformula(n ~ exercise + SU + (1 | gr(Detector, by = exercise)))

default_prior(rich_formula, data = richness_2025)
```

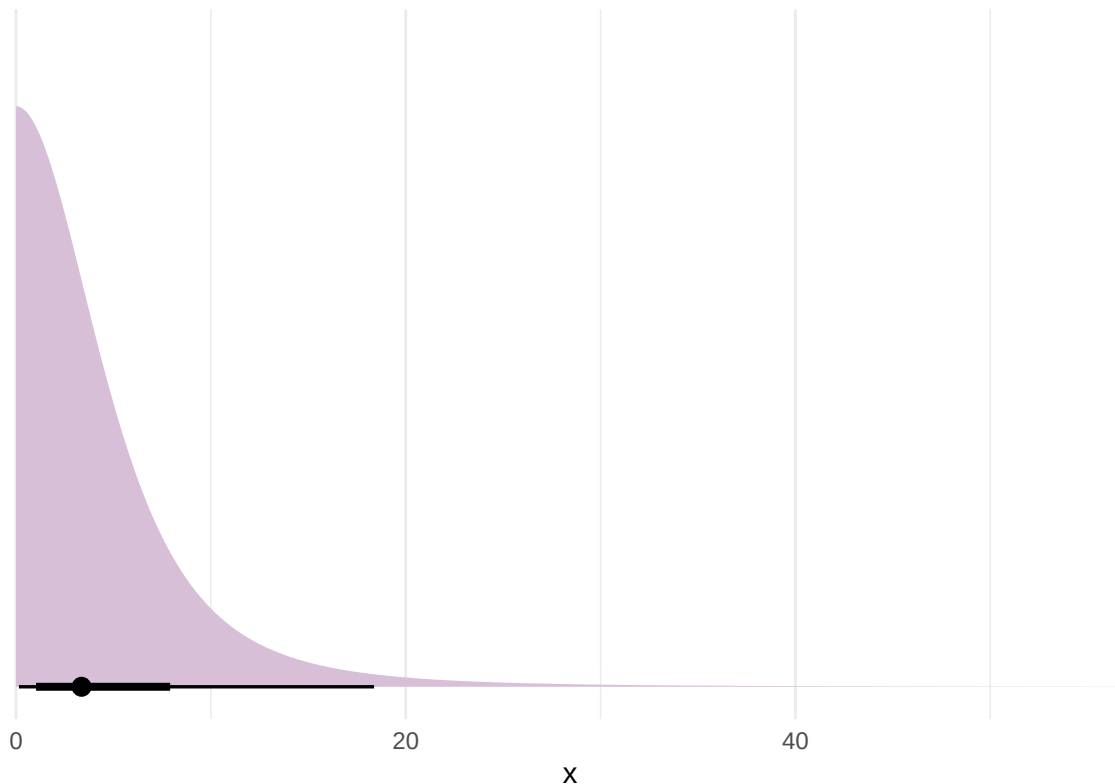
```
##           prior      class      coef      group resp dpar nlpar lb ub tag
##           (flat)         b           exercise2
##           (flat)         b           SU2
##           (flat)         b           SU3
##           (flat)         b           SU4
##           (flat)         b           SU5
##           (flat)         b           SU6
##           (flat)         b           SU7
##           (flat)         b           SU8
## student_t(3, 17, 4.4) Intercept
## student_t(3, 0, 4.4)      sd
## student_t(3, 0, 4.4)      sd      Detector
## student_t(3, 0, 4.4)      sd Intercept Detector
## student_t(3, 0, 4.4)      sigma
##           source
##           default
## (vectorized)
## (vectorized)
```

```
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## default
## default
## (vectorized)
## (vectorized)
## default
```

```
## priors-----

#default prior
ggplot()+
  stat_halfeye(aes(xdist = dist_truncated(dist_student_t(3, 0, 4.4), lower = 0)),
    fill = "thistle", .width = c(0.66, 0.95))+
  scale_y_continuous(NULL, breaks = NULL)+
  ggtitle(expression("Default prior for"~sigma))
```

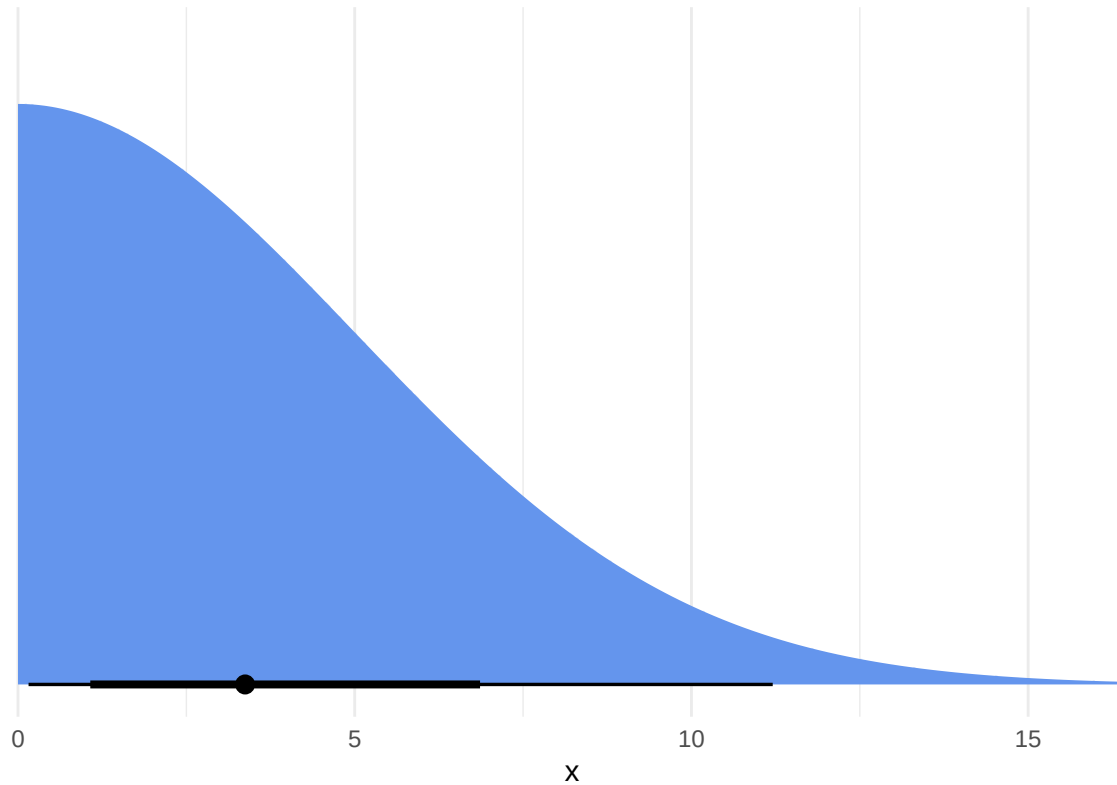
Default prior for σ



```
#adjusted priors
ggplot()+
  stat_halfeye(aes(xdist = dist_truncated(dist_normal(0, 5), lower = 0)),
    fill = "cornflowerblue", .width = c(0.66, 0.95))+
```

```
scale_y_continuous(NULL, breaks = NULL)+
ggtitle(expression("Adjusted prior for"~sigma))
```

Adjusted prior for σ



```
ggplot()+
  stat_halfeye(aes(xdist = dist_normal(0, 5)),
    fill = "chocolate1", .width = c(0.66, 0.95))+
  scale_y_continuous(NULL, breaks = NULL)+
  ggtitle(expression("Adjusted prior for"~beta))
```

Adjusted prior for β



```
prior_rich <- c(prior(normal(0, 5), class = "b"), #effect of exercise and SU
               prior(normal(0, 5), class = "sd", lb = 0)) #multilevel SD
```

```
## 2025 -----
```

```
brm_rich_2025 <- brm(formula = rich_formula,
                    data = richness_2025,
                    prior = prior_rich,
                    file = "fits/rich_2025",
                    sample_prior = "yes",
                    seed = 2958)
```

```
#check prior used
```

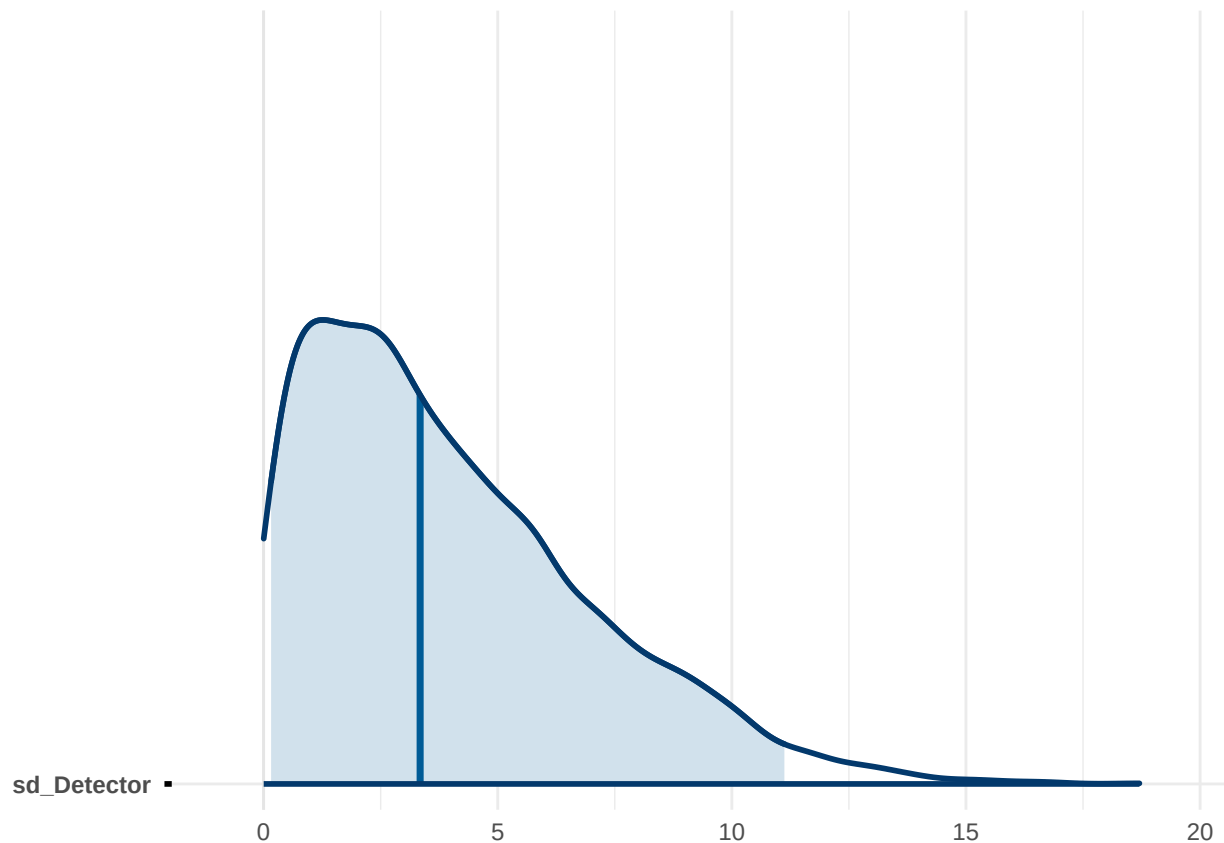
```
prior_summary(brm_rich_2025)
```

##	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
##	normal(0, 5)	b								
##	normal(0, 5)	b	exercise2							
##	normal(0, 5)	b	SU2							
##	normal(0, 5)	b	SU3							
##	normal(0, 5)	b	SU4							
##	normal(0, 5)	b	SU5							
##	normal(0, 5)	b	SU6							
##	normal(0, 5)	b	SU7							

```
##          normal(0, 5)          b          SU8
## student_t(3, 17, 4.4) Intercept
##          normal(0, 5)          sd          0
##          normal(0, 5)          sd          Detector 0
##          normal(0, 5)          sd Intercept Detector 0
## student_t(3, 0, 4.4)          sigma          0
##          source
##          user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
##          default
##          user
## (vectorized)
## (vectorized)
##          default
```

#prior predictive check for sd

```
prior_draws(brm_rich_2025) |>
  select(sd_Detector) |>
  mcmc_areas(border_size = 1, prob = 0.95, point_est = "median")
```



```
#posterior summary
summary(brm_rich_2025)
```

```
## Family: gaussian
## Links: mu = identity
## Formula: n ~ exercise + SU + (1 | gr(Detector, by = exercise))
## Data: richness_2025 (Number of observations: 240)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Detector (Number of levels: 30)
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(Intercept:exercise1)	1.37	0.51	0.47	2.49	1.00	1086
sd(Intercept:exercise2)	1.00	0.48	0.12	2.04	1.00	1213

```
##
```

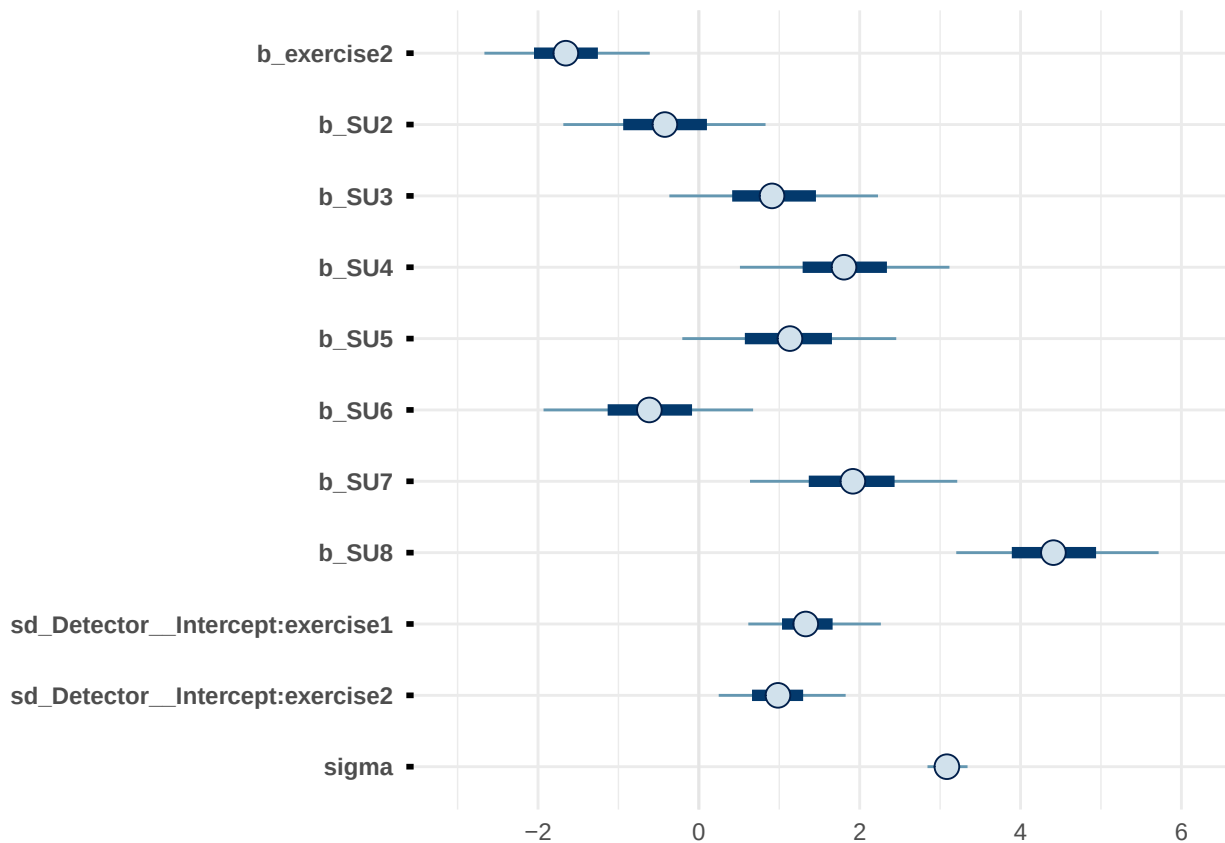
	Tail_ESS
sd(Intercept:exercise1)	1112
sd(Intercept:exercise2)	1173

```
##
## Regression Coefficients:
##
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	17.32	0.70	15.90	18.62	1.00	1383	1972
exercise2	-1.65	0.62	-2.87	-0.41	1.00	2044	2213
SU2	-0.42	0.78	-1.94	1.11	1.00	2217	2406
SU3	0.92	0.79	-0.64	2.49	1.00	1948	2689

```
## SU4          1.81      0.78      0.29      3.34 1.00      2167      2663
## SU5          1.12      0.80     -0.45      2.72 1.00      2112      2614
## SU6         -0.62      0.78     -2.20      0.89 1.00      2172      2666
## SU7          1.91      0.78      0.37      3.46 1.00      2239      2895
## SU8          4.42      0.76      2.96      5.90 1.00      1969      2558
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      3.09      0.15      2.80      3.40 1.00      3518      3202
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
brm_rich_2025 |>
  mcmc_intervals(pars = vars(b_exercise2:sigma))
```



```
#calculate contrast
posterior_2025 <- as_draws_df(brm_rich_2025) |>
  select(sd_before = 'sd_Detector__Intercept:exercise1',
         sd_after = 'sd_Detector__Intercept:exercise2') |>
  mutate(contrast = sd_after - sd_before,
         ratio = sd_after^2/sd_before^2,
         logratio = 2*(log(sd_after) - log(sd_before))) |>
  select(contrast, ratio, logratio) |>
```

```

pivot_longer(everything())

# Mean and 95% credible intervals for contrasts

posterior_2025 |>
  group_by(name) |>
  median_qi()

## # A tibble: 3 x 7
##   name      value  .lower .upper .width .point .interval
##   <chr>      <dbl>   <dbl> <dbl> <dbl> <chr>  <chr>
## 1 contrast -0.370 -1.79    1.02  0.95 median qi
## 2 logratio -0.652 -4.78    1.93  0.95 median qi
## 3 ratio     0.521  0.00838  6.91  0.95 median qi

# Posterior probability that sd is lower in exercise 2

posterior_2025 |>
  mutate(thres = if_else(name == "ratio", 1, 0)) |>
  group_by(name) |>
  summarise(mean(value < thres))

## # A tibble: 3 x 2
##   name      `mean(value < thres)`
##   <chr>                <dbl>
## 1 contrast              0.705
## 2 logratio              0.705
## 3 ratio                 0.705

#or alternatively
hypothesis(brm_rich_2025,
  "Detector__Intercept:exercise1^2 > Detector__Intercept:exercise2^2",
  class = "sd")

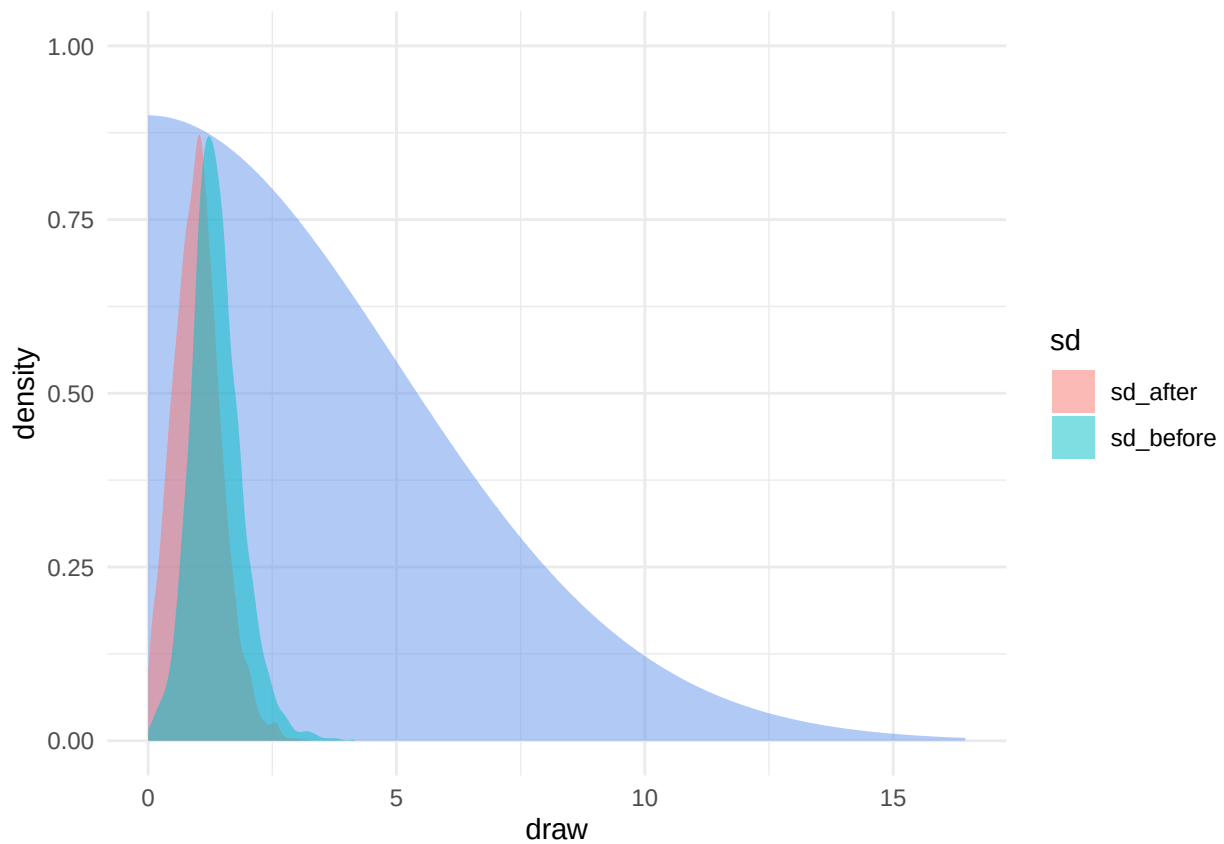
## Hypothesis Tests for class sd:
##           Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio
## 1 (Detector__Intercept... > 0      0.92      1.98   -1.92     4.2      2.39
##   Post.Prob Star
## 1      0.7
## ---
## 'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
## '*': For one-sided hypotheses, the posterior probability exceeds 95%;
## for two-sided hypotheses, the value tested against lies outside the 95%-CI.
## Posterior probabilities of point hypotheses assume equal prior probabilities.

### plots -----

```

From this we can see that the prior for SD is not informative, and that the posterior has converged to a tighter probability region.

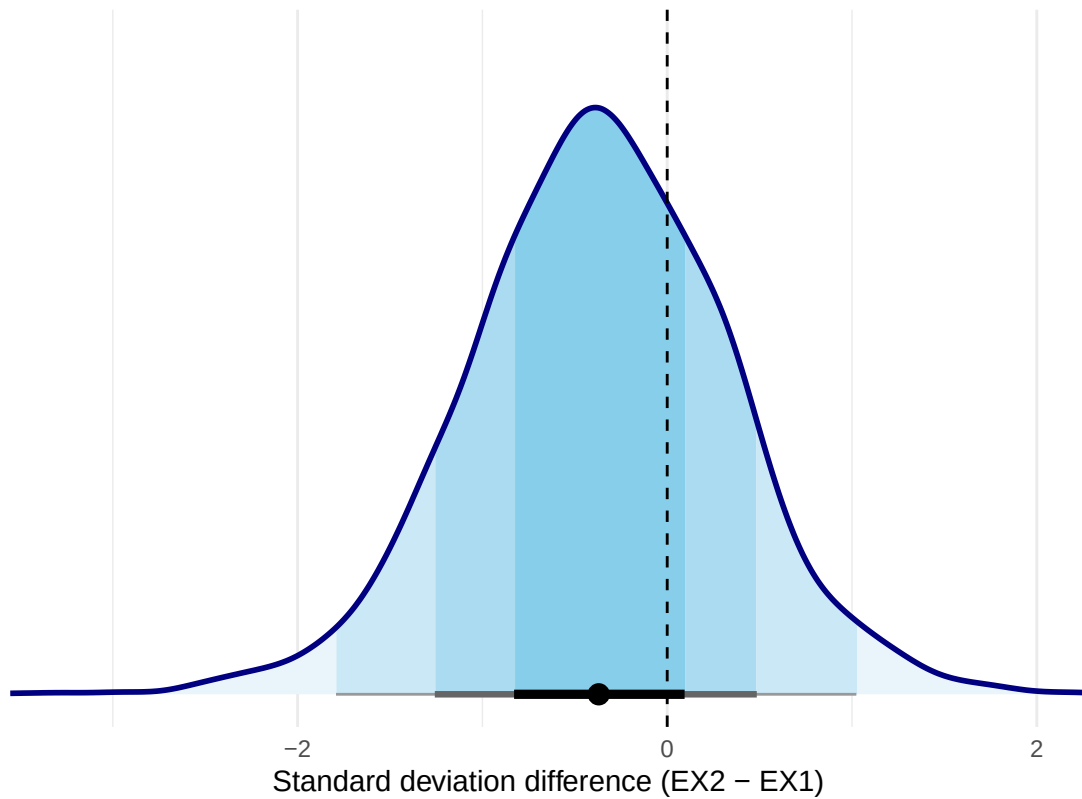
```
#prior for sd vs posteriors
brm_rich_2025 |>
  as_draws_df() |>
  select(sd_before = 'sd_Detector__Intercept:exercise1',
         sd_after = 'sd_Detector__Intercept:exercise2') |>
  pivot_longer(1:2, names_to = "sd", values_to = "draw") |>
  ggplot(aes(x = draw, fill = sd)) +
  stat_slab(aes(xdist = dist_truncated(dist_normal(0, 5), lower = 0)),
           fill = "cornflowerblue", alpha = 0.5, inherit.aes = F,
           data = tibble()) +
  geom_density(color = NA, alpha = 0.5)
```



These are alternative visualization of the contrast between exercises.

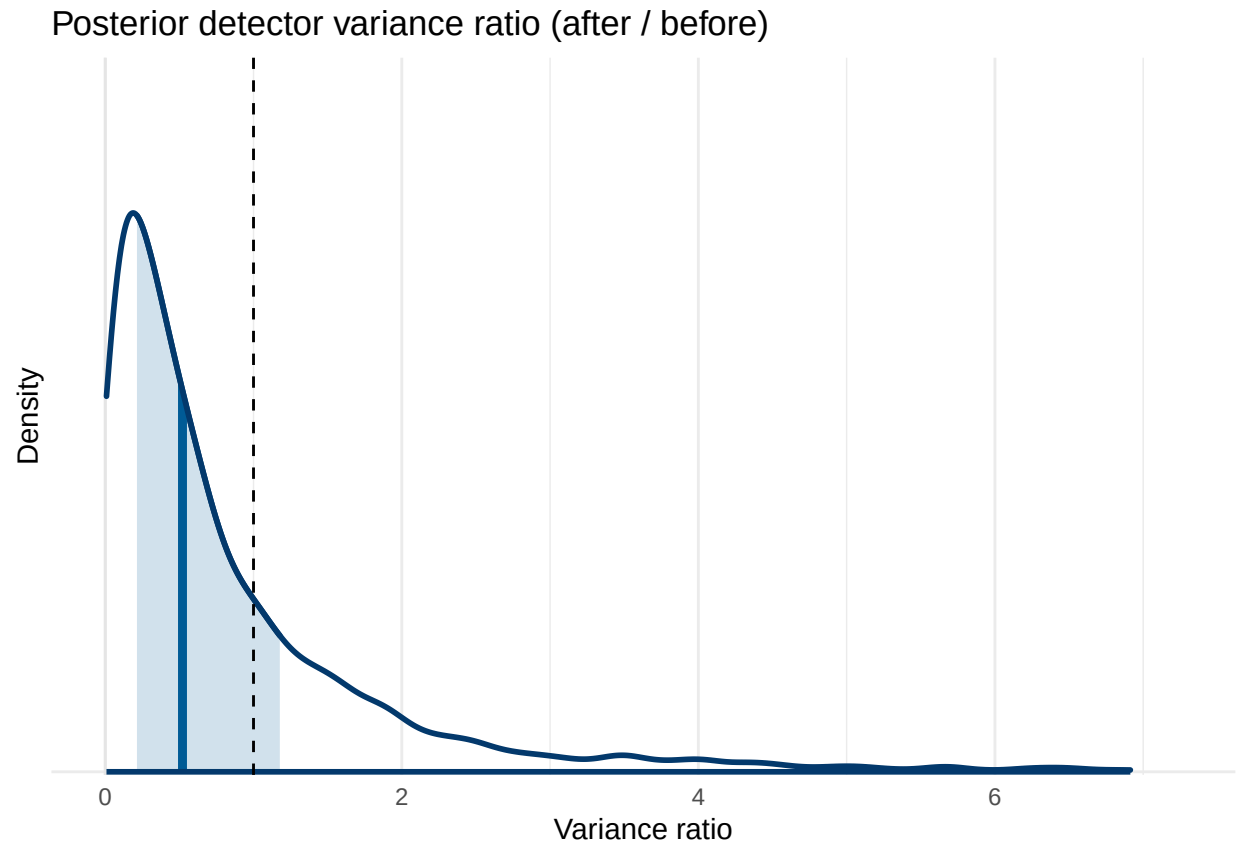
```
#difference
plot_rich_25 <- posterior_2025 |>
  filter(name == "contrast") |>
  plot_post() +
  labs(title = "2025",
       x = "Standard deviation difference (EX2 - EX1)",
       y = "")
plot_rich_25
```

2025



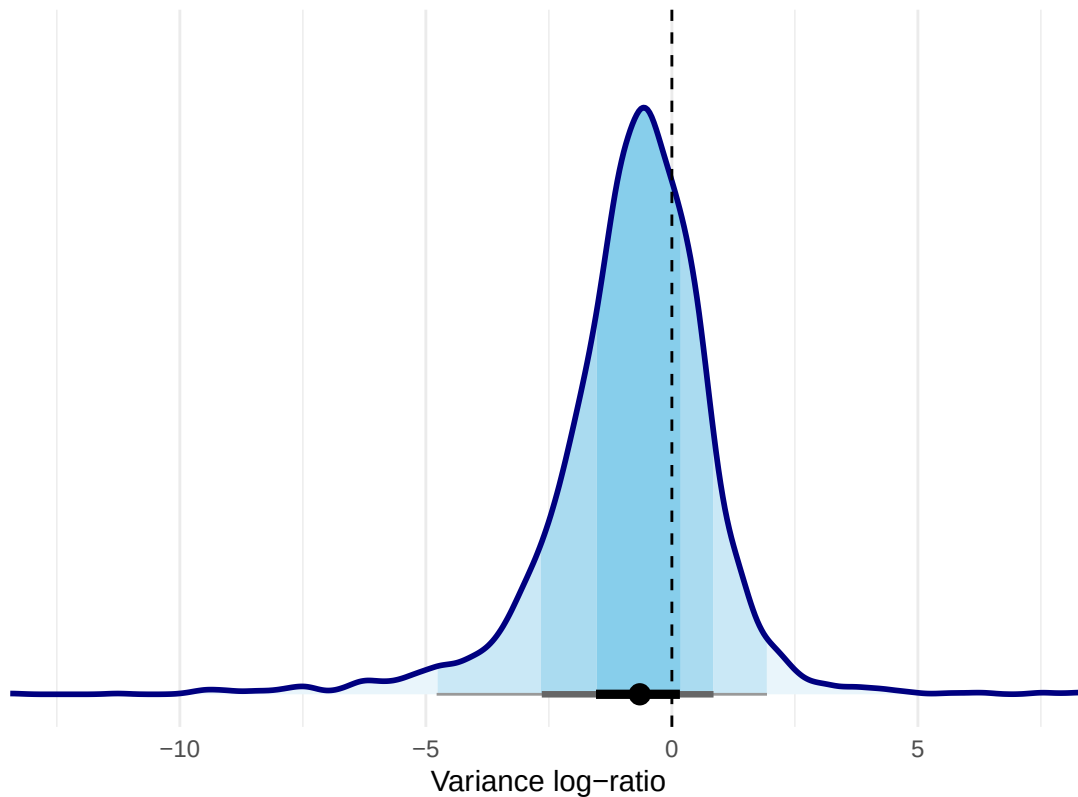
```
#ratio
```

```
posterior_2025 |>
  filter(name == "ratio") |>
  select(value) |>
  mcmc_areas(border_size = 1, prob = 0.5, prob_outer = 0.95, point_est = "median")+
  geom_vline(xintercept = 1, linetype = 2)+
  coord_cartesian(xlim = c(0, NA), ylim = c(1, 1.1))+
  labs(title = "Posterior detector variance ratio (after / before)",
       x = "Variance ratio", y = "Density")+
  theme_minimal()+
  theme(axis.text.y = element_blank())
```



```
#logratio  
  
posterior_2025 |>  
  filter(name == "logratio") |>  
  plot_post()+  
  labs(title = "Posterior detector variance log-ratio log(after / before)",  
        x = "Variance log-ratio", y = "")
```

Posterior detector variance log-ratio log(after / before)



```
## 2023 -----
brm_rich_2023 <- brm(formula = rich_formula,
  data = richness_2023,
  prior = prior_rich,
  file = "fits/rich_2023",
  seed = 967812)

prior_summary(brm_rich_2023)
```

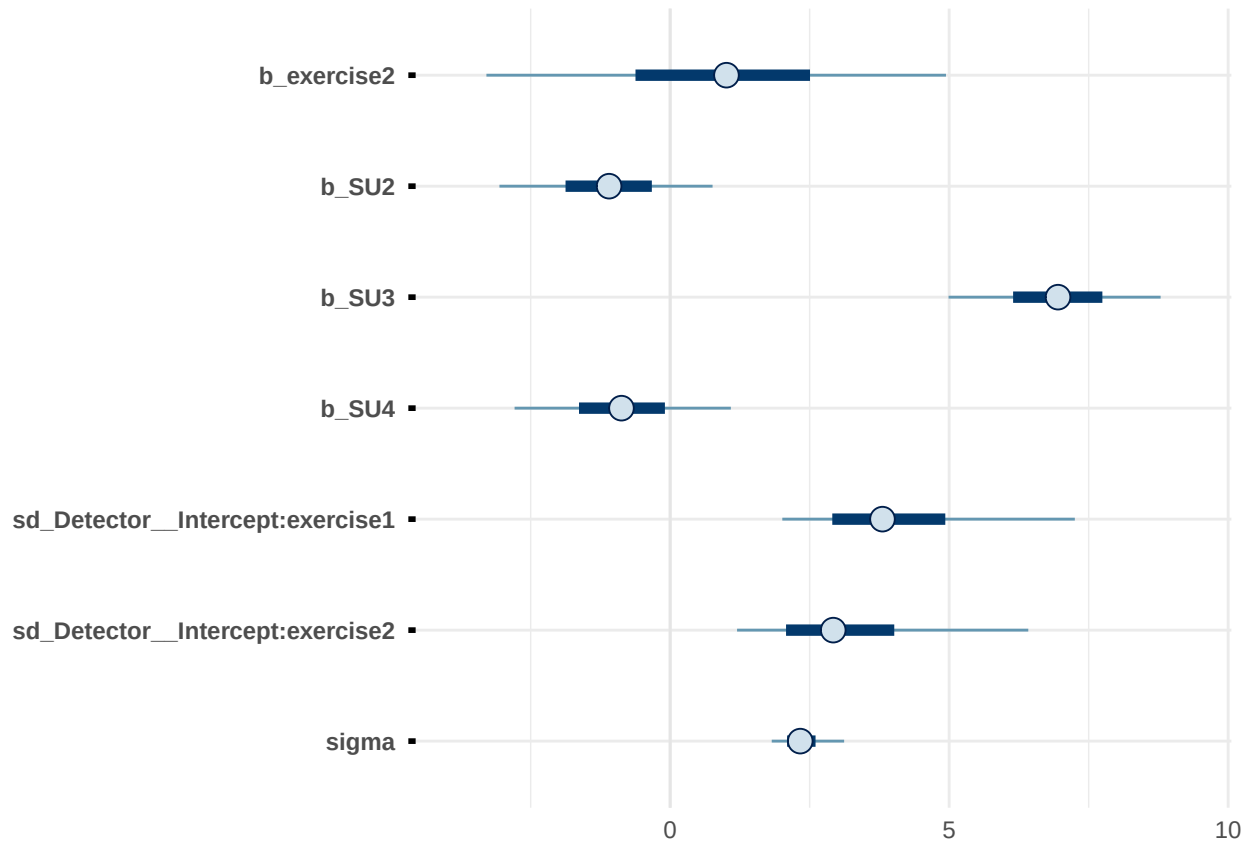
##	prior	class	coef	group	resp	dpar	nlpar	lb	ub	tag
##	normal(0, 5)	b								
##	normal(0, 5)	b	exercise2							
##	normal(0, 5)	b	SU2							
##	normal(0, 5)	b	SU3							
##	normal(0, 5)	b	SU4							
##	student_t(3, 25, 3.7)	Intercept								
##	normal(0, 5)	sd						0		
##	normal(0, 5)	sd	Detector					0		
##	normal(0, 5)	sd	Intercept	Detector				0		
##	student_t(3, 0, 3.7)	sigma						0		
##	source									
##	user									
##	(vectorized)									
##	(vectorized)									

```
## (vectorized)
## (vectorized)
##      default
##      user
## (vectorized)
## (vectorized)
##      default
```

```
#posterior summary
summary(brm_rich_2023)
```

```
## Family: gaussian
## Links: mu = identity
## Formula: n ~ exercise + SU + (1 | gr(Detector, by = exercise))
## Data: richness_2023 (Number of observations: 32)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##      total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Detector (Number of levels: 8)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS
## sd(Intercept:exercise1)    4.10      1.66    1.76    8.17 1.00    1561
## sd(Intercept:exercise2)    3.24      1.66    0.88    7.42 1.00    1390
##      Tail_ESS
## sd(Intercept:exercise1)    2181
## sd(Intercept:exercise2)    1464
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    23.33      2.03   19.21   27.38 1.00    1722    2036
## exercise2     0.92      2.52   -4.42    5.79 1.00    1624    1798
## SU2          -1.11      1.16   -3.46    1.16 1.00    3105    2621
## SU3           6.92      1.17    4.54    9.18 1.00    2860    2913
## SU4          -0.86      1.18   -3.19    1.49 1.00    3111    2654
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma      2.38      0.40    1.73    3.33 1.00    2051    1954
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
brm_rich_2023 |>
  mcmc_intervals(pars = vars(b_exercise2:sigma))
```



```
posterior_2023 <- as_draws_df(brm_rich_2023) |>
  select(sd_before = 'sd_Detector__Intercept:exercise1',
         sd_after = 'sd_Detector__Intercept:exercise2') |>
  mutate(contrast = sd_after - sd_before,
         ratio = sd_after^2/sd_before^2,
         logratio = 2*(log(sd_after) - log(sd_before))) |>
  select(contrast, ratio, logratio) |>
  pivot_longer(everything())
```

Mean and 95% credible intervals

```
posterior_2023 |>
  group_by(name) |>
  median_qi()
```

```
## # A tibble: 3 x 7
##   name      value .lower .upper .width .point .interval
##   <chr>      <dbl>   <dbl> <dbl>   <dbl> <chr>   <chr>
## 1 contrast -0.854 -5.47    3.89   0.95 median qi
## 2 logratio -0.550 -3.21    1.79   0.95 median qi
## 3 ratio     0.577  0.0404  5.99   0.95 median qi
```

Posterior probability that sd is lower in exercise 2

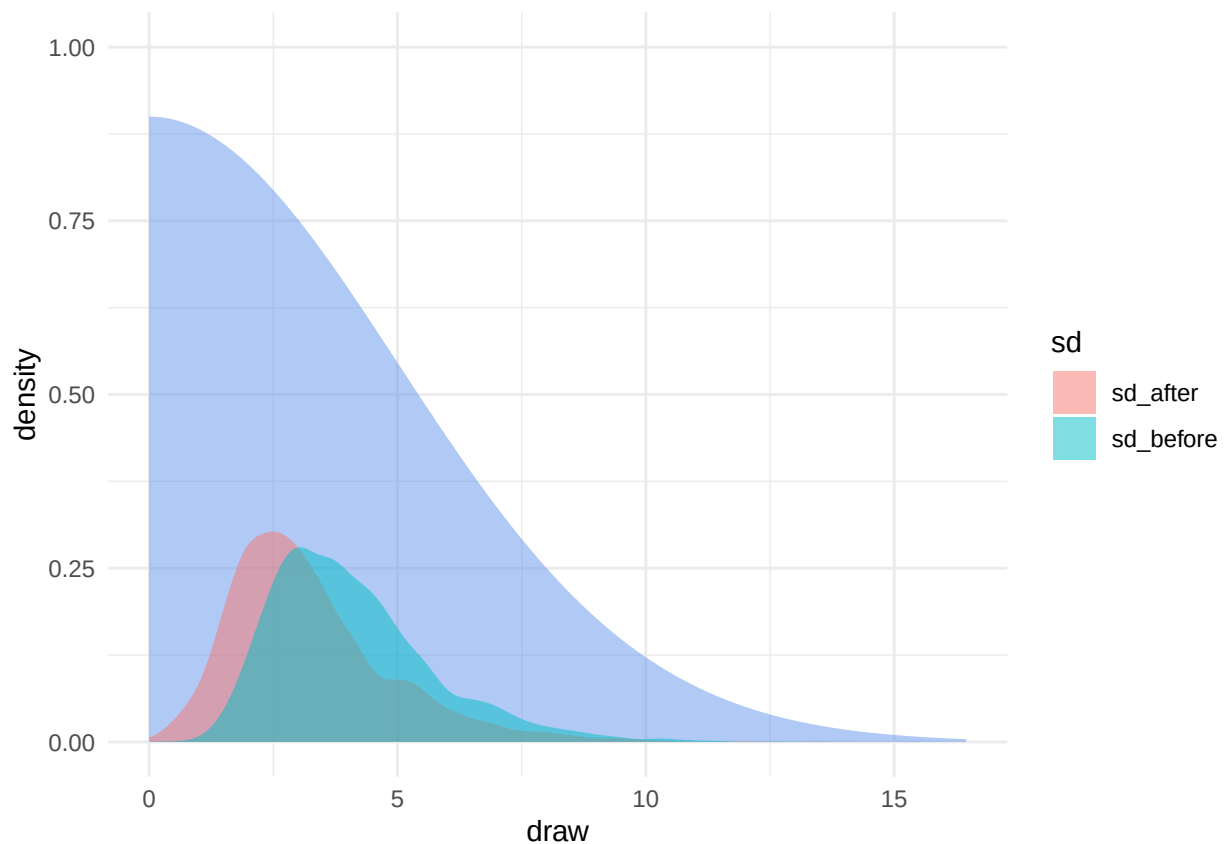
```
posterior_2023 |>
```

```
mutate(thres = if_else(name == "ratio", 1, 0)) |>
group_by(name) |>
summarise(mean(value < thres))
```

```
## # A tibble: 3 x 2
##   name      `mean(value < thres)`
##   <chr>                <dbl>
## 1 contrast              0.666
## 2 logratio              0.666
## 3 ratio                 0.666
```

```
### plots-----

#prior for sd vs posteriors
brm_rich_2023 |>
  as_draws_df() |>
  select(sd_before = 'sd_Detector__Intercept:exercise1',
         sd_after = 'sd_Detector__Intercept:exercise2') |>
  pivot_longer(1:2, names_to = "sd", values_to = "draw") |>
  ggplot(aes(x = draw, fill = sd)) +
  stat_slab(aes(xdist = dist_truncated(dist_normal(0, 5), lower = 0)),
           fill = "cornflowerblue", alpha = 0.5, inherit.aes = F,
           data = tibble()) +
  geom_density(color = NA, alpha = 0.5)
```

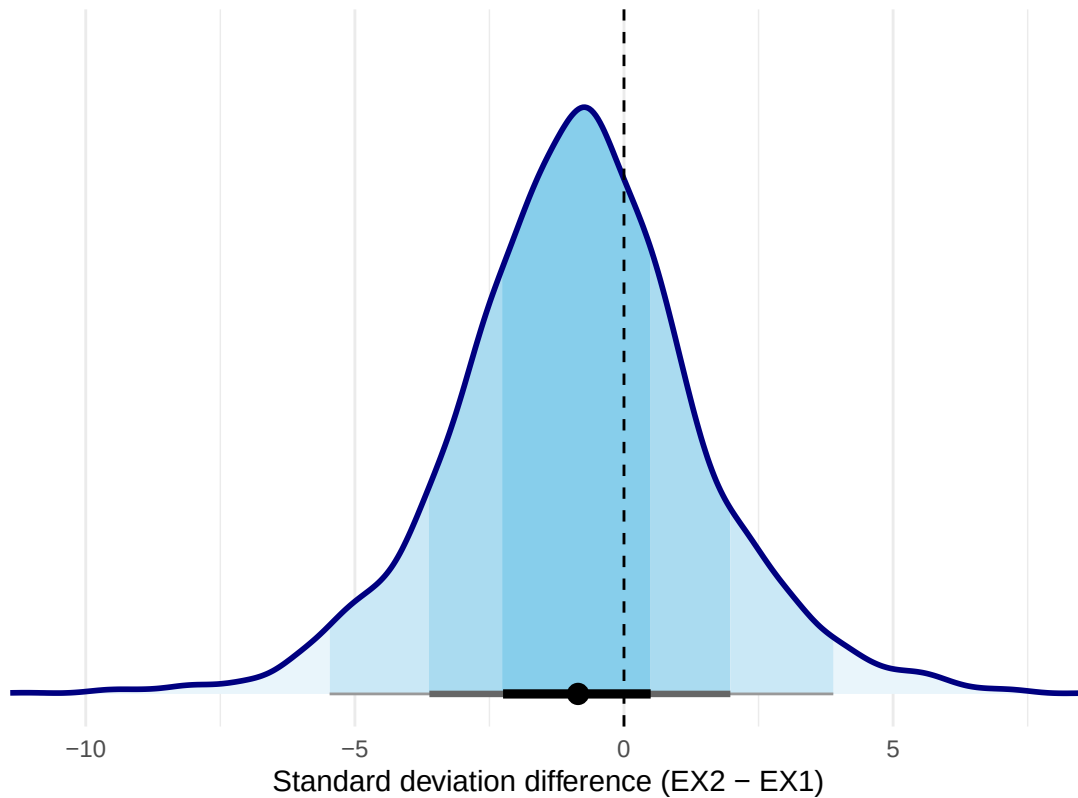


```
#difference
```

```
plot_rich_23 <- posterior_2023 |>  
  filter(name == "contrast") |>  
  plot_post() +  
  labs(title = "2023",  
        x = "Standard deviation difference (EX2 - EX1)",  
        y = "")
```

```
plot_rich_23
```

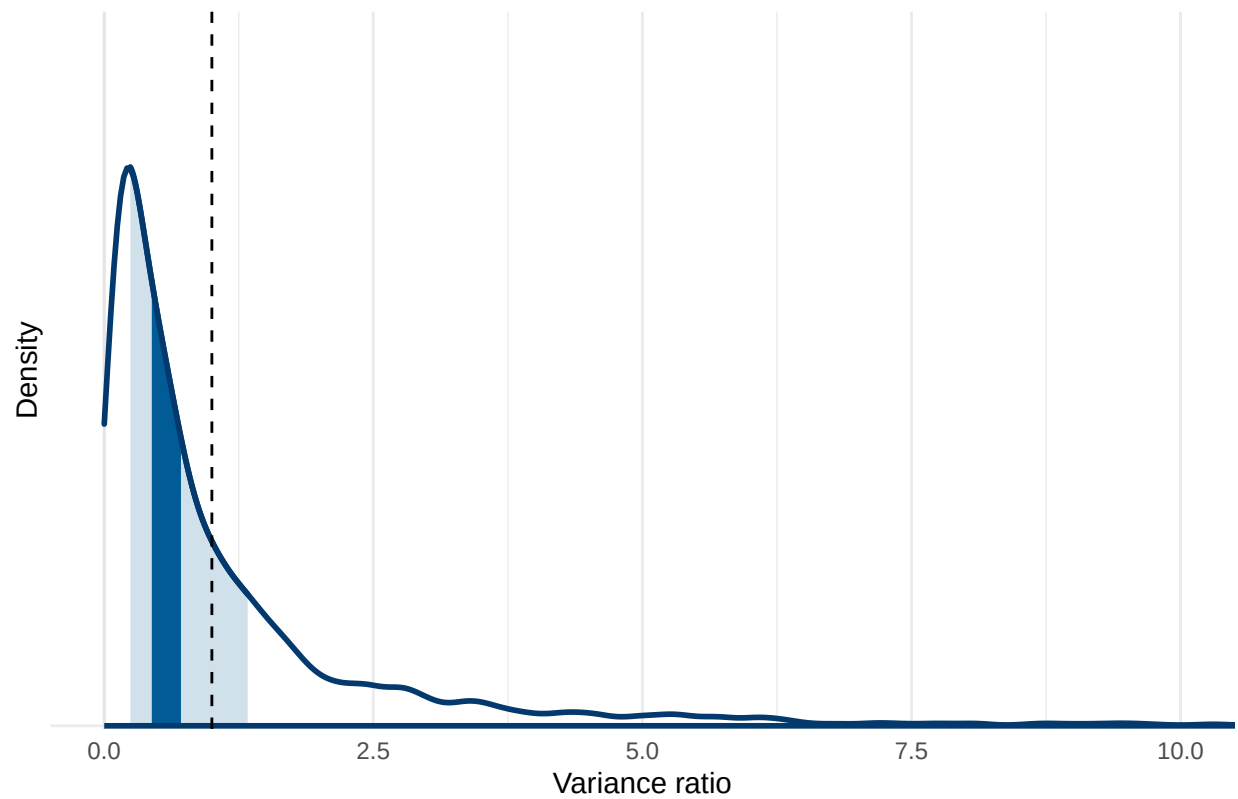
2023



```
#ratio
```

```
posterior_2023 |>  
  filter(name == "ratio") |>  
  select(value) |>  
  mcmc_areas(border_size = 1, point_est = "median")+  
  geom_vline(xintercept = 1, linetype = 2)+  
  coord_cartesian(xlim = c(0, 10), ylim = c(1, 1.1))+  
  labs(title = "Posterior detector variance ratio (after / before)",  
        x = "Variance ratio", y = "Density")+  
  theme_minimal()+  
  theme(axis.text.y = element_blank())
```

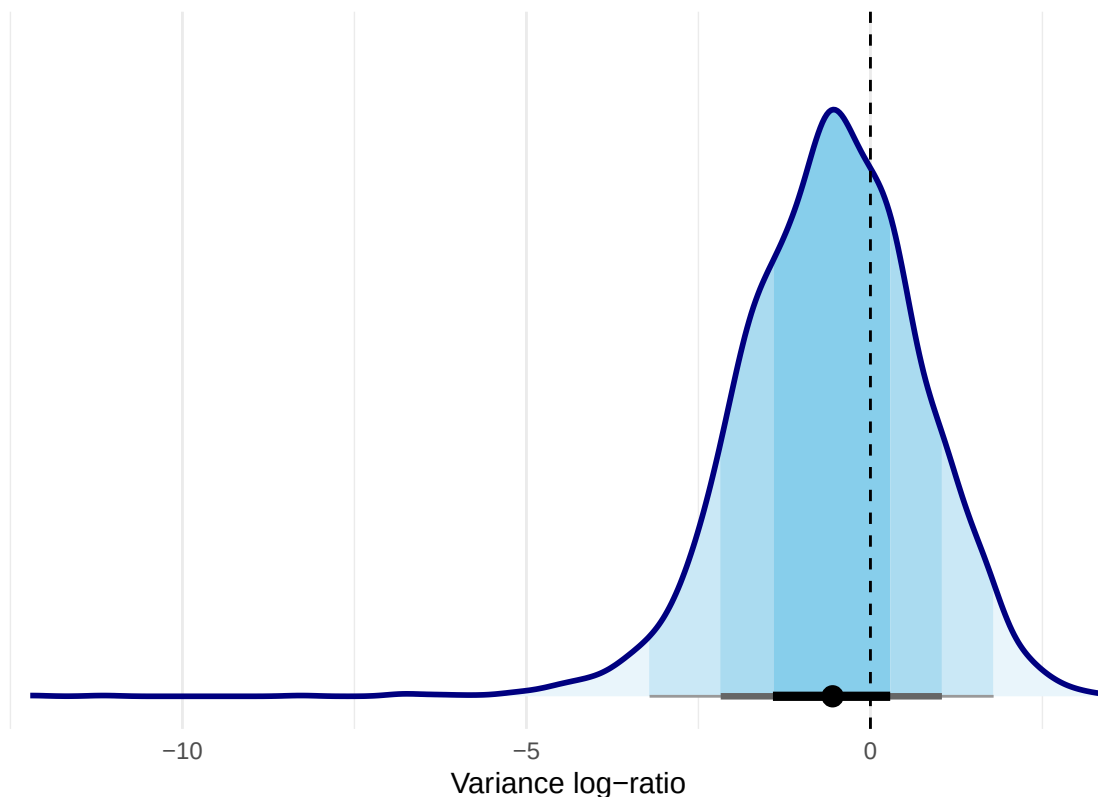
Posterior detector variance ratio (after / before)



```
#logratio

posterior_2023 |>
  filter(name == "logratio") |>
  plot_post() +
  labs(title = "Detector variance log-ratio log(after / before)",
        x = "Variance log-ratio", y = "")
```

Detector variance log-ratio log(after / before)



Models based on dissimilarity

Jaccard and Bray-Curtis dissimilarities are bounded between 0 and 1. Euclidean dissimilarity is also bounded in practice, although the upper bound depends on the data. In this case, it is also scaled in the (0, 1) range so that the same models can be used for all three. The likelihood of choice is beta, with logit link to the predictors.

```
# beta models -----
beta_formula <- brmsformula(dissimilarity ~ exercise + (1|SU) + (1|detector),
                             family = Beta(link_phi = "identity"))
get_prior(beta_formula, data = jaccard_2025)
```

```
##           prior      class      coef      group resp dpar nlpar lb ub tag
##           (flat)         b
##           (flat)         b exercise2
## student_t(3, 0, 2.5) Intercept
##   gamma(0.01, 0.01)      phi                0
## student_t(3, 0, 2.5)      sd                0
## student_t(3, 0, 2.5)      sd      detector    0
## student_t(3, 0, 2.5)      sd Intercept detector 0
## student_t(3, 0, 2.5)      sd              SU    0
```

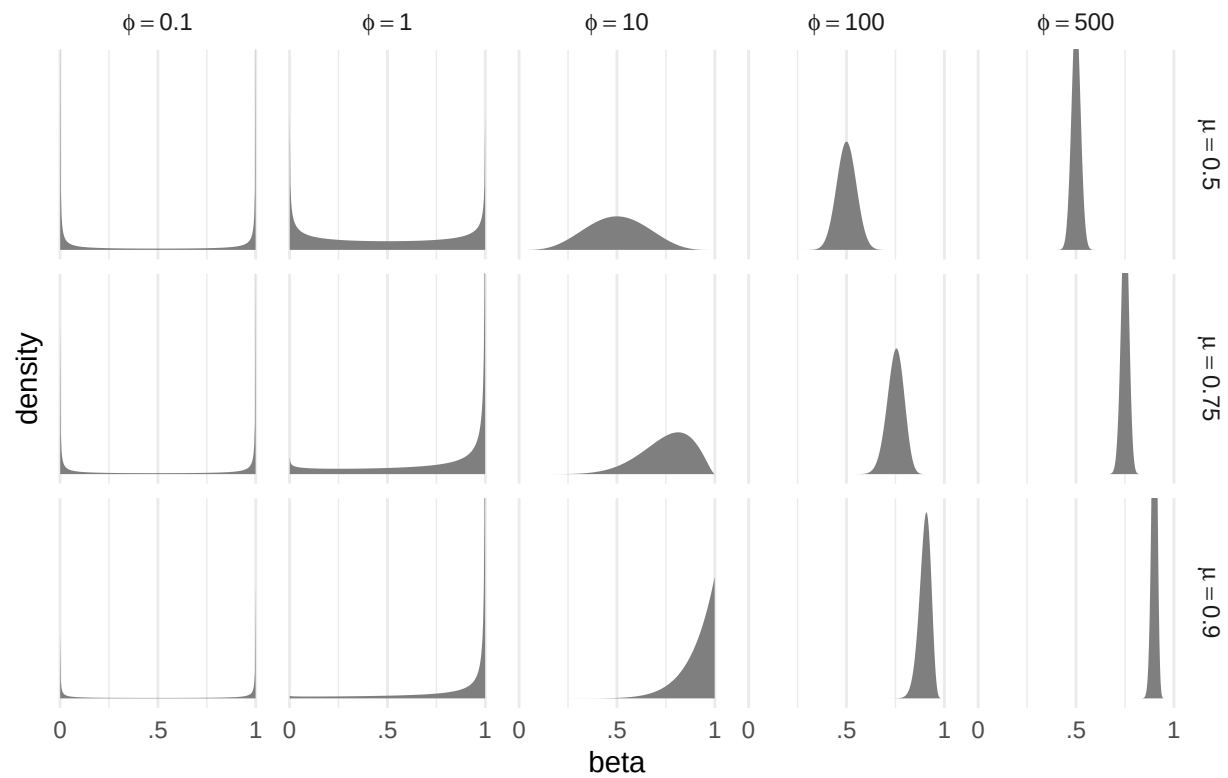
```
## student_t(3, 0, 2.5)      sd Intercept      SU      0
##      source
##      default
## (vectorized)
##      default
##      default
##      default
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

```
## priors-----

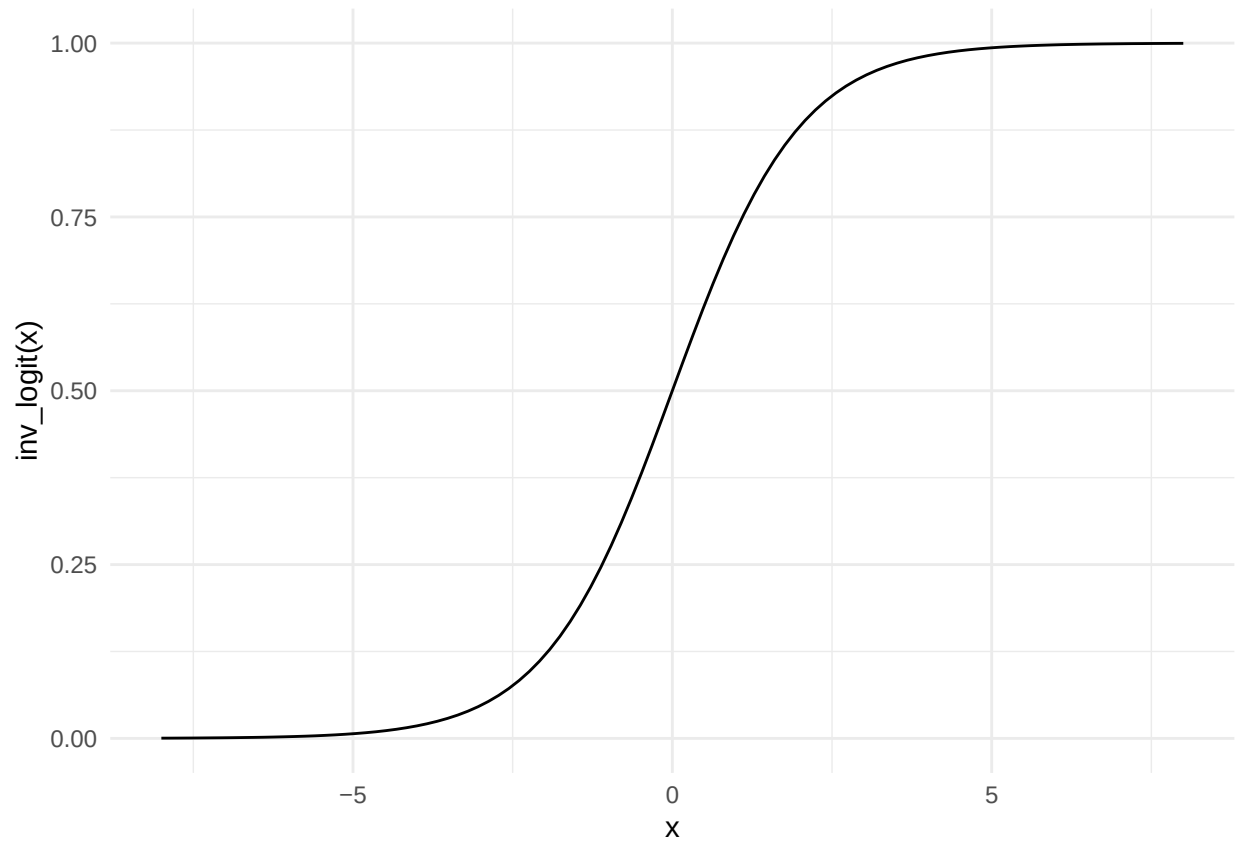
### visualize beta -----
#adapted from https://solomonkurz.netlify.app/blog/2023-06-25-causal-inference-with-beta-regression/

expand_grid(mu = c(.5, .75, .9),
            phi = c(0.1, 1, 10, 100, 500),
            beta = seq(from = .001, to = .999, length.out = 300)) |>
  mutate(d = dbeta(beta, shape1 = mu * phi, shape2 = phi - (mu * phi)),
         mu = str_c("mu==", mu),
         phi = str_c("phi==", phi)) %>%
  mutate(phi = factor(phi, levels = str_c("phi==", c(0.1, 1, 10, 100, 500)))) |>
  ggplot(aes(x = beta, y = d)) +
  geom_area(fill = "grey50") +
  scale_x_continuous(breaks = 0:2 / 2, labels = c("0", ".5", "1")) +
  scale_y_continuous("density", breaks = NULL) +
  ggtitle(expression("Beta given different combinations of \"*mu*\" and \"*phi\"")) +
  coord_cartesian(ylim = c(0, 14)) +
  facet_grid(mu ~ phi, labeller = label_parsed)
```

Beta given different combinations of μ and ϕ



```
### set priors -----
# the inverse logit function
ggplot()+
  geom_function(fun = plogis)+
  xlim(-8, 8)+
  ylab("inv_logit(x)")
```



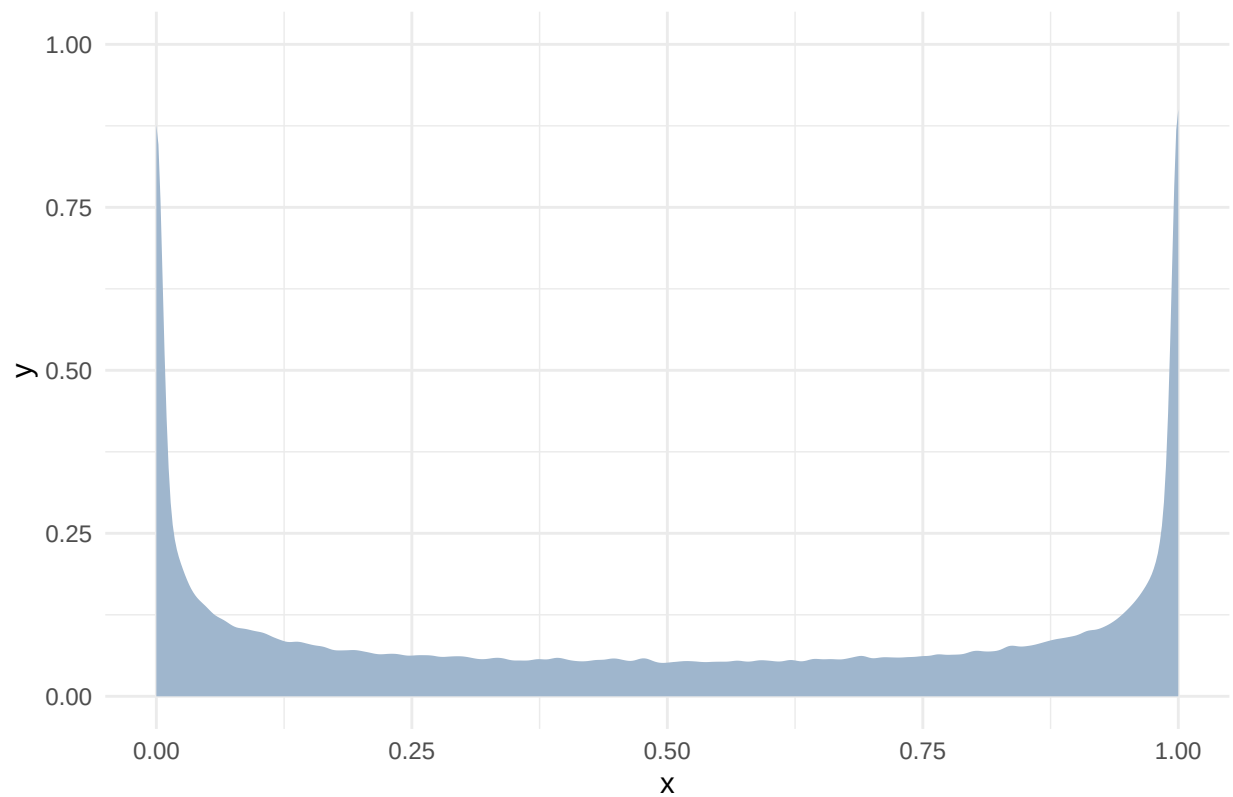
```
#intercept
```

A prior that is too wide results in too much probability mass on extreme values in the outcome scale because of the link function.

```
# default prior
```

```
tibble(x = rstudent_t(1e5, 3, 0, 2.5) |>  
  plogis()) |>  
ggplot(aes(x = x))+  
stat_slab(fill = "slategray3")+  
ggtitle("Default prior for Intercept on outcome scale")
```

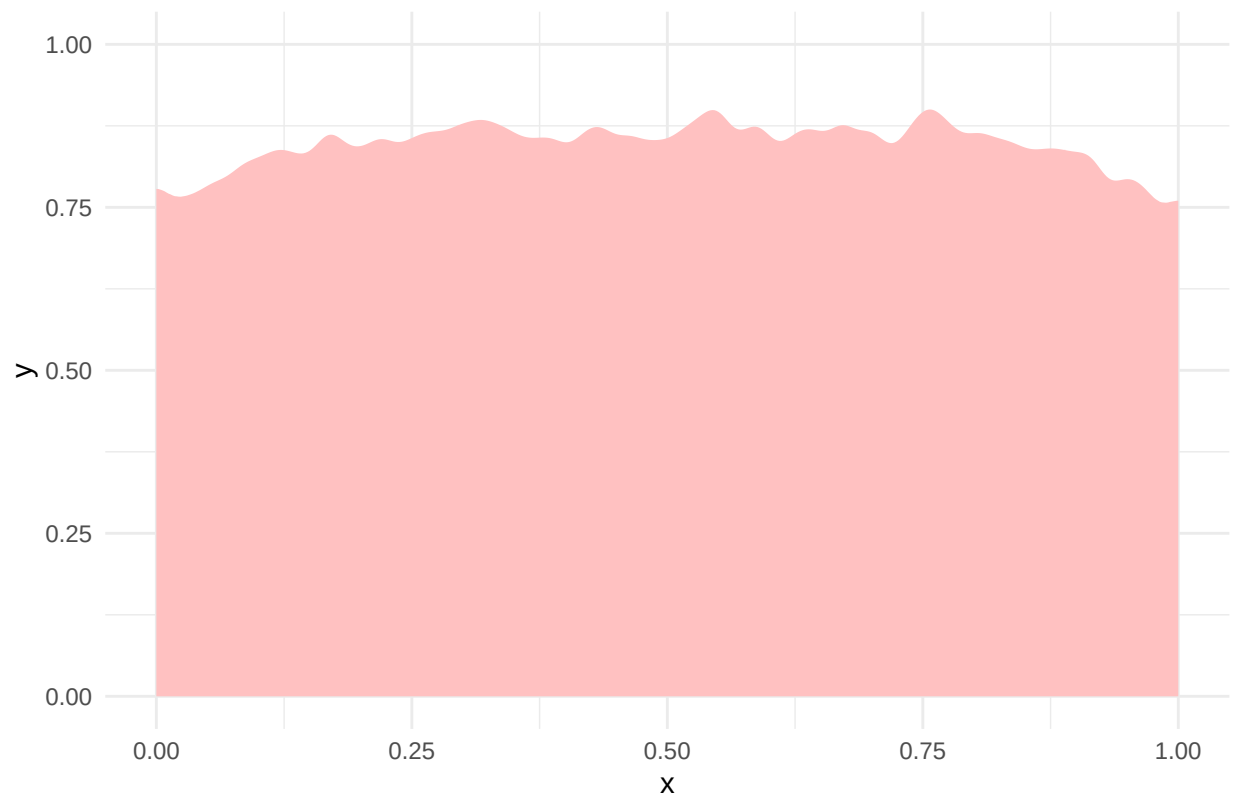
Default prior for Intercept on outcome scale



```
# adjusted prior
```

```
tibble(x = rstudent_t(1e5, 7, 0, 1.5) |>  
  plogis()) |>  
ggplot(aes(x = x))+  
stat_slab(fill = "rosybrown1")+  
ggtitle("Adjusted prior for Intercept on outcome scale")
```

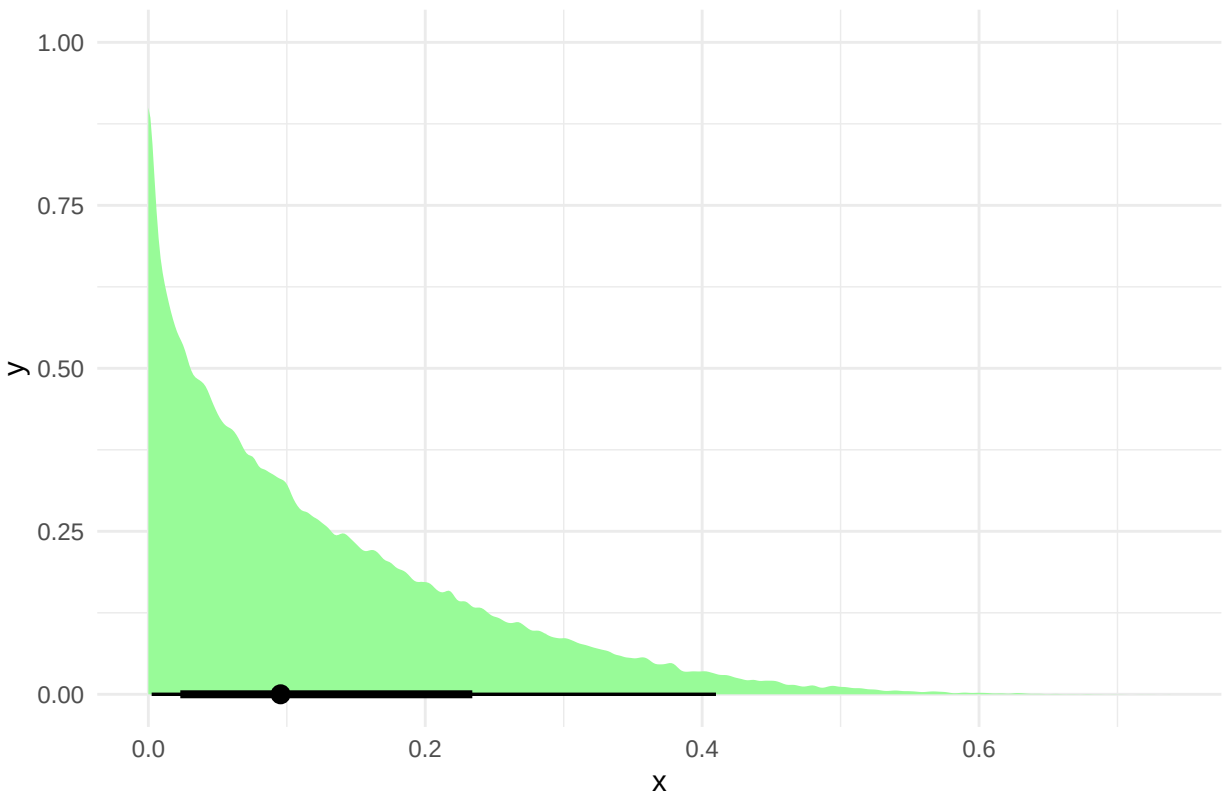
Adjusted prior for Intercept on outcome scale



exercise

```
tibble(int = rstudent_t(1e5, 7, 0, 1.5),  
       b0 = plogis(int),  
       b1 = plogis(int + rnorm(1e5, 0, 1)),  
       x = abs(b1 - b0)) |>  
ggplot(aes(x = x))+  
stat_halfeye(fill = "palegreen")+  
ggtitle("Prior for absolute difference of exercise on outcome scale")
```

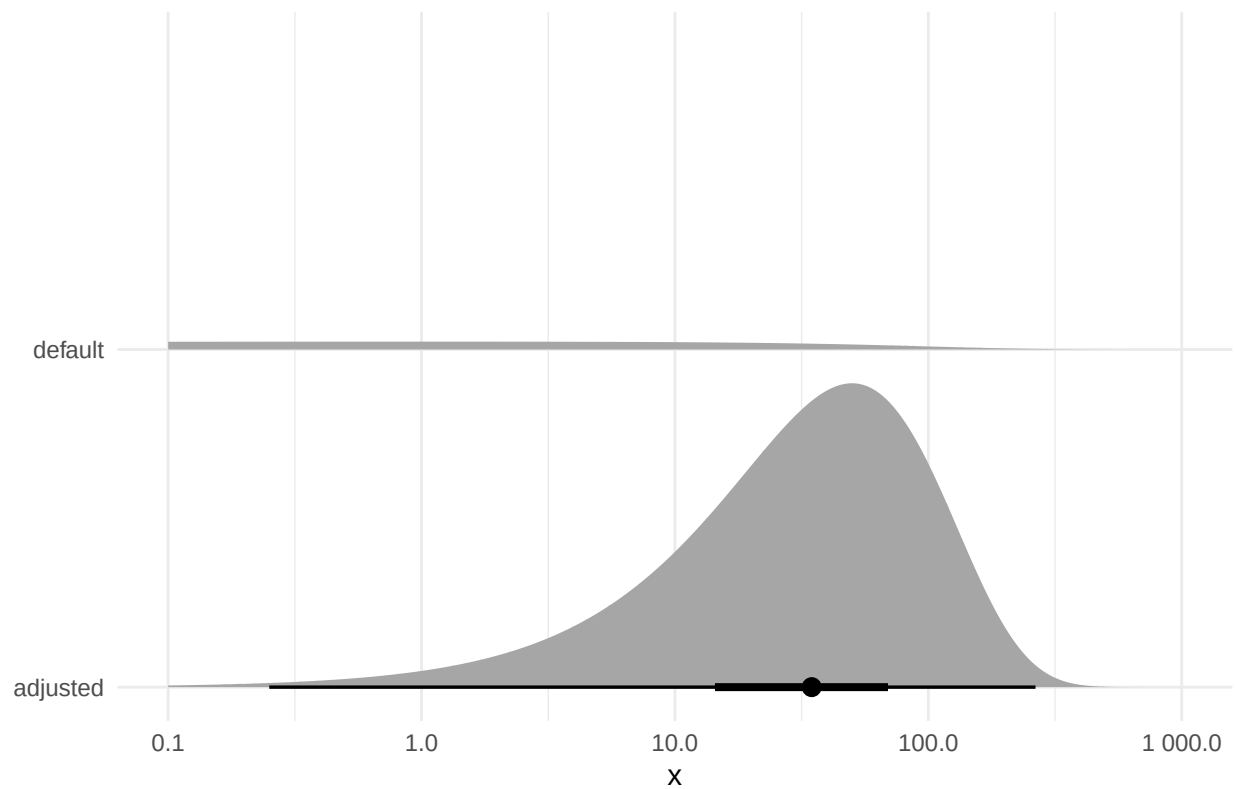
Prior for absolute difference of exercise on outcome scale



The default prior for ϕ puts most of the probability on values smaller than 1, which result in a bimodal distribution. The adjusted prior still allows this, but is concentrated on more reasonable distributions.

```
#  $\phi$ 
tibble(prior = c("default", "adjusted"),
       dist = c(dist_gamma(0.01, 0.01), dist_gamma(1, 0.02))) |>
  ggplot(aes(xdist = dist, y = prior)) +
  stat_halfeye(.width = c(.5, .99)) +
  scale_x_log10(limits = c(0.1, 1000),
               breaks = 10^(-1:3),
               labels = scales::label_number()) +
  scale_y_discrete(NULL, expand = expansion(add = 0.1)) +
  labs(title = expression("Prior for"~phi))
```

Prior for ϕ



```
#check some quantiles
```

```
qgamma(0.01, 1, 0.02)
```

```
## [1] 0.5025168
```

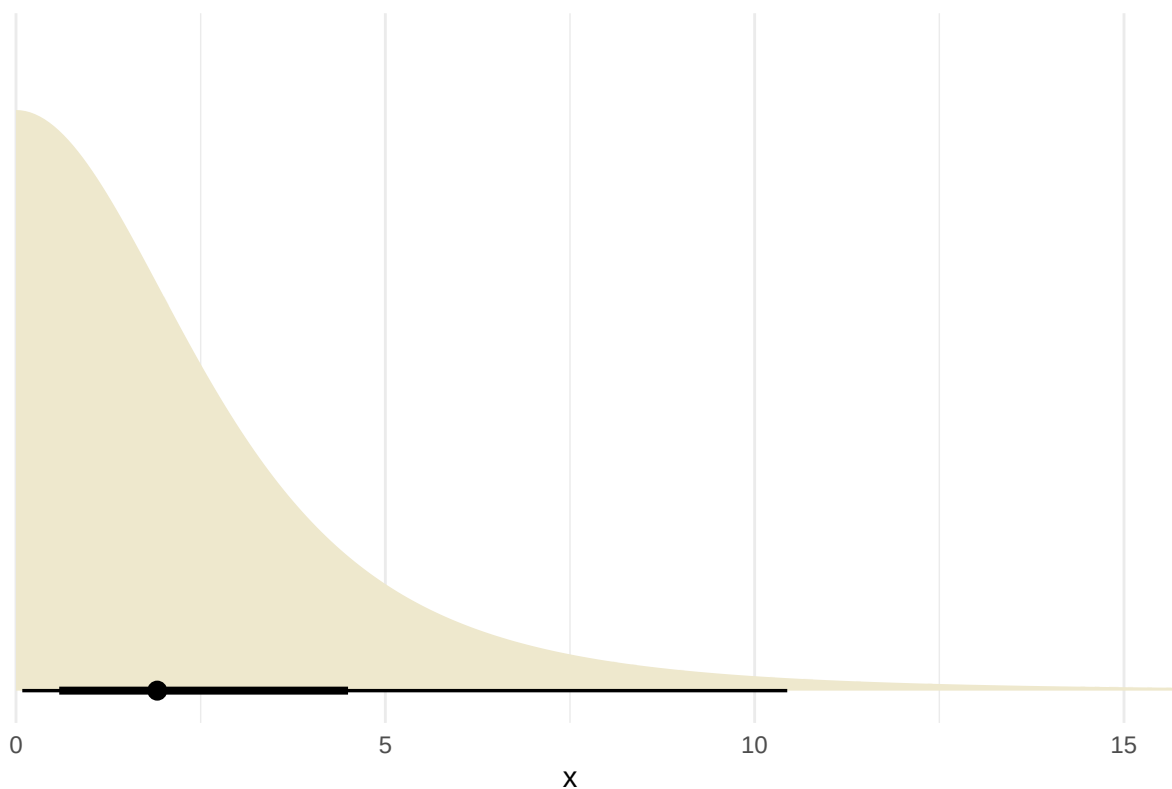
```
qgamma(0.99, 1, 0.02)
```

```
## [1] 230.2585
```

```
# sd
```

```
ggplot()+
  stat_halfeye(aes(xdist = dist_truncated(dist_student_t(3, 0, 2.5), lower = 0)),
    fill = "cornsilk2", .width = c(0.66, 0.95))+
  scale_y_continuous(NULL, breaks = NULL)+
  coord_cartesian(xlim = c(0, 15))+
  ggtitle(expression("Default prior for"~sigma))
```

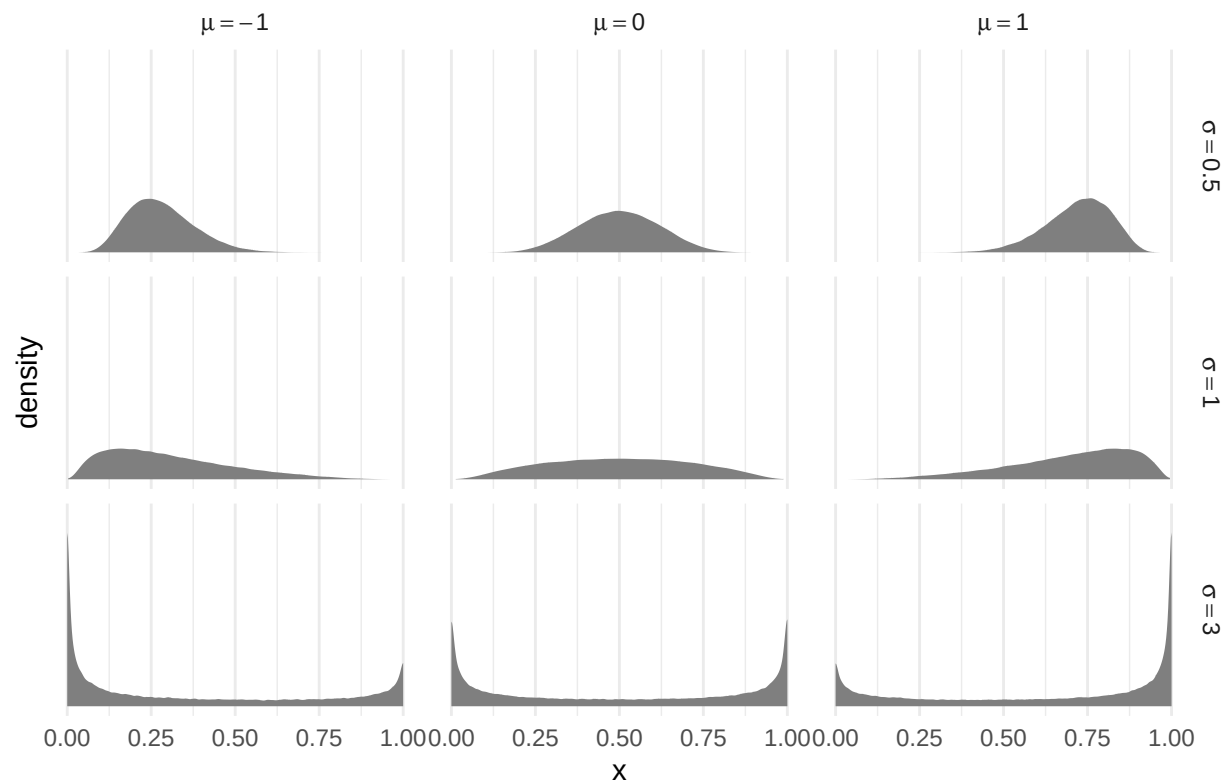
Default prior for σ



With this default prior, a SD of 3 is very likely. What does it mean to have a standard deviation of 3 on the outcome scale?

```
expand_grid(mu = c(-1, 0, 1),
            sd = c(0.5, 1, 3)) |>
  mutate(x = map2(mu, sd, \(x, y) rnorm(1e5, x, y))) |>
  unnest(x) |>
  mutate(mu = str_c("mu==", mu),
         sd = str_c("sigma==", sd),
         x = plogis(x)) |>
  ggplot(aes(x = x)) +
  stat_slab(fill = "grey50") +
  scale_y_continuous("density", breaks = NULL) +
  facet_grid(sd ~ mu, labeller = label_parsed) +
  ggtitle("Effect of different random SDs on outcome scale")
```

Effect of different random SDs on outcome scale



It is clear that SDs larger than 1 should be avoided, since they can shift the intercept to the extremes of the scale, regardless of the mean value. Here is then a more reasonable prior:

```
ggplot()+
  stat_halfeye(aes(xdist = dist_truncated(dist_student_t(5, 0, 0.5), lower = 0)),
    fill = "cornsilk2", .width = c(0.66, 0.95))+
  scale_y_continuous(NULL, breaks = NULL)+
  ggtitle(expression("Default prior for"~sigma))
```

Default prior for σ



```
prior_beta <- c(prior(student_t(7, 0, 1.5), class = "Intercept"),
  prior(normal(0, 1), class = "b"),
  prior(gamma(1, 0.02), class = "phi", lb = 0),
  prior(student_t(5, 0, 0.5), lb = 0, class = "sd"))

# 2023 has fewer levels, requiring a tighter prior

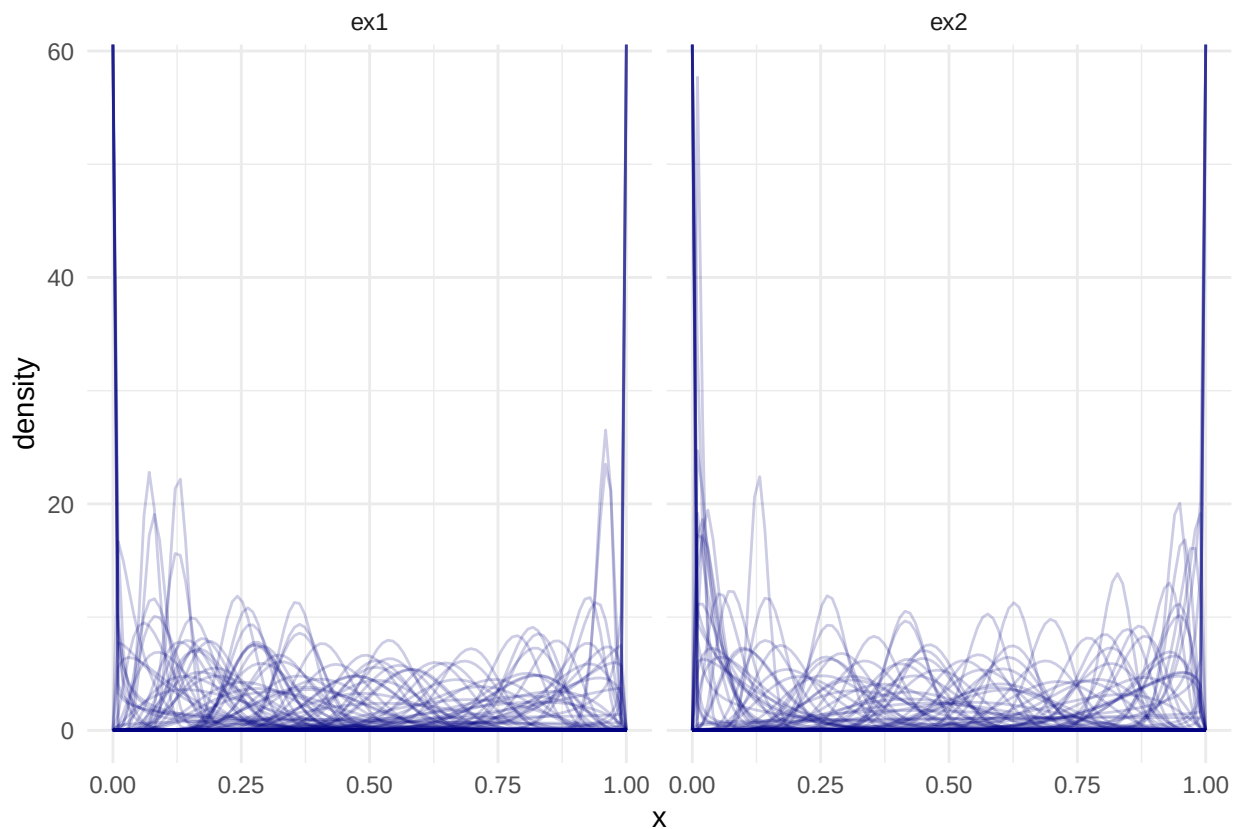
prior_23 <- c(prior(student_t(7, 0, 1.5), class = "Intercept"),
  prior(normal(0, 1), class = "b"),
  prior(gamma(1, 0.02), class = "phi", lb = 0),
  prior(student_t(3, 0, 0.25), lb = 0, class = "sd"))

data_prior <- data.frame(dissimilarity = 0.5,
  exercise = factor(1:2),
  SU = factor(1:2),
  detector = factor(1:2))

prior_pred <- brm(beta_formula,
  data = data_prior,
  sample_prior = "only",
  prior = prior_beta,
  file = "fits/beta_prior",
  seed = 9512)
```

Here's a prior predictive check for beta distributions.

```
prior_pred |>
  spread_draws(b_Intercept, b_exercise2, sd_SU__Intercept, sd_detector__Intercept, phi) |>
  slice_sample(n = 100) |>
  mutate(su = rnorm(n(), 0, sd_SU__Intercept),
         det = rnorm(n(), 0, sd_detector__Intercept),
         ex1 = plogis(b_Intercept + su + det),
         ex2 = plogis(b_Intercept + b_exercise2, su + det)) |>
  pivot_longer(c(ex1, ex2)) |>
  expand_grid(x = seq(0, 1, length.out = 100)) |>
  mutate(density = dbeta(x, value*phi, (1-value)*phi)) |>
  ggplot(aes(x = x, y = density, group = .draw)) +
  geom_line(alpha = 0.2, color = "navy")+
  facet_wrap(~ name)
```



```
## 2025 -----
```

```
### jaccard -----
```

```
brm_jac_2025 <- brm(formula = beta_formula,
  data = jaccard_2025,
  prior = prior_beta,
  file = "fits/jac_2025",
  seed = 85628)
```

```
prior_summary(brm_jac_2025)
```

```
##           prior      class      coef      group resp dpar nlpar lb ub tag
##      normal(0, 1)         b
##      normal(0, 1)         b exercise2
## student_t(7, 0, 1.5) Intercept
##      gamma(1, 0.02)      phi                                0
## student_t(5, 0, 0.5)      sd                                0
## student_t(5, 0, 0.5)      sd      detector                  0
## student_t(5, 0, 0.5)      sd Intercept detector            0
## student_t(5, 0, 0.5)      sd      SU                        0
## student_t(5, 0, 0.5)      sd Intercept SU                    0
##      source
##      user
## (vectorized)
##      user
##      user
##      user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

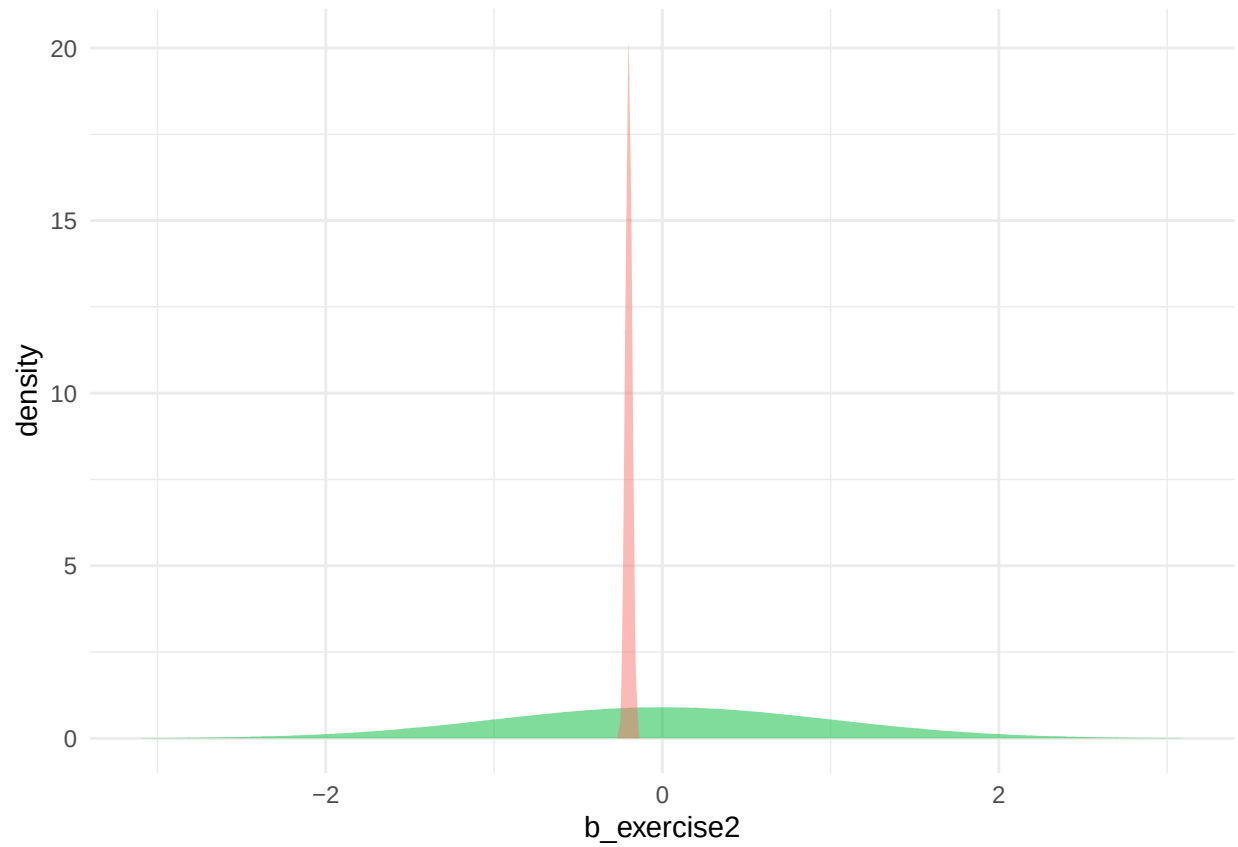
```
summary(brm_jac_2025)
```

```
## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: jaccard_2025 (Number of observations: 3304)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##      total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 16)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.26    0.05    0.18    0.39 1.00      729    1336
##
## ~SU (Number of levels: 8)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.19    0.06    0.11    0.35 1.00      856    1820
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    -0.08    0.09   -0.27    0.11 1.00      709    1243
## exercise2    -0.20    0.02   -0.24   -0.16 1.00     3055    2709
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi    13.01    0.32   12.41   13.65 1.00     3417    2701
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

It is clear from the following that the prior is very uninformative. This is on the logit scale.

```
#posterior vs prior for exercise
```

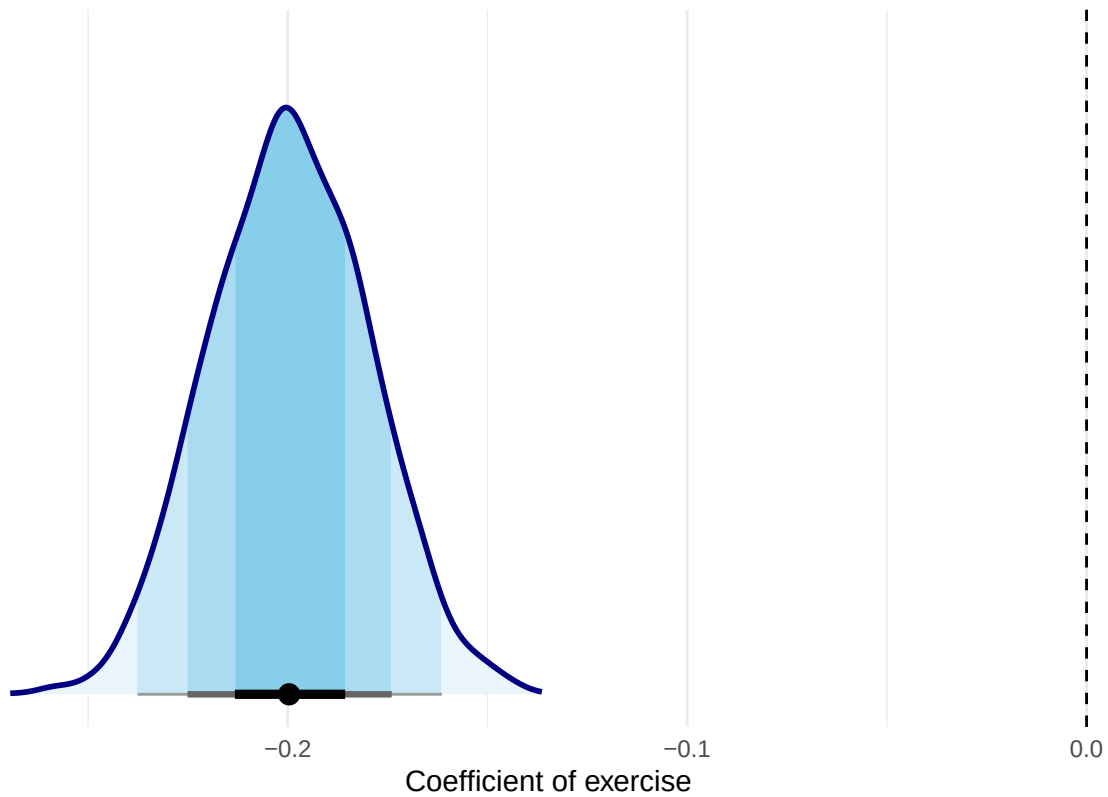
```
brm_jac_2025 |>  
  post_prior_beta()
```



```
#on the logit scale
```

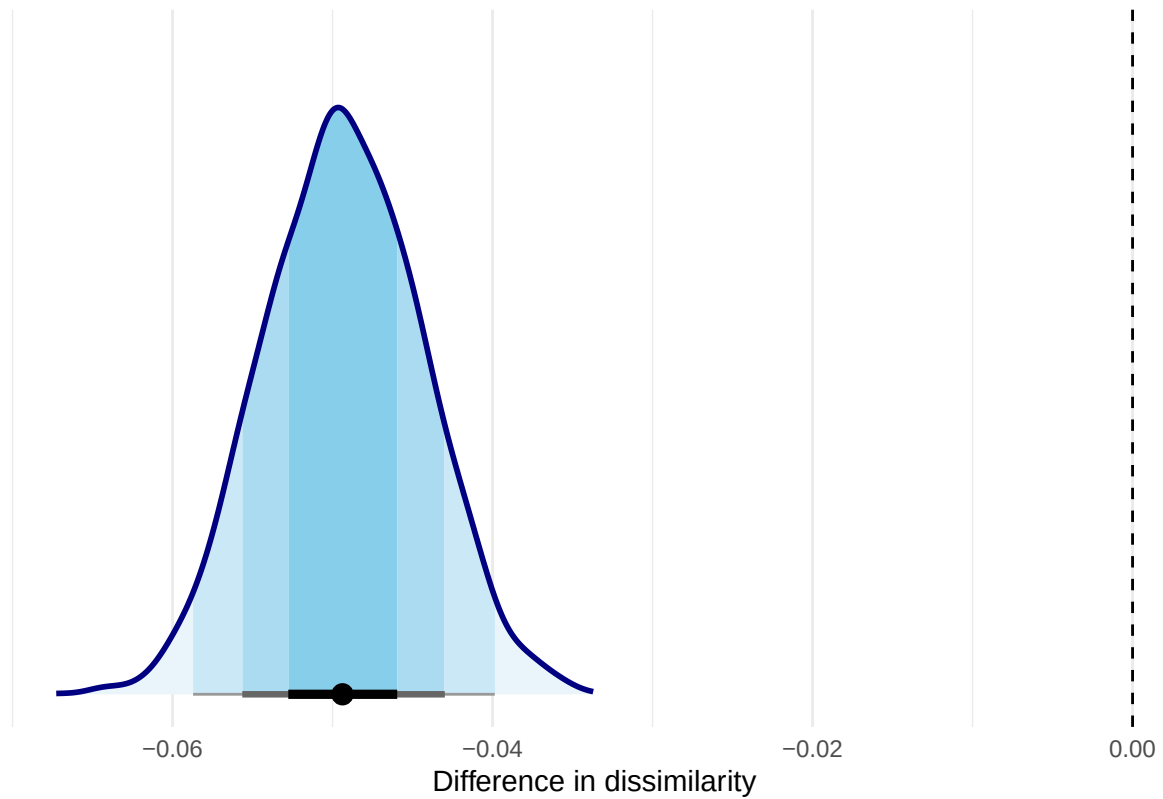
```
brm_jac_2025 |>  
  plot_post(fit = T) +  
  labs(title = "Posterior distribution for Jaccard dissimilarity",  
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Jaccard dissimilarity



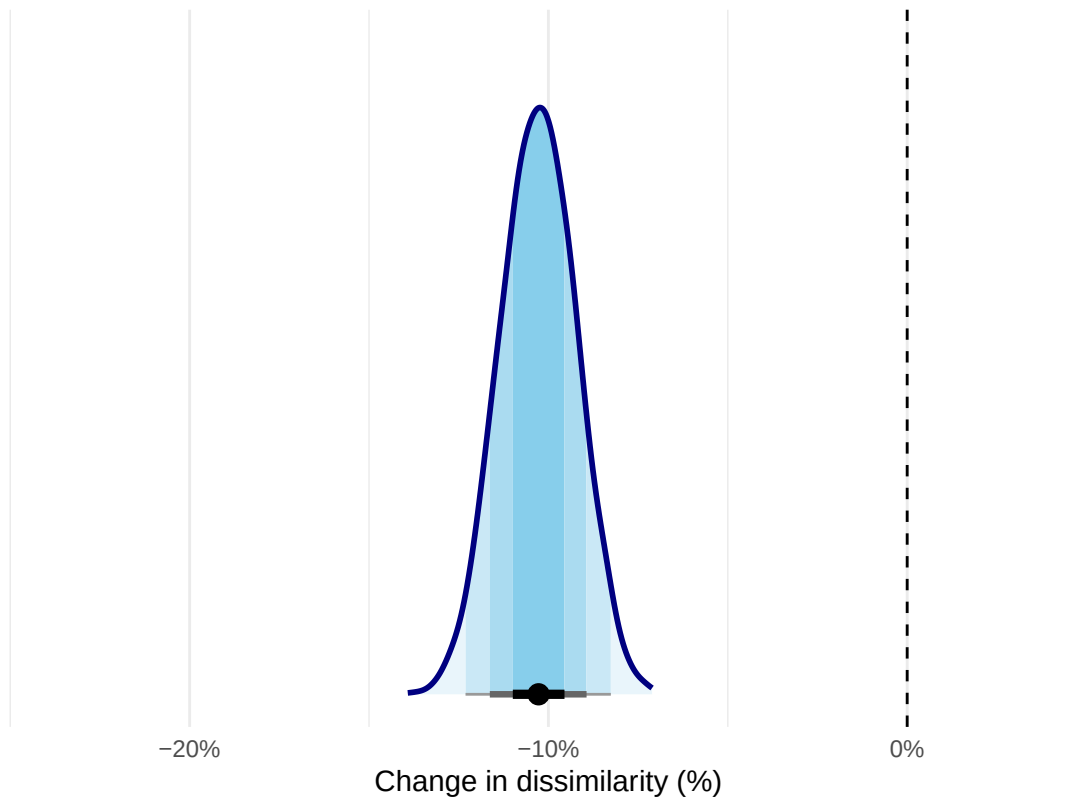
```
#on the outcome scale  
#as difference  
brm_jac_2025 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Jaccard dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

Effect of training on Jaccard dissimilarity

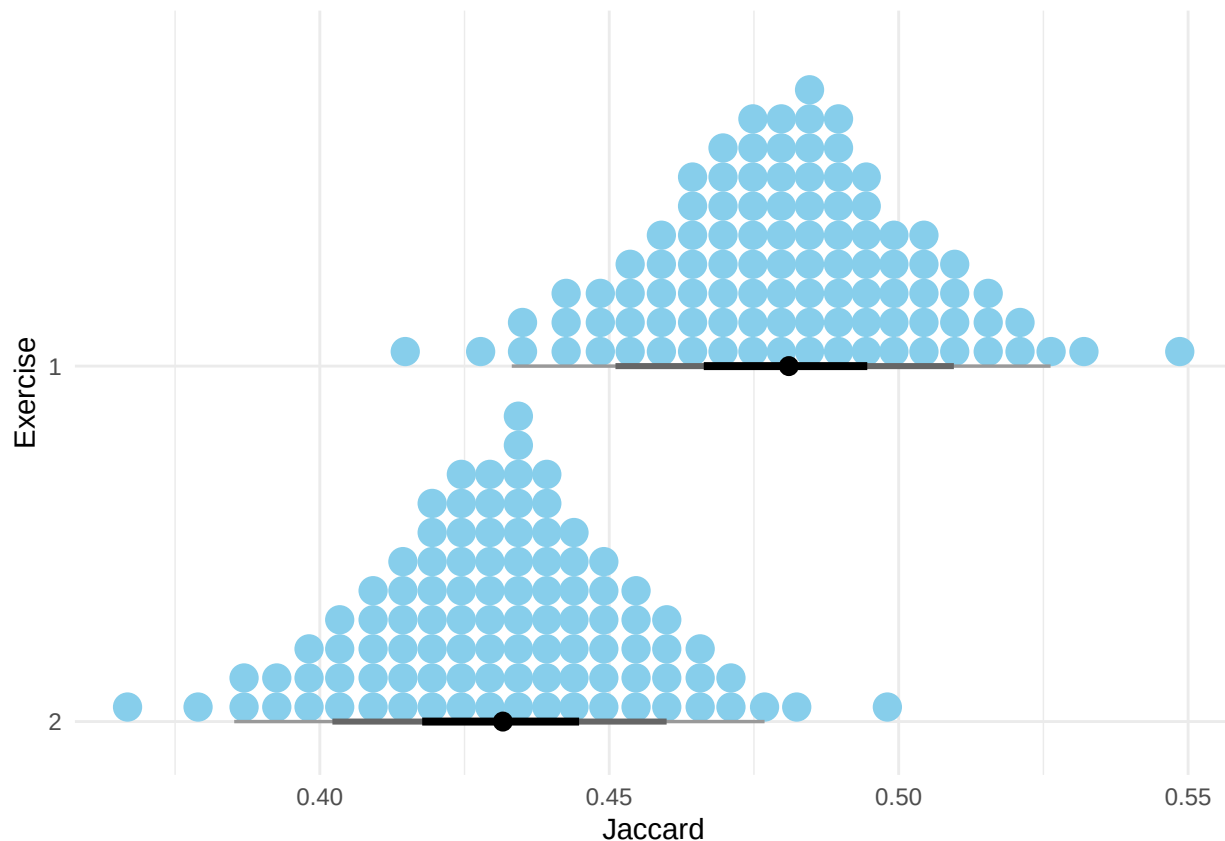


```
#as percent lift
plot_lift_jac_25 <- brm_jac_2025 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(),
                     limits = c(-0.25, 0.05))+
  labs(title = "Jaccard", x = "Change in dissimilarity (%)", y = "")
plot_lift_jac_25
```

Jaccard



```
#quantile dotplot of predictions  
plot_pred_jac_25 <- brm_jac_2025 |>  
  plot_pred() +  
  labs(x = "Jaccard", y = "Exercise")  
plot_pred_jac_25
```



```
### bray-curtis ----
```

```
brm_bray_2025 <- brm(formula = beta_formula,
  data = bray_2025,
  prior = prior_beta,
  file = "fits/bray_2025",
  seed = 61278)
```

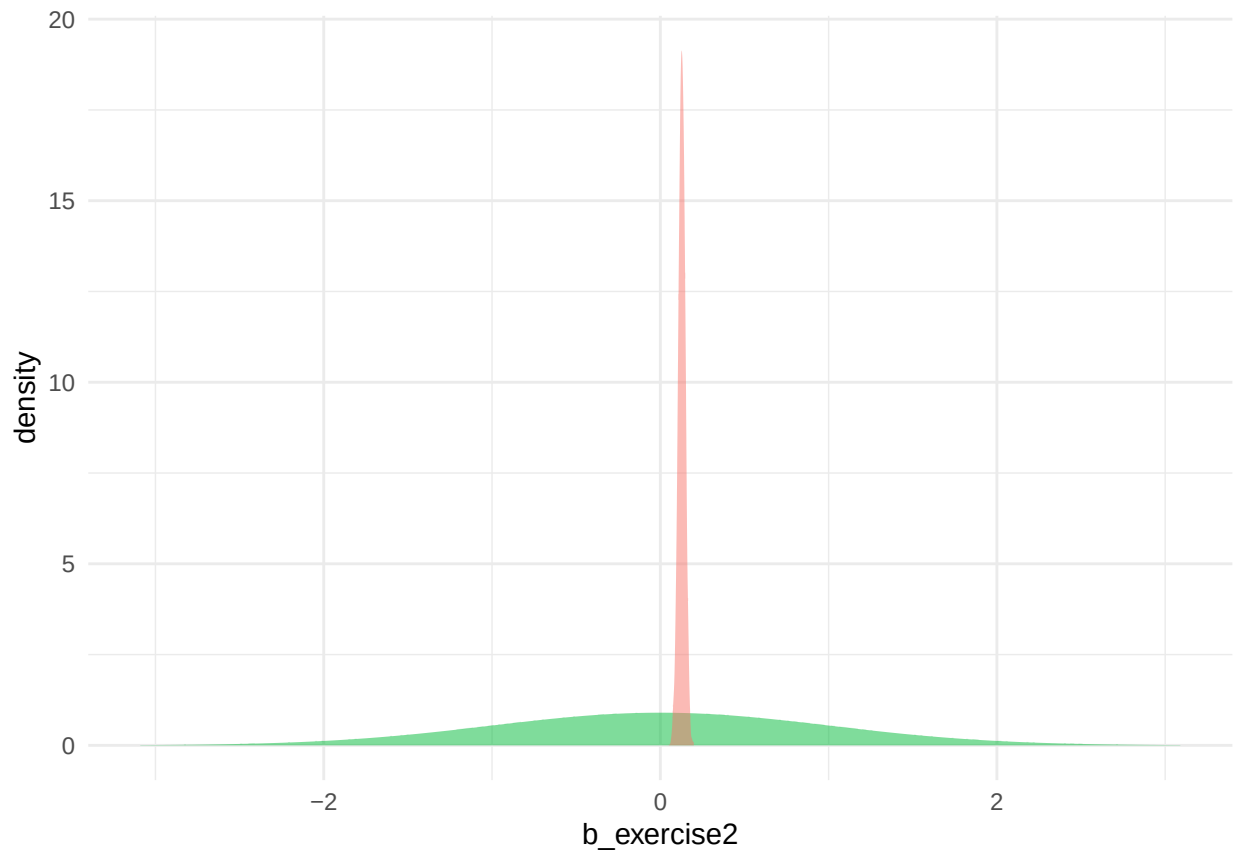
```
summary(brm_bray_2025)
```

```
## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: bray_2025 (Number of observations: 3304)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 16)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.23     0.05    0.16    0.33 1.00     985     1583
##
## ~SU (Number of levels: 8)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.13     0.05    0.07    0.25 1.00     812     1246
```

```
##
## Regression Coefficients:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    -0.48     0.08   -0.65   -0.33 1.00     795    1122
## exercise2     0.13     0.02    0.09    0.17 1.00    2960    2129
##
## Further Distributional Parameters:
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi    12.87     0.30   12.28   13.46 1.00    3315    2620
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#posterior vs. prior
```

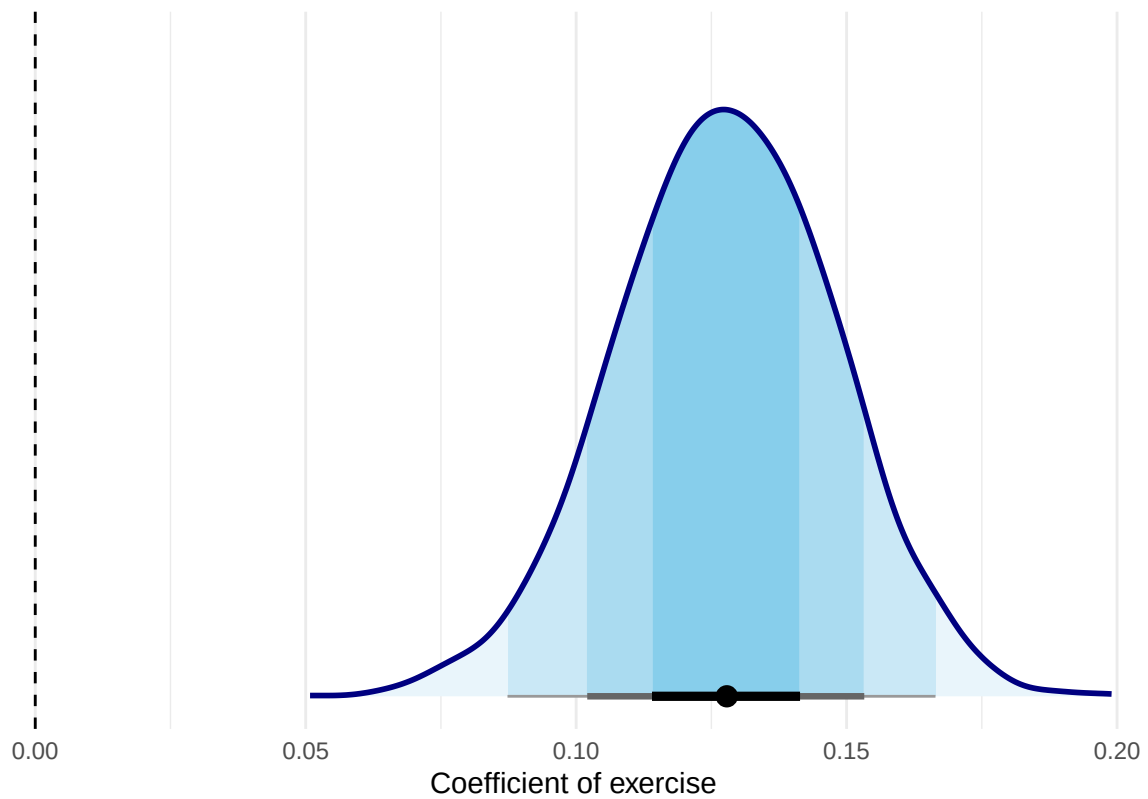
```
brm_bray_2025 |>
  post_prior_beta()
```



```
#on the logit scale
```

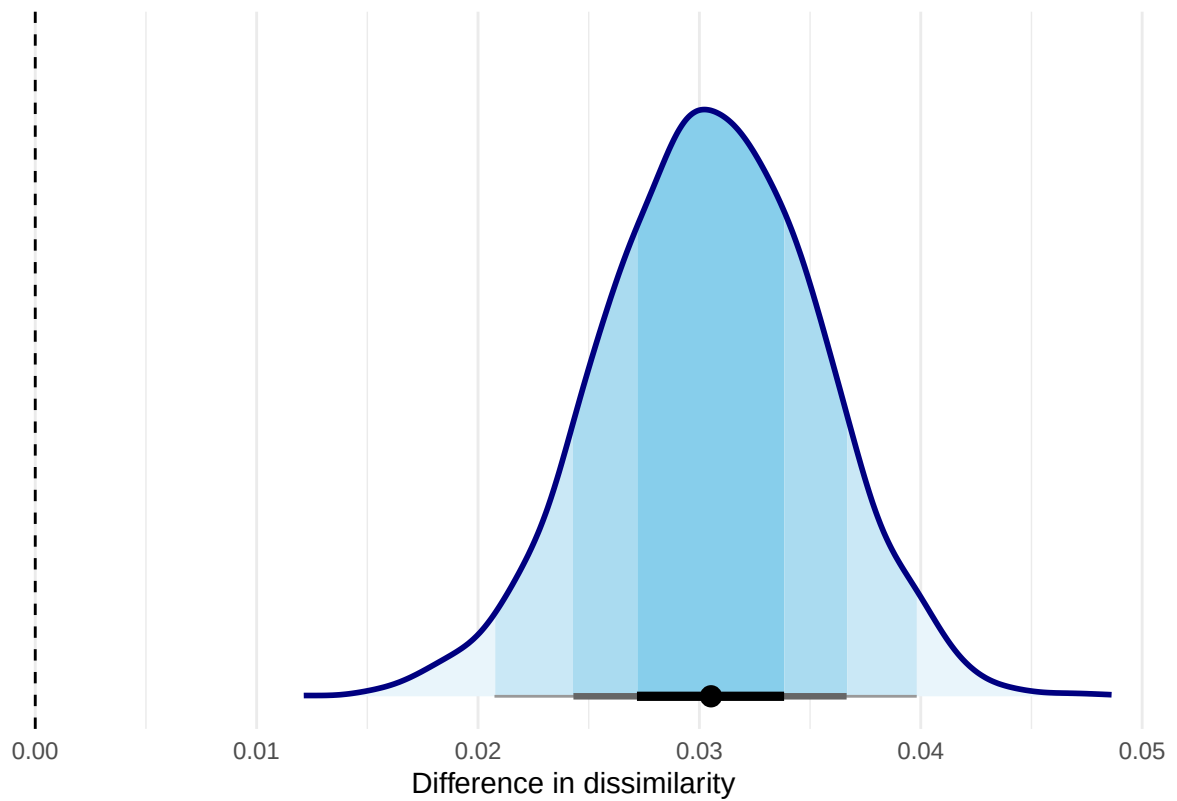
```
brm_bray_2025 |>
  plot_post(fit = T) +
  labs(title = "Posterior distribution for Bray-Curtis dissimilarity",
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Bray–Curtis dissimilarity



```
#on the outcome scale  
#as difference  
brm_bray_2025 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Bray-Curtis dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

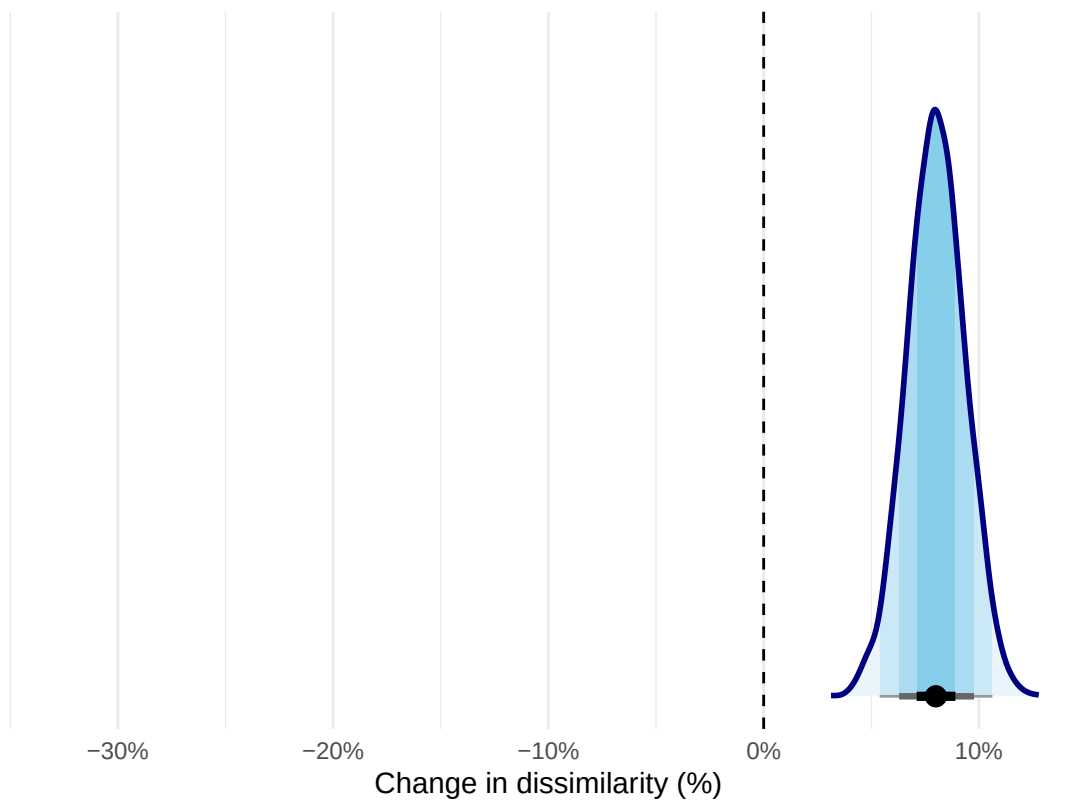
Effect of training on Bray–Curtis dissimilarity



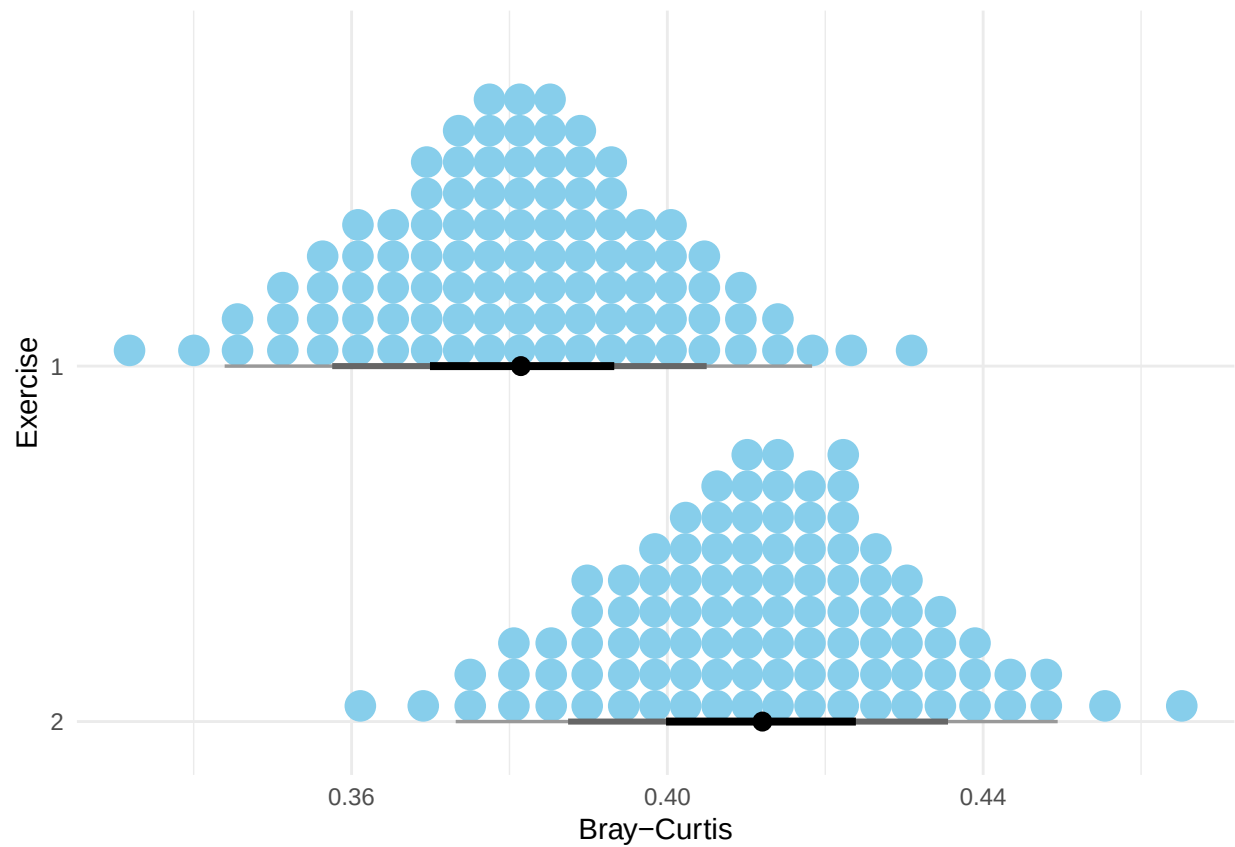
```
#as percent lift
plot_lift_bray_25 <- brm_bray_2025 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(), limits = c(-.35, .15))+
  labs(title = "Bray-Curtis", x = "Change in dissimilarity (%)", y = "")

plot_lift_bray_25
```

Bray-Curtis

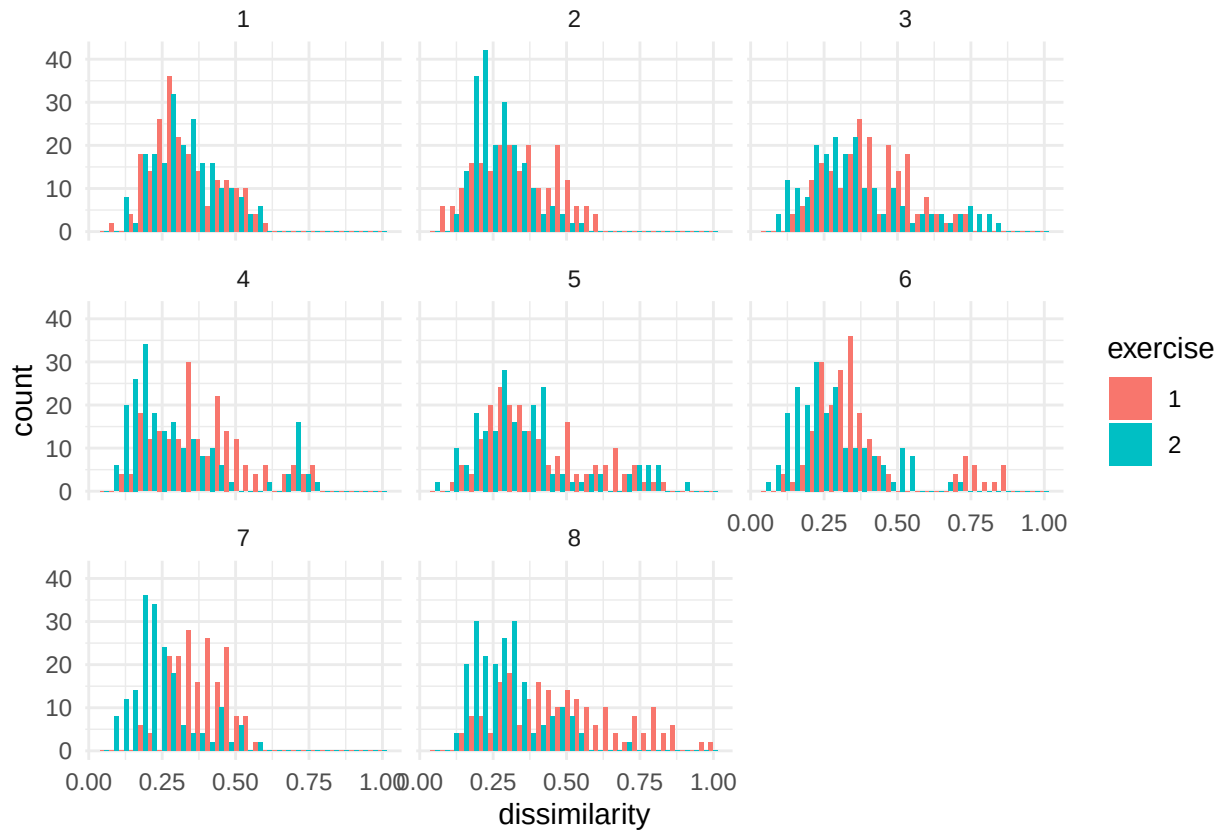


```
#quantile dotplot of predictions  
plot_pred_bray_25 <- brm_bray_2025 |>  
  plot_pred() +  
  labs(x = "Bray-Curtis", y = "Exercise")  
plot_pred_bray_25
```



```
### euclidean -----

euc_2025 |>
  ggplot(aes(x = dissimilarity, fill = exercise))+
  geom_histogram(position = "dodge")+
  facet_wrap(~SU)
```



```
#model
brm_euc_2025 <- brm(formula = beta_formula,
  data = euc_2025,
  prior = prior_beta,
  file = "fits/euc_2025",
  seed = 21856)

summary(brm_euc_2025)
```

```
## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: euc_2025 (Number of observations: 3304)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 16)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.32     0.06    0.22    0.46 1.00     688    1465
##
## ~SU (Number of levels: 8)
##           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.18     0.06    0.10    0.34 1.00     899    1609
##
```

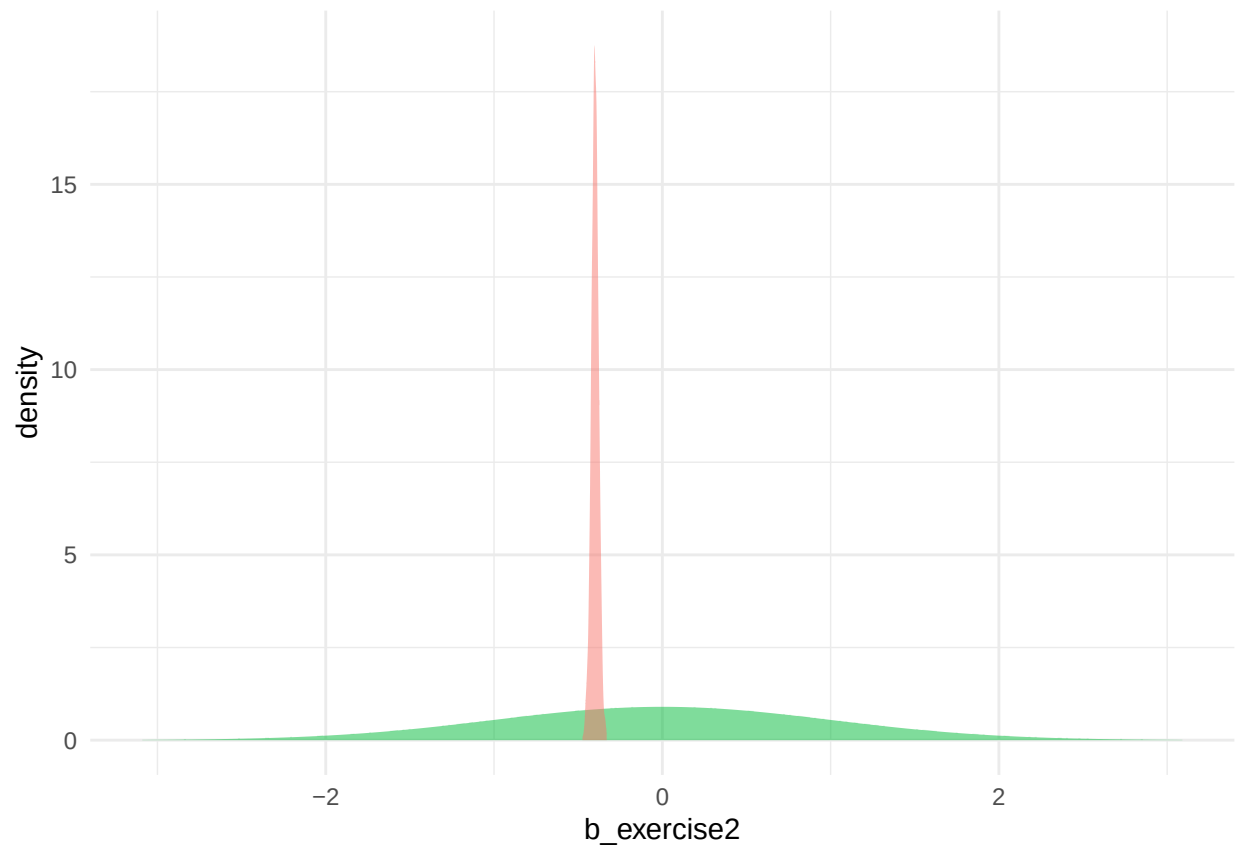
```
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept    -0.42      0.10   -0.63   -0.22 1.01      868     1323
## exercise2    -0.40      0.02   -0.44   -0.36 1.00     3431     2578
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi    10.97      0.26    10.45    11.50 1.00     3002     1829
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
prior_summary(brm_euc_2025)
```

```
##      prior      class      coef      group resp dpar nlpar lb ub tag
##      normal(0, 1)      b
##      normal(0, 1)      b exercise2
## student_t(7, 0, 1.5) Intercept
##      gamma(1, 0.02)      phi      0
## student_t(5, 0, 0.5)      sd      0
## student_t(5, 0, 0.5)      sd      detector      0
## student_t(5, 0, 0.5)      sd Intercept detector      0
## student_t(5, 0, 0.5)      sd      SU      0
## student_t(5, 0, 0.5)      sd Intercept      SU      0
##      source
##      user
## (vectorized)
##      user
##      user
##      user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

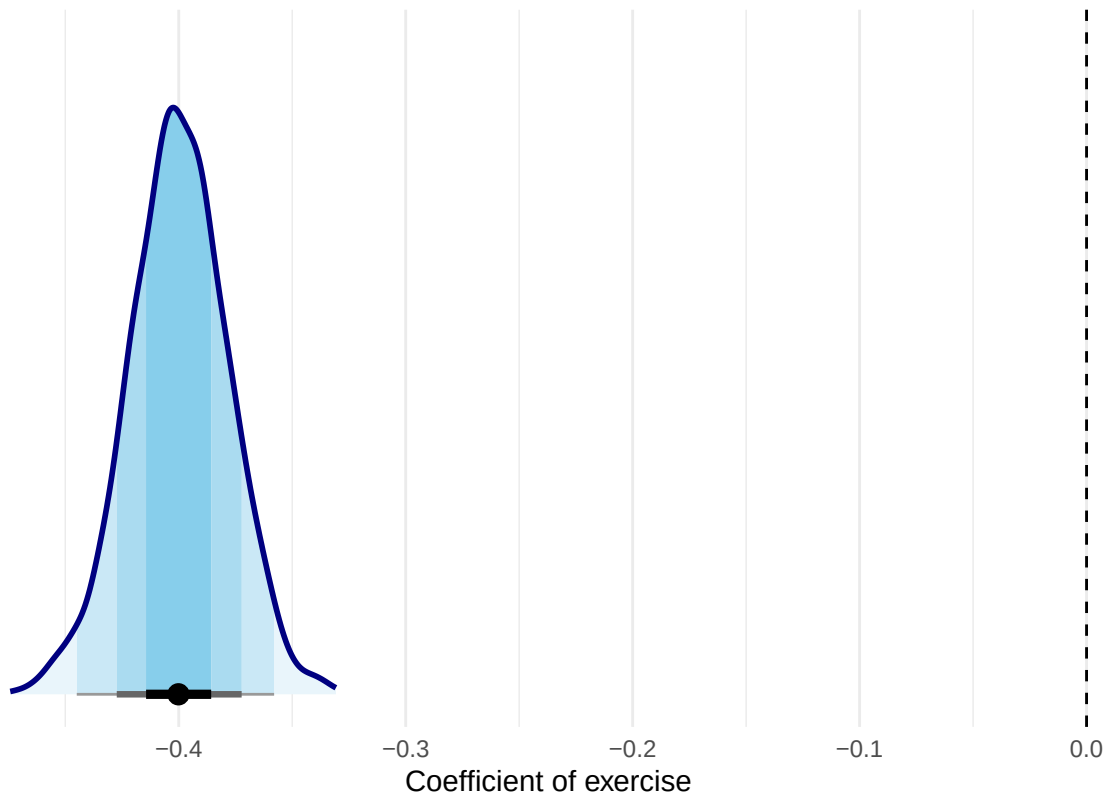
```
#posterior vs. prior
```

```
brm_euc_2025 |>
  post_prior_beta()
```



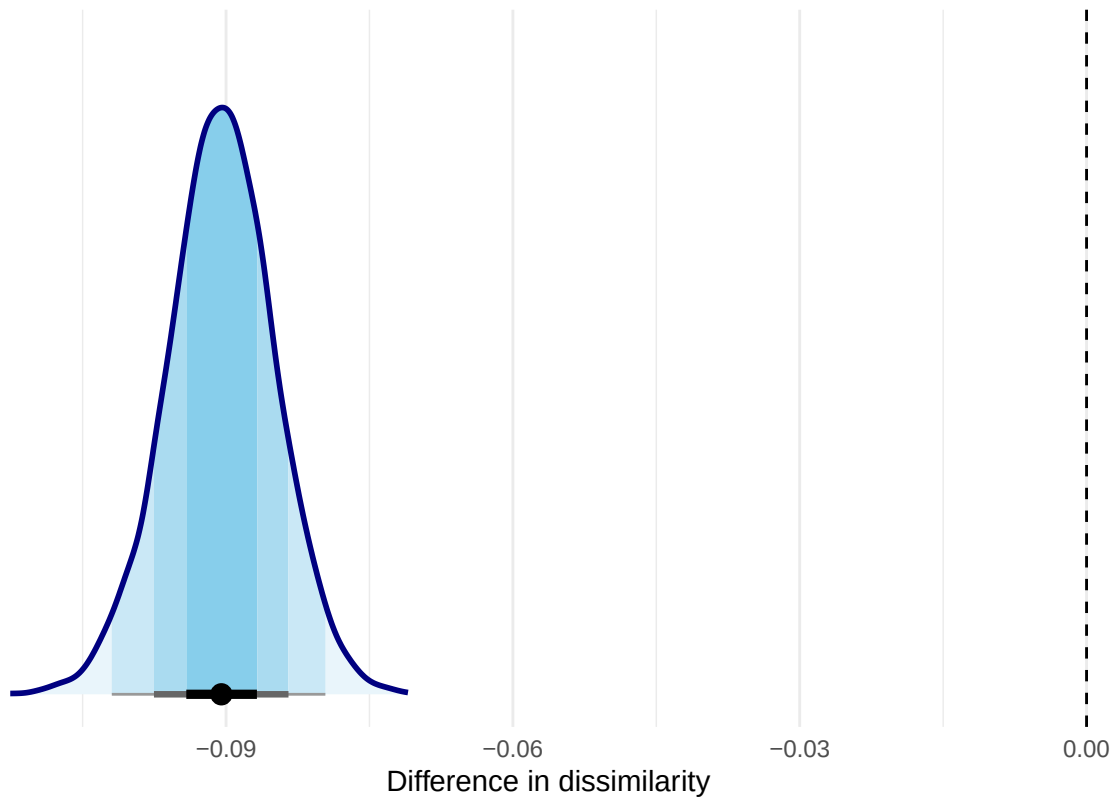
```
#on the logit scale  
brm_euc_2025 |>  
  plot_post(fit = T) +  
  labs(title = "Posterior distribution for Euclidean dissimilarity",  
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Euclidean dissimilarity



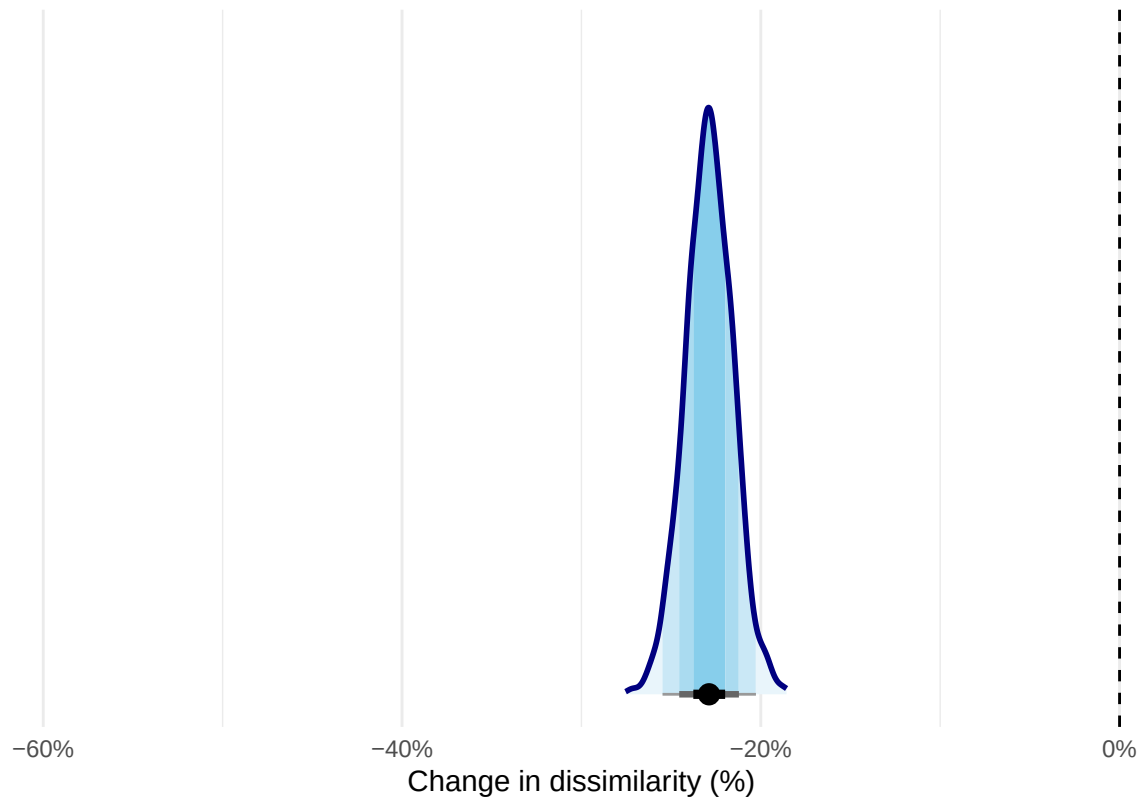
```
#on the outcome scale  
#as difference  
brm_euc_2025 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Euclidean dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

Effect of training on Euclidean dissimilarity

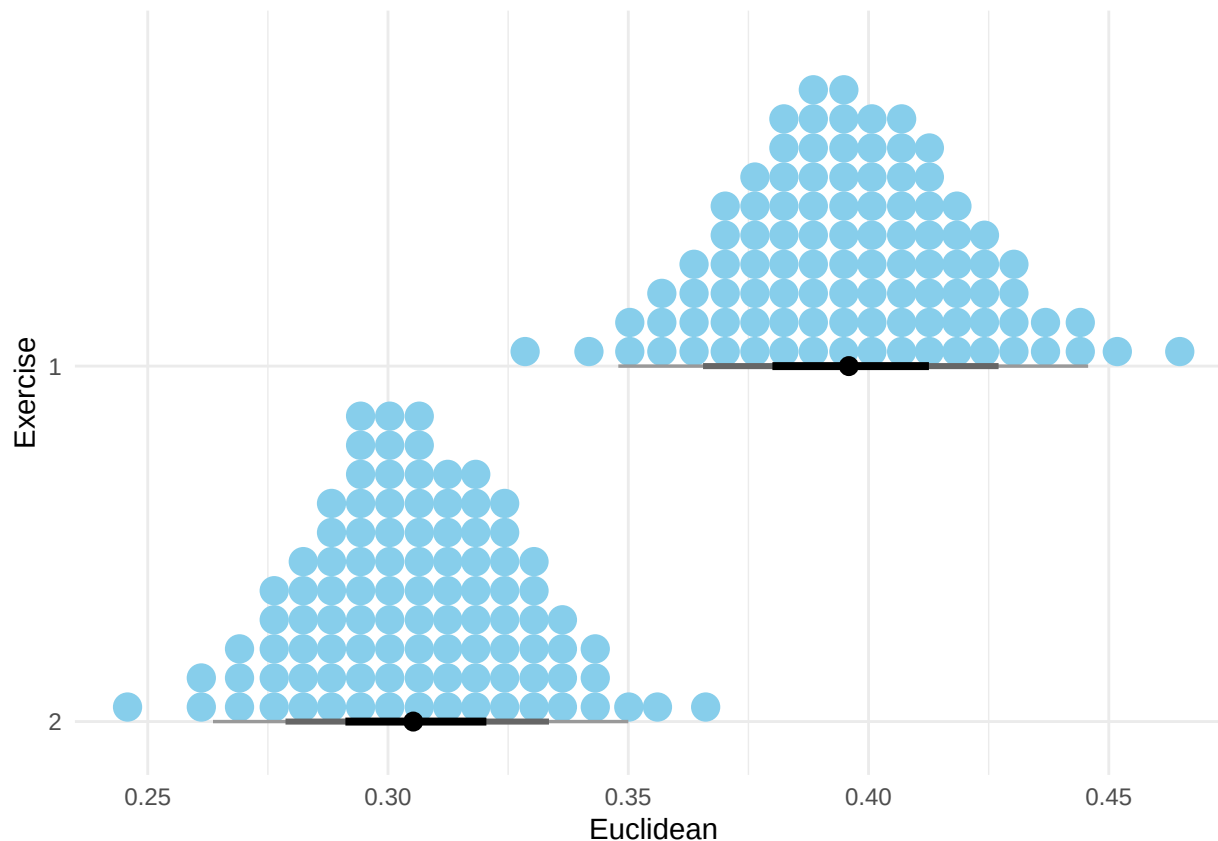


```
#as percent lift
plot_lift_euc_25 <- brm_euc_2025 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(), limits = c(-.6, 0)) +
  labs(title = "Euclidean", x = "Change in dissimilarity (%)", y = "")
plot_lift_euc_25
```

Euclidean



```
#quantile dotplot of predictions  
plot_pred_euc_25 <- brm_euc_2025 |>  
  plot_pred() +  
  labs(x = "Euclidean", y = "Exercise")  
plot_pred_euc_25
```



```
## 2023 -----

### jaccard -----

brm_jac_2023 <- brm(formula = beta_formula,
  data = jaccard_2023,
  prior = prior_23,
  file = "fits/jac_2023",
  seed = 59293,
  control = list(adapt_delta = 0.9))

summary(brm_jac_2023)

## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: jaccard_2023 (Number of observations: 96)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 4)
##
```

	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.27	0.12	0.11	0.58	1.00	1178	2059

```
##
```

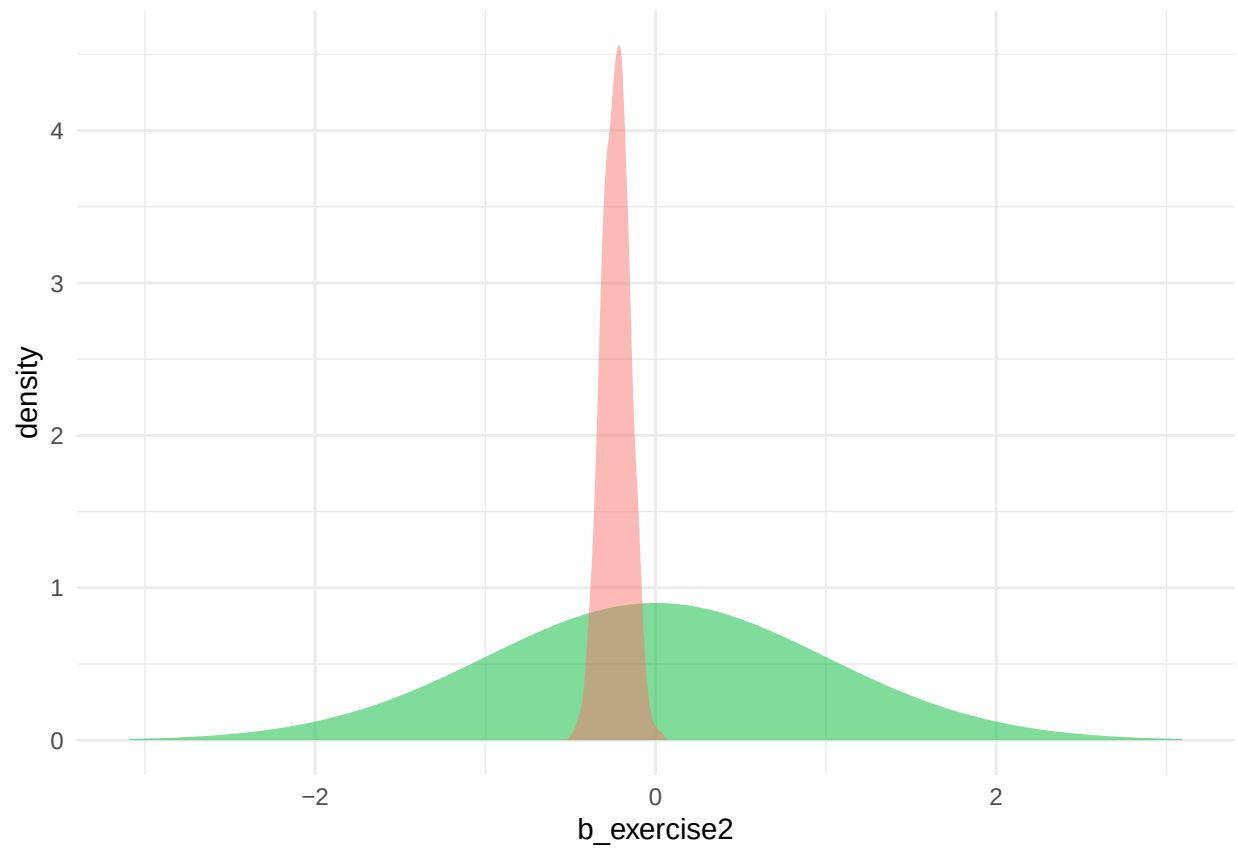
```
## ~SU (Number of levels: 4)
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.09      0.08    0.00    0.30 1.00    1329    1698
##
## Regression Coefficients:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.43      0.16    0.11    0.78 1.01    1260    1087
## exercise2     -0.23      0.08   -0.39   -0.08 1.00    3748    2749
##
## Further Distributional Parameters:
##           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi      24.82      3.76   18.22   32.78 1.00    2356    1368
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
prior_summary(brm_jac_2023)
```

```
##           prior      class      coef      group resp dpar nlpar lb ub tag
##           normal(0, 1)      b
##           normal(0, 1)      b exercise2
## student_t(7, 0, 1.5) Intercept
##           gamma(1, 0.02)      phi                                0
## student_t(3, 0, 0.25)      sd                                0
## student_t(3, 0, 0.25)      sd      detector                                0
## student_t(3, 0, 0.25)      sd Intercept detector                                0
## student_t(3, 0, 0.25)      sd      SU                                0
## student_t(3, 0, 0.25)      sd Intercept      SU                                0
##           source
##           user
## (vectorized)
##           user
##           user
##           user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

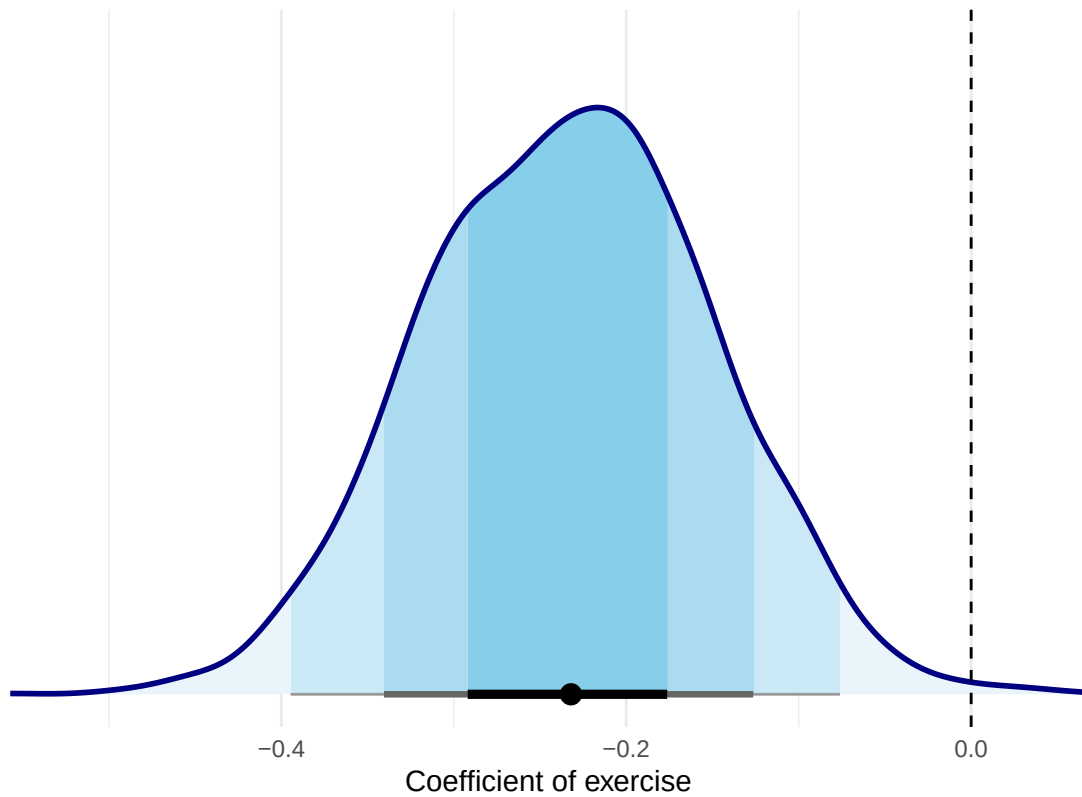
```
#posterior vs. prior
```

```
brm_jac_2023 |>
  post_prior_beta()
```



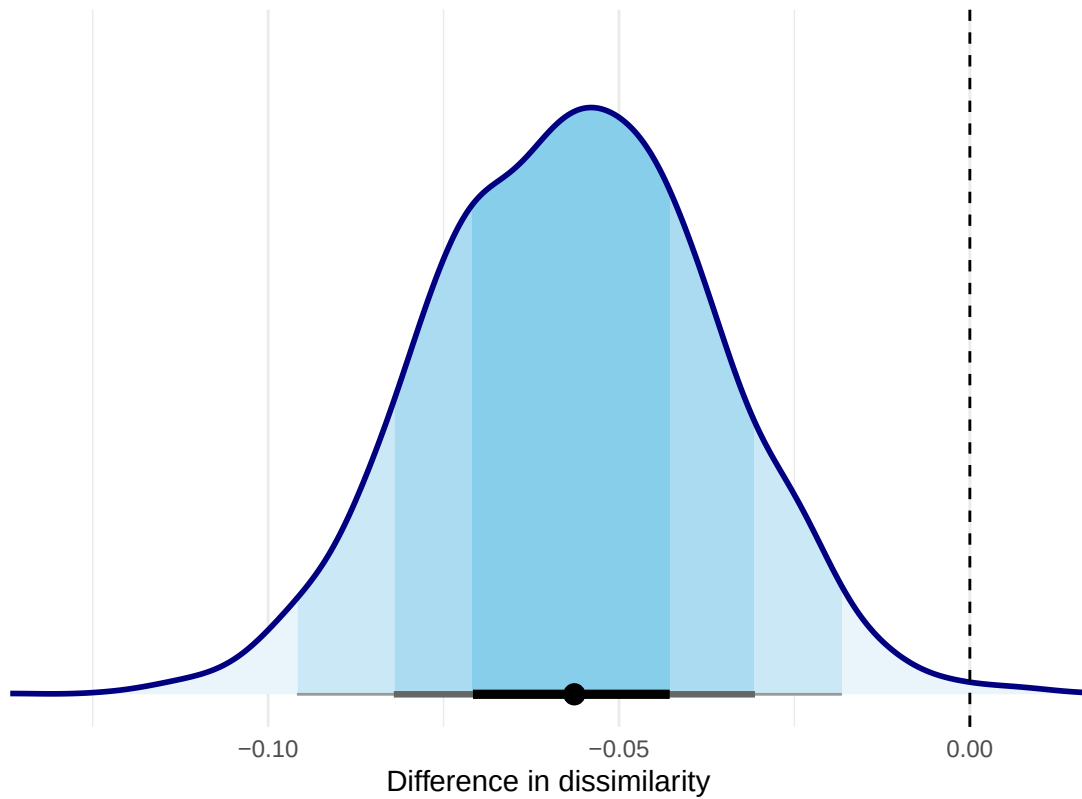
```
#on the logit scale  
brm_jac_2023 |>  
  plot_post(fit = T) +  
  labs(title = "Posterior distribution for Jaccard dissimilarity",  
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Jaccard dissimilarity



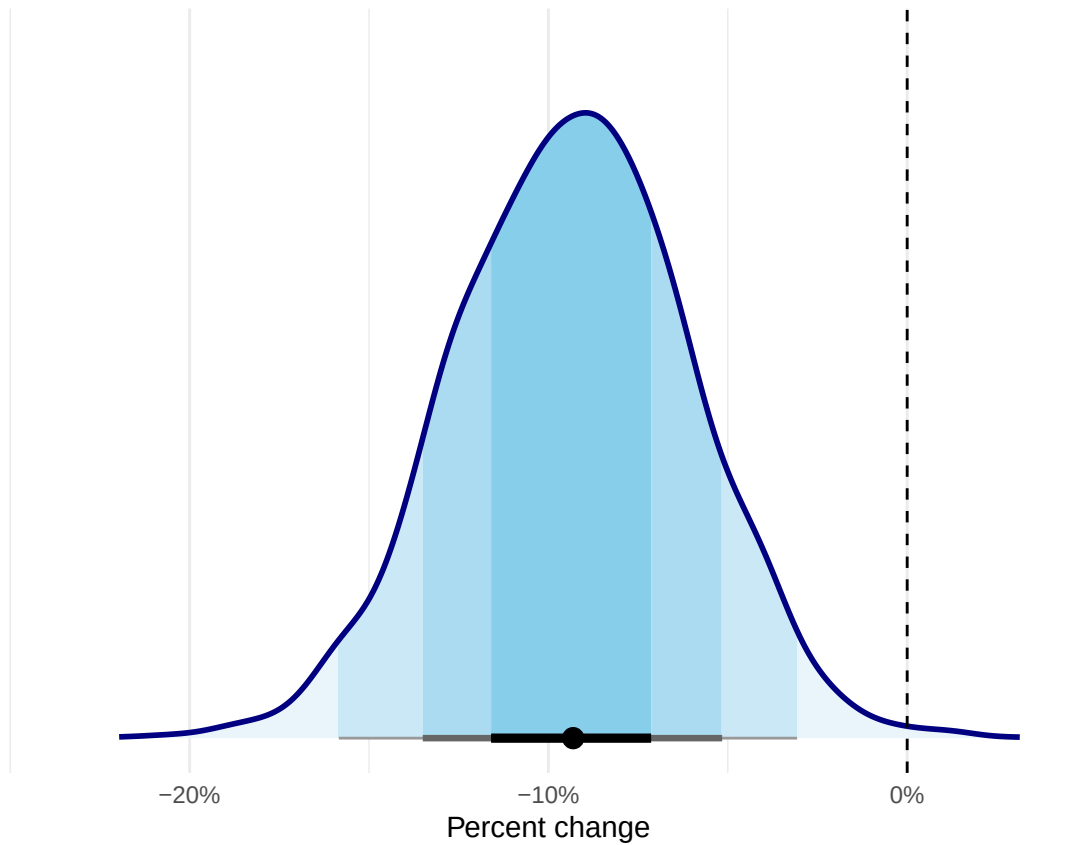
```
#on the outcome scale  
#as difference  
brm_jac_2023 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Jaccard dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

Effect of training on Jaccard dissimilarity

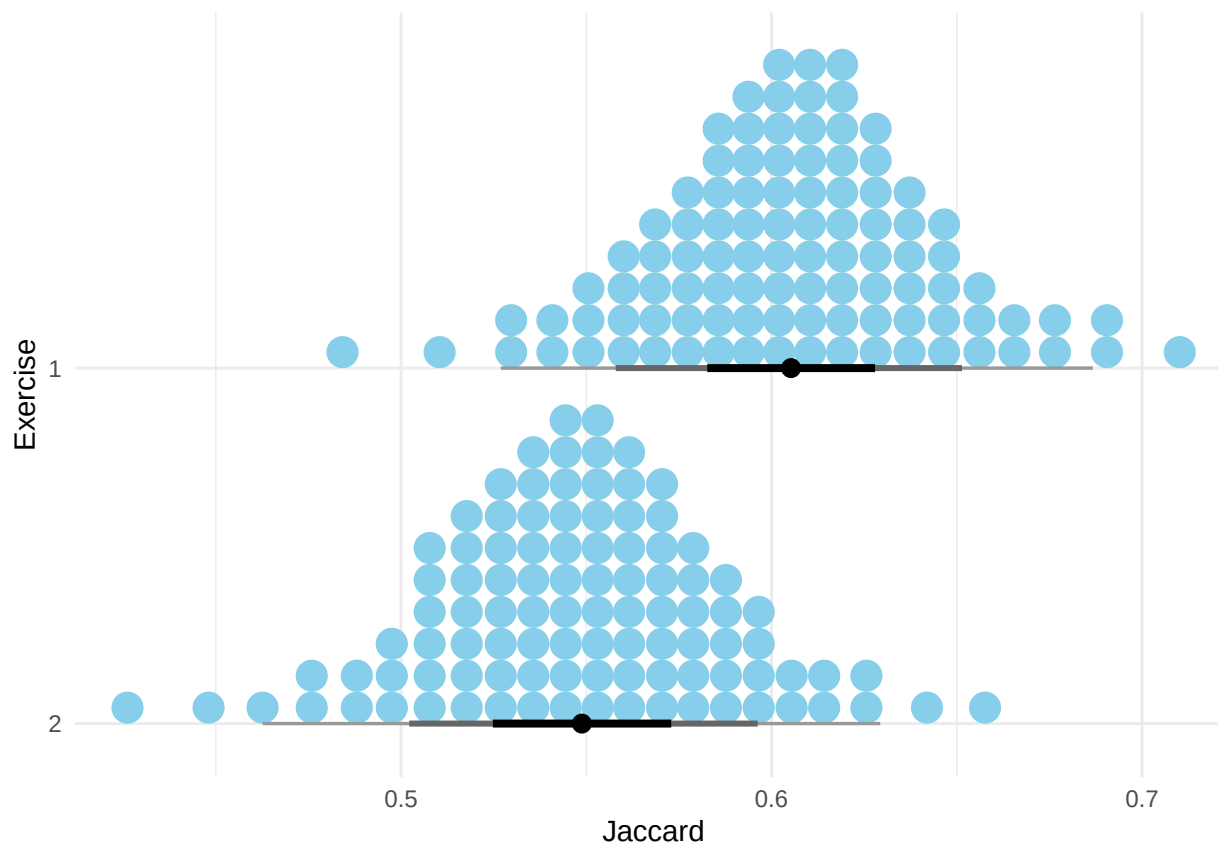


```
#as percent lift
plot_lift_jac_23 <- brm_jac_2023 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(), limits = c(-0.25, 0.05)) +
  labs(x = "Percent change", y = "")

plot_lift_jac_23
```



```
#quantile dotplot of predictions  
plot_pred_jac_23 <- brm_jac_2023 |>  
  plot_pred() +  
  labs(x = "Jaccard", y = "Exercise")  
plot_pred_jac_23
```



```
### bray-curtis -----
```

```
brm_bray_2023 <- brm(formula = beta_formula,
  data = bray_2023,
  prior = prior_23,
  file = "fits/bray_2023",
  seed = 61278,
  control = list(adapt_delta = 0.9))
```

```
summary(brm_bray_2023)
```

```
## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: bray_2023 (Number of observations: 96)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 4)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.14    0.10    0.01    0.39 1.00    1202    1215
##
## ~SU (Number of levels: 4)
##      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

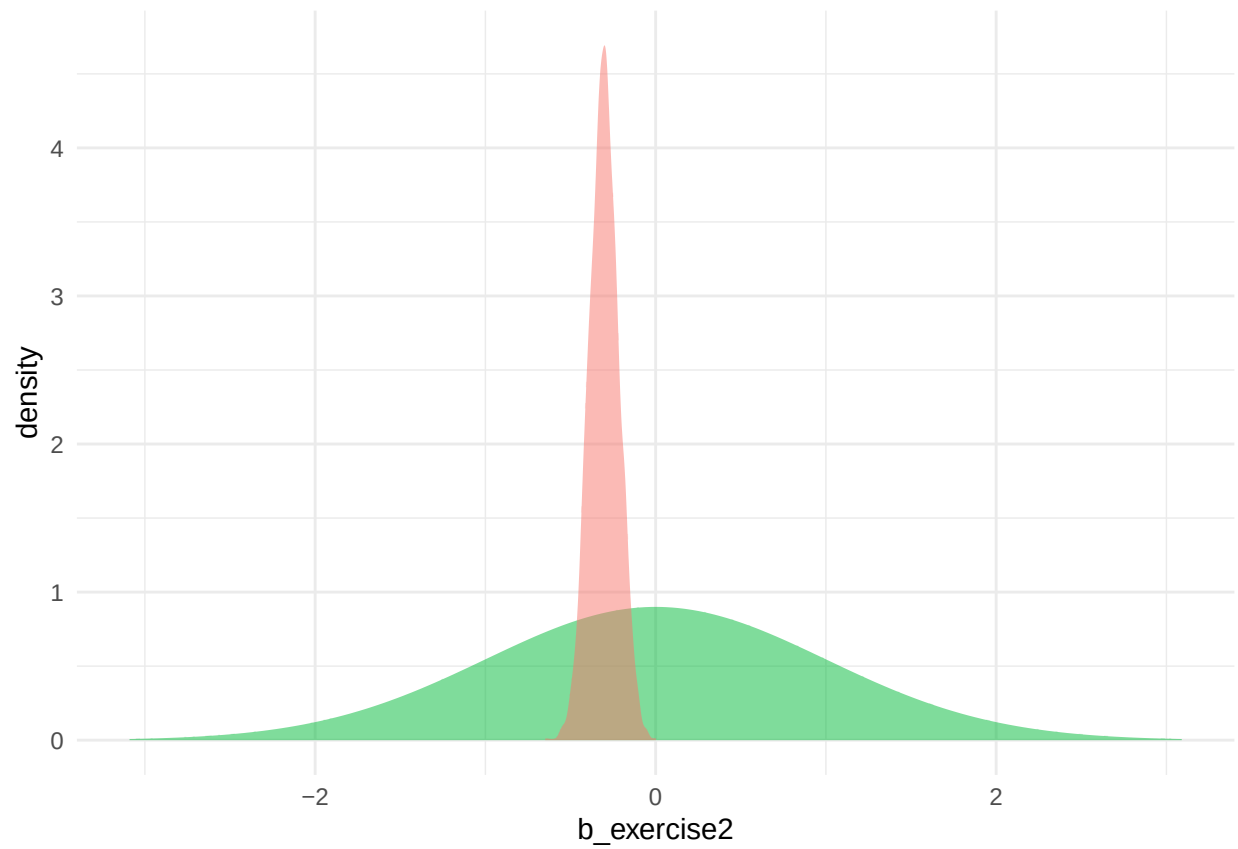
```
## sd(Intercept)      0.30      0.13      0.13      0.65 1.01      1667      2110
##
## Regression Coefficients:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      -0.30      0.19    -0.66     0.10 1.00      1470      1270
## exercise2      -0.30      0.09    -0.47    -0.14 1.00      4319      2737
##
## Further Distributional Parameters:
##      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi      25.56      3.79     18.95     33.45 1.00      4151      3110
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
prior_summary(brm_bray_2023)
```

```
##           prior      class      coef      group resp dpar nlpar lb ub tag
##      normal(0, 1)         b
##      normal(0, 1)         b exercise2
## student_t(7, 0, 1.5) Intercept
##      gamma(1, 0.02)      phi                                0
## student_t(3, 0, 0.25)      sd                                0
## student_t(3, 0, 0.25)      sd      detector                0
## student_t(3, 0, 0.25)      sd Intercept detector          0
## student_t(3, 0, 0.25)      sd      SU                    0
## student_t(3, 0, 0.25)      sd Intercept      SU            0
##      source
##      user
## (vectorized)
##      user
##      user
##      user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

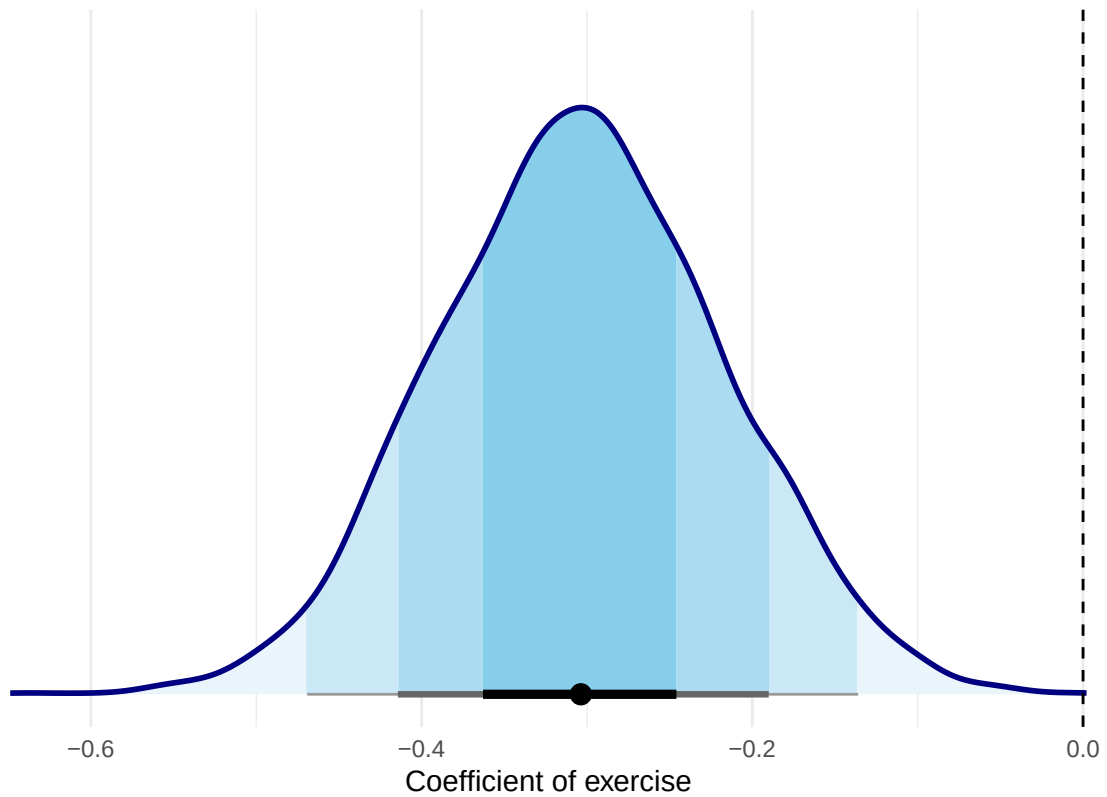
```
#posterior vs. prior
```

```
brm_bray_2023 |>
  post_prior_beta()
```



```
#on the logit scale  
brm_bray_2023 |>  
  plot_post(fit = T) +  
  labs(title = "Posterior distribution for Bray-Curtis dissimilarity",  
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Bray–Curtis dissimilarity

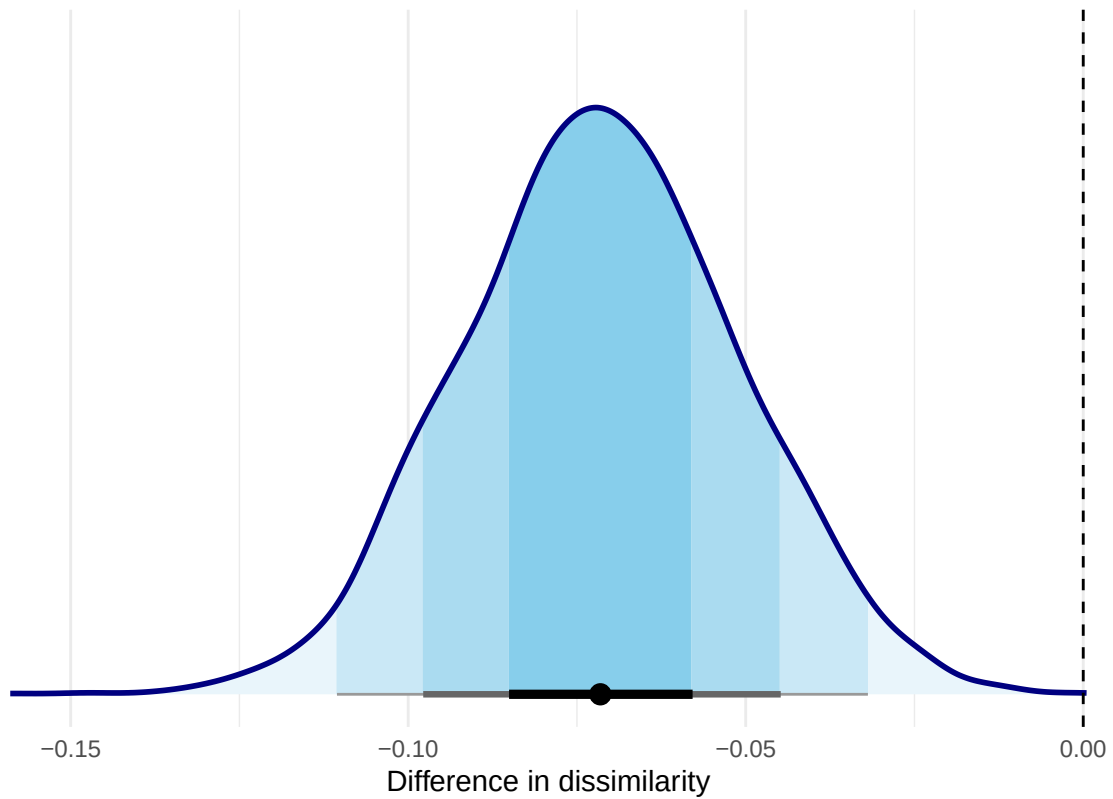


```
#on the outcome scale
```

```
#as difference
```

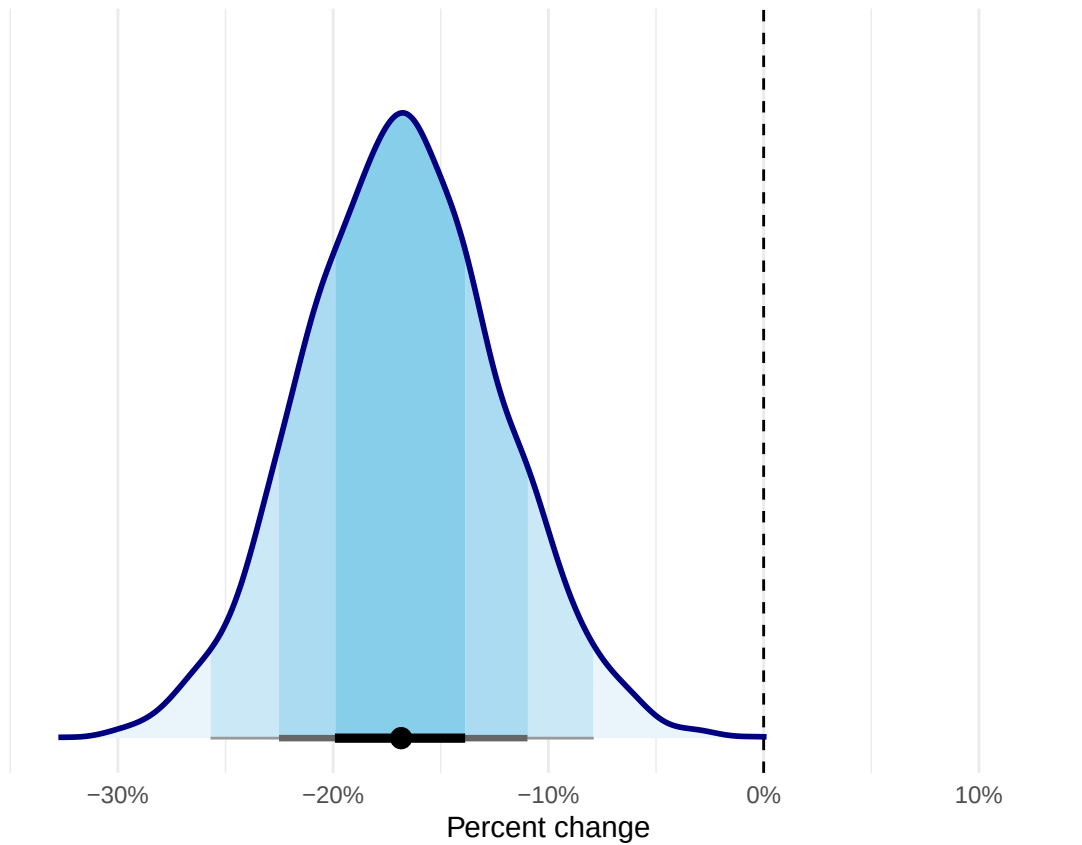
```
brm_bray_2023 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Bray-Curtis dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

Effect of training on Bray–Curtis dissimilarity

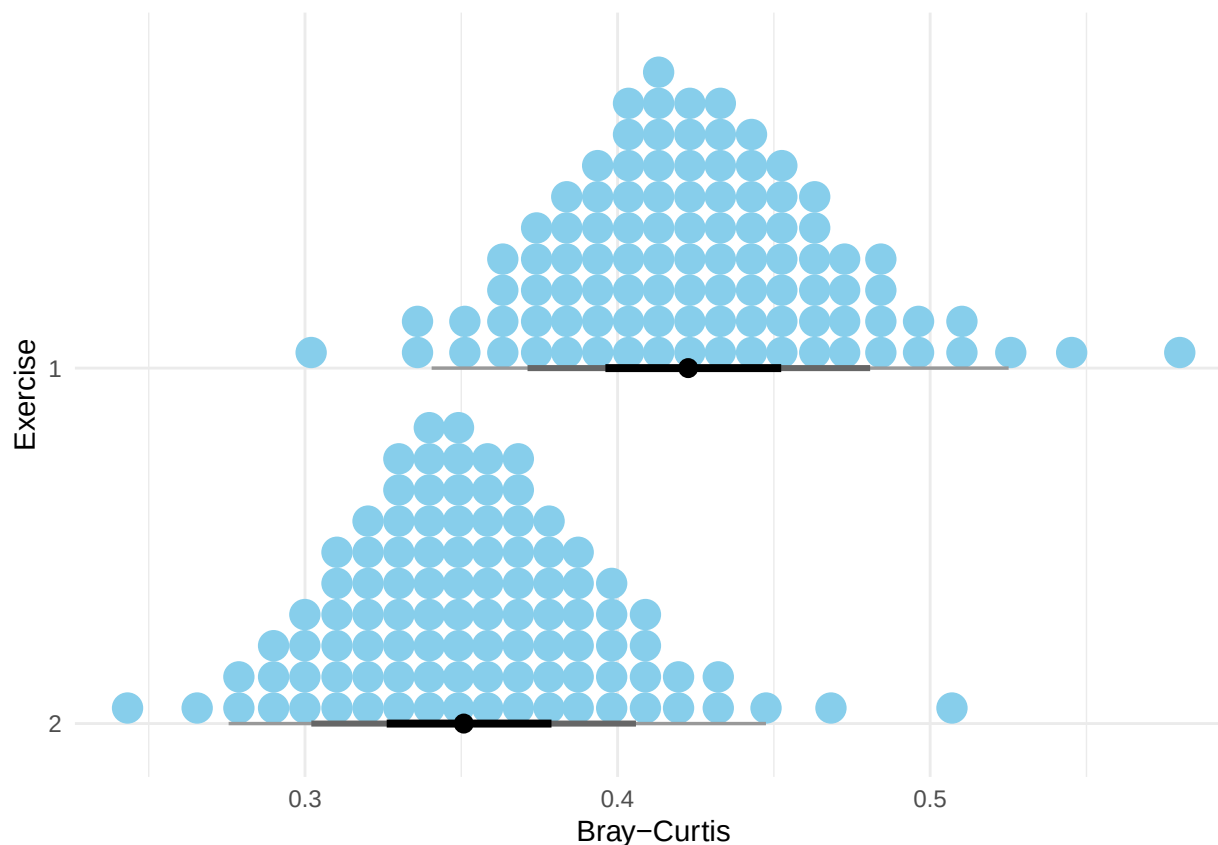


```
#as percent lift
plot_lift_bray_23 <- brm_bray_2023 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(), limits = c(-.35, .15))+
  labs(x = "Percent change", y = "")

plot_lift_bray_23
```



```
#quantile dotplot of predictions  
plot_pred_bray_23 <- brm_bray_2023 |>  
  plot_pred() +  
  labs(x = "Bray-Curtis", y = "Exercise")  
plot_pred_bray_23
```



```
### euclidean -----
```

```
#model
```

```
brm_euc_2023 <- brm(formula = beta_formula,
  data = euc_2023,
  prior = prior_23,
  file = "fits/euc_2023",
  seed = 21856,
  control = list(adapt_delta = 0.9))
```

```
summary(brm_euc_2023)
```

```
## Family: beta
## Links: mu = logit
## Formula: dissimilarity ~ exercise + (1 | SU) + (1 | detector)
## Data: euc_2023 (Number of observations: 96)
## Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
## total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~detector (Number of levels: 4)
## Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept) 0.24 0.15 0.02 0.60 1.00 1572 1380
##
## ~SU (Number of levels: 4)
```

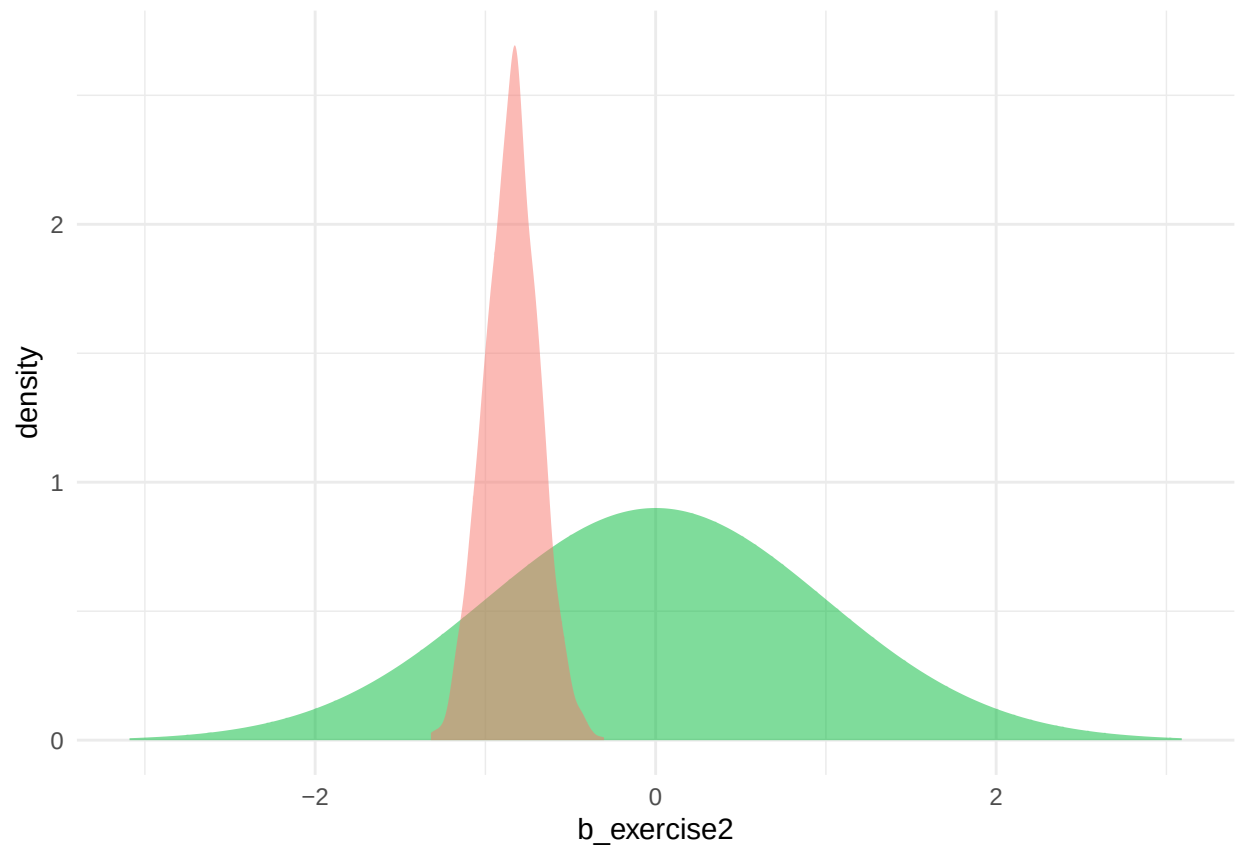
```
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)    0.50      0.20    0.24    1.04 1.00    1704    2059
##
## Regression Coefficients:
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept      0.45      0.31   -0.16    1.08 1.00    1750    1837
## exercise2     -0.84      0.16   -1.15   -0.53 1.00    4078    2441
##
## Further Distributional Parameters:
##               Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## phi          6.55      0.91    4.86    8.45 1.00    3085    2639
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
prior_summary(brm_euc_2023)
```

```
##               prior      class      coef      group resp dpar nlpar lb ub tag
##          normal(0, 1)         b
##          normal(0, 1)         b exercise2
## student_t(7, 0, 1.5) Intercept
##          gamma(1, 0.02)      phi                                0
## student_t(3, 0, 0.25)      sd                                0
## student_t(3, 0, 0.25)      sd      detector                    0
## student_t(3, 0, 0.25)      sd Intercept detector              0
## student_t(3, 0, 0.25)      sd      SU                        0
## student_t(3, 0, 0.25)      sd Intercept SU                    0
##          source
##          user
## (vectorized)
##          user
##          user
##          user
## (vectorized)
## (vectorized)
## (vectorized)
## (vectorized)
```

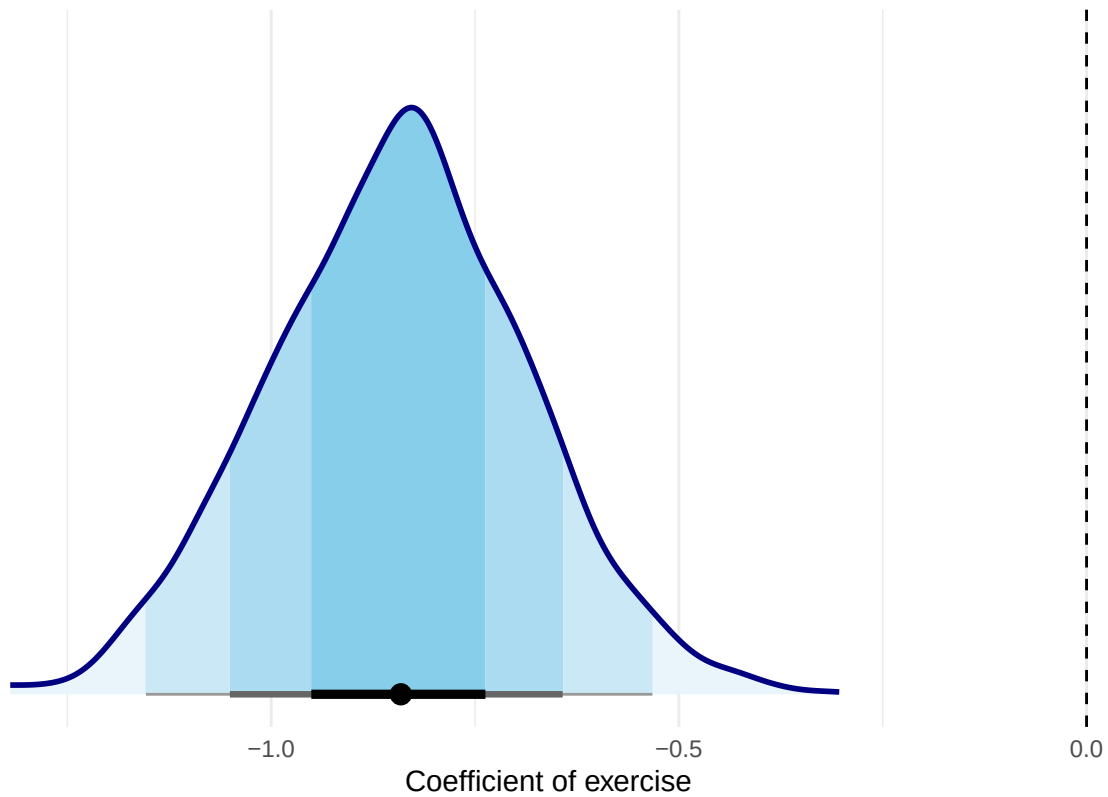
```
#posterior vs. prior
```

```
brm_euc_2023 |>
  post_prior_beta()
```



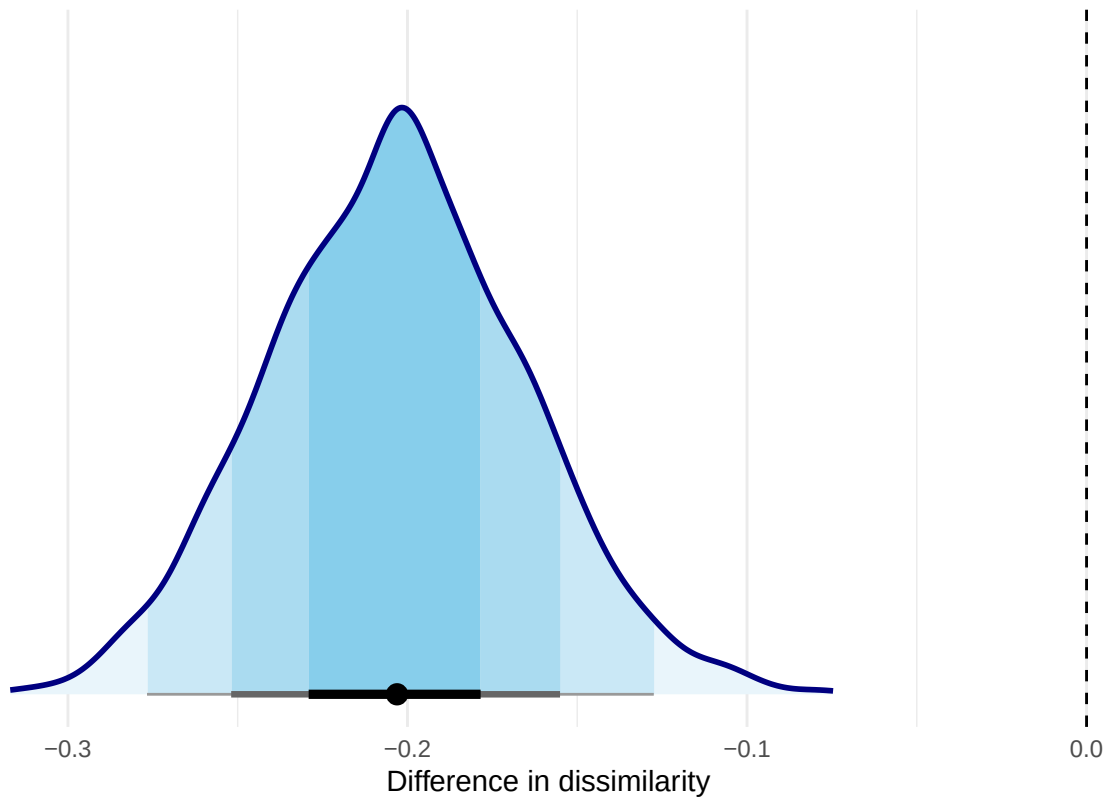
```
#on the logit scale  
brm_euc_2023 |>  
  plot_post(fit = T) +  
  labs(title = "Posterior distribution for Euclidean dissimilarity",  
        x = "Coefficient of exercise", y = "")
```

Posterior distribution for Euclidean dissimilarity



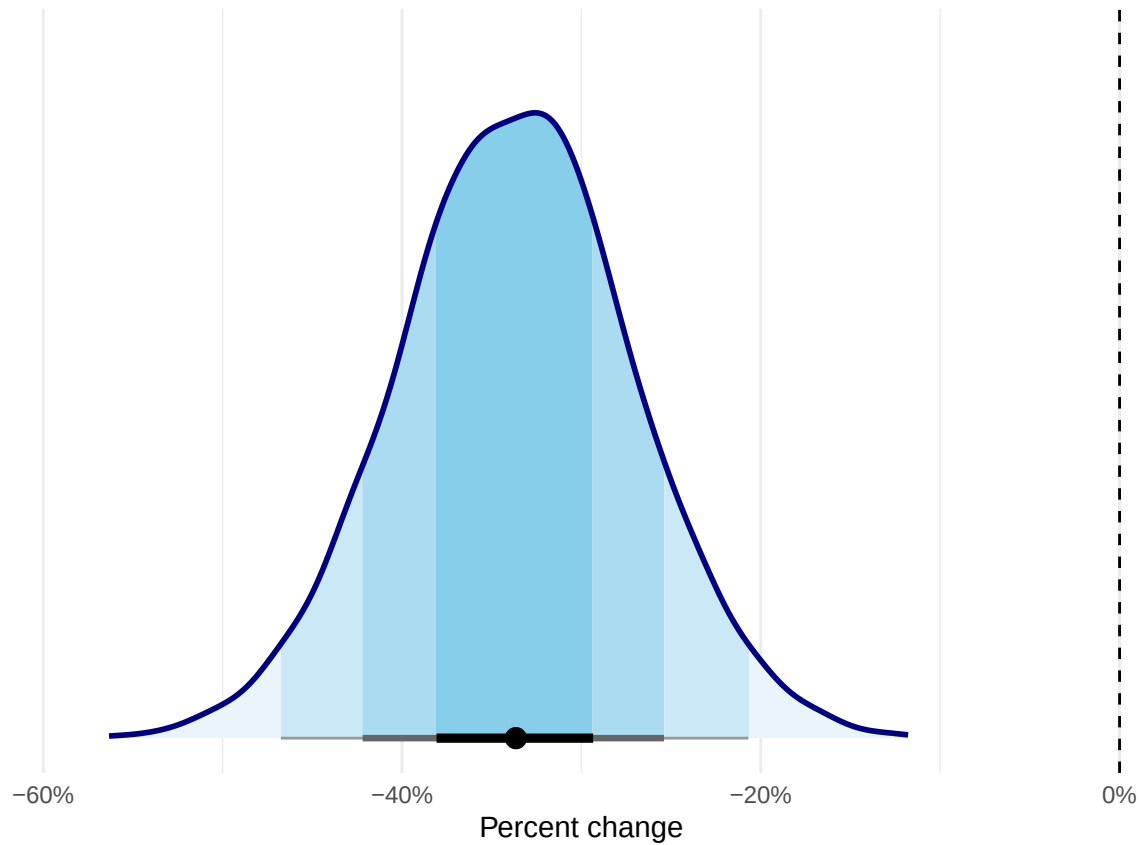
```
#on the outcome scale  
#as difference  
brm_euc_2023 |>  
  plot_comp(contrast) +  
  labs(title = "Effect of training on Euclidean dissimilarity",  
        x = "Difference in dissimilarity", y = "")
```

Effect of training on Euclidean dissimilarity

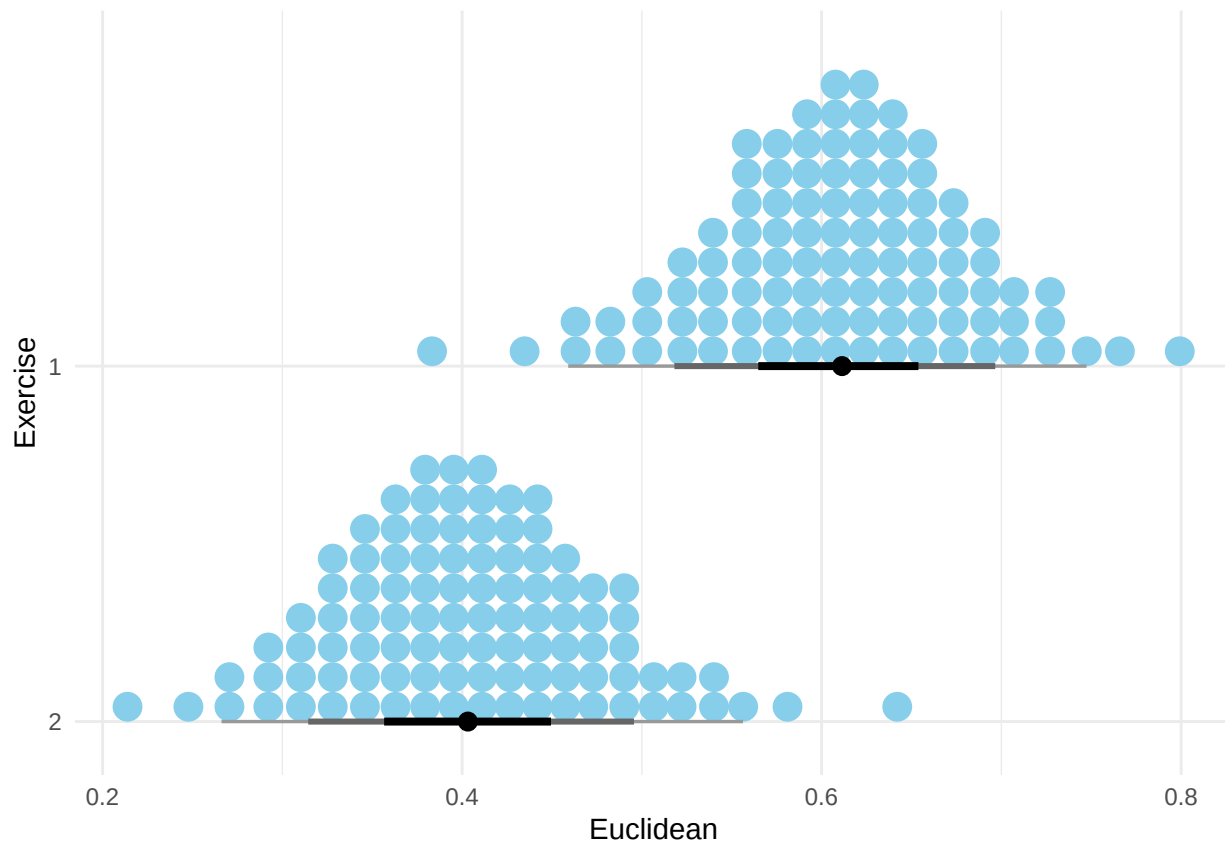


```
#as percent lift
plot_lift_euc_23 <- brm_euc_2023 |>
  plot_comp(lift) +
  scale_x_continuous(labels = scales::label_percent(), limits = c(-.6, 0)) +
  labs(x = "Percent change", y = "")

plot_lift_euc_23
```



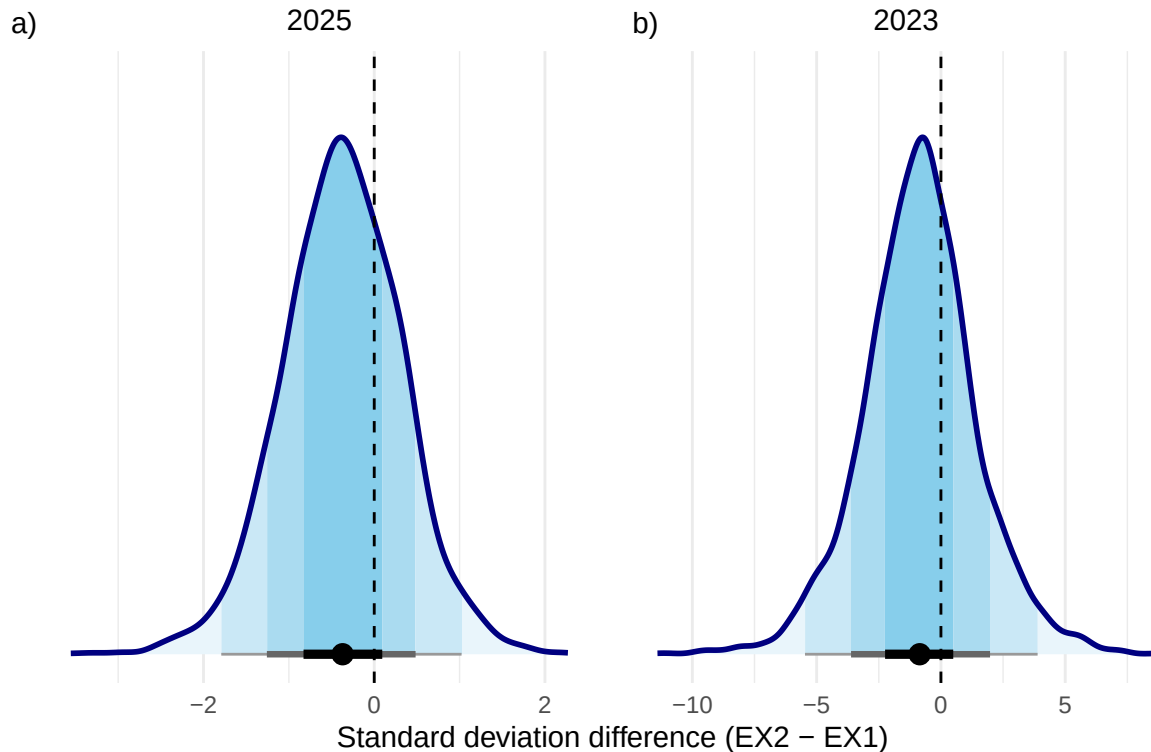
```
#quantile dotplot of predictions  
plot_pred_euc_23 <- brm_euc_2023 |>  
  plot_pred() +  
  labs(x = "Euclidean", y = "Exercise")  
plot_pred_euc_23
```



```
# Building the figures -----
#richness

wrap_plots(grid::textGrob("Observer richness variability before and after training",
  gp = grid::gpar(fontsize = 12)),
  plot_rich_25, plot_rich_23) +
  plot_layout(axes = 'collect',
    ncol = 2, nrow = 2, byrow = T,
    heights = c(0, 1, 1),
    design = "AA
      BC") +
  plot_annotation(tag_levels = list(c("", paste0(letters[1:2], ")")))) &
  # plot_annotation("Observer richness variability before and after training",
  #   tag_levels = 'a',
  #   tag_suffix = ')')
  theme(plot.title = element_text(size = 11, hjust = 0.5),
    plot.tag = element_text(size = 11),
    plot.tag.location = 'plot',
    margins = margin(t = 10, b = 5, l = 10, r = 5))
```

Observer richness variability before and after training



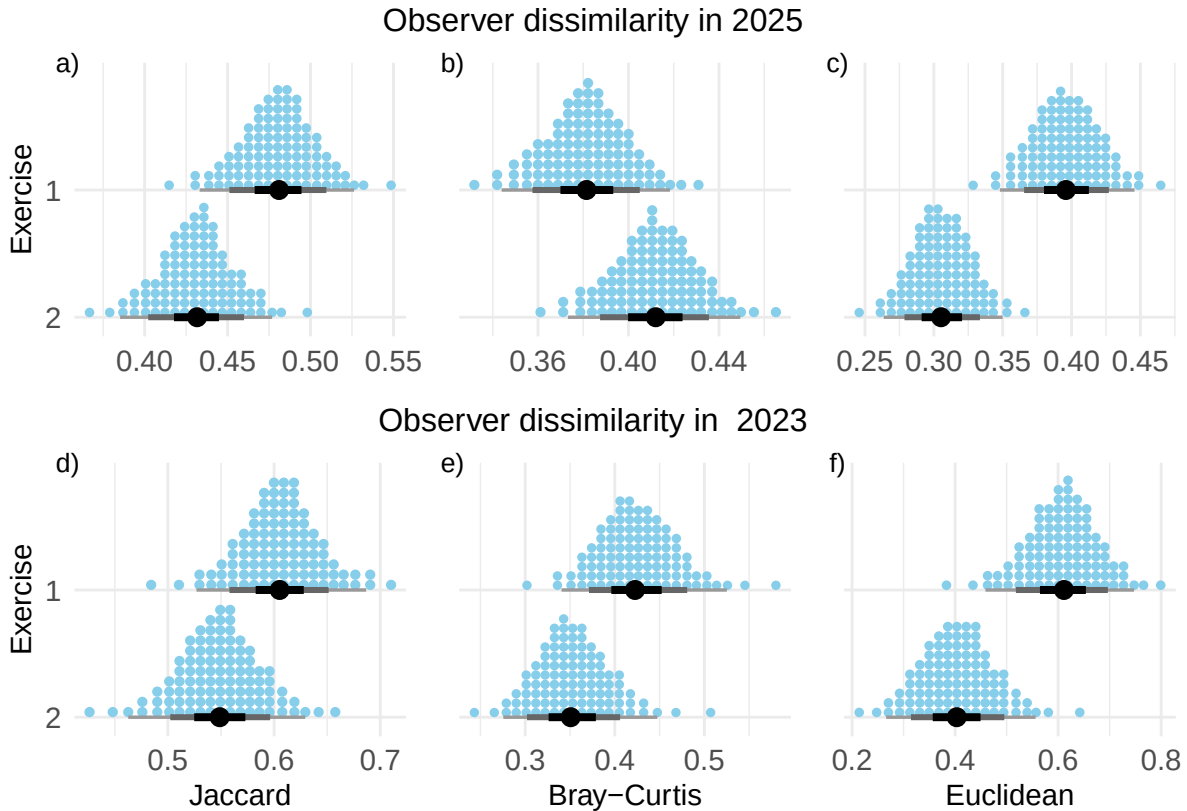
```
ggsave('output/richness.png', bg = 'white', scale = 1.1)

#beta predictions

r1_pred <- wrap_plots(plot_pred_jac_25, plot_pred_bray_25, plot_pred_euc_25,
                      axes = 'collect') &
  theme(axis.title.x = element_blank(),
        margins = margin_part(l = 10, r = 10))

r2_pred <- wrap_plots(plot_pred_jac_23, plot_pred_bray_23, plot_pred_euc_23,
                      axes = 'collect') &
  theme(margins = margin_part(l = 10, r = 10))

wrap_plots(grid::textGrob("Observer dissimilarity in 2025", gp = grid::gpar(fontsize = 12)),
            r1_pred,
            grid::textGrob("Observer dissimilarity in 2023", gp = grid::gpar(fontsize = 12)),
            r2_pred,
            ncol = 1, heights = rep(c(1, 10), times = 2)) +
  plot_annotation(tag_levels = list(c("", paste0(letters[1:3], ")),
                                     "", paste0(letters[4:6], ")))) &
  theme(axis.text = element_text(size = 11),
        plot.tag = element_text(size = 10),
        plot.tag.location = 'panel',
        plot.tag.position = c(-.02, 1))
```



```
ggsave('output/beta_pred.png', bg = 'white', scale = 1.1)

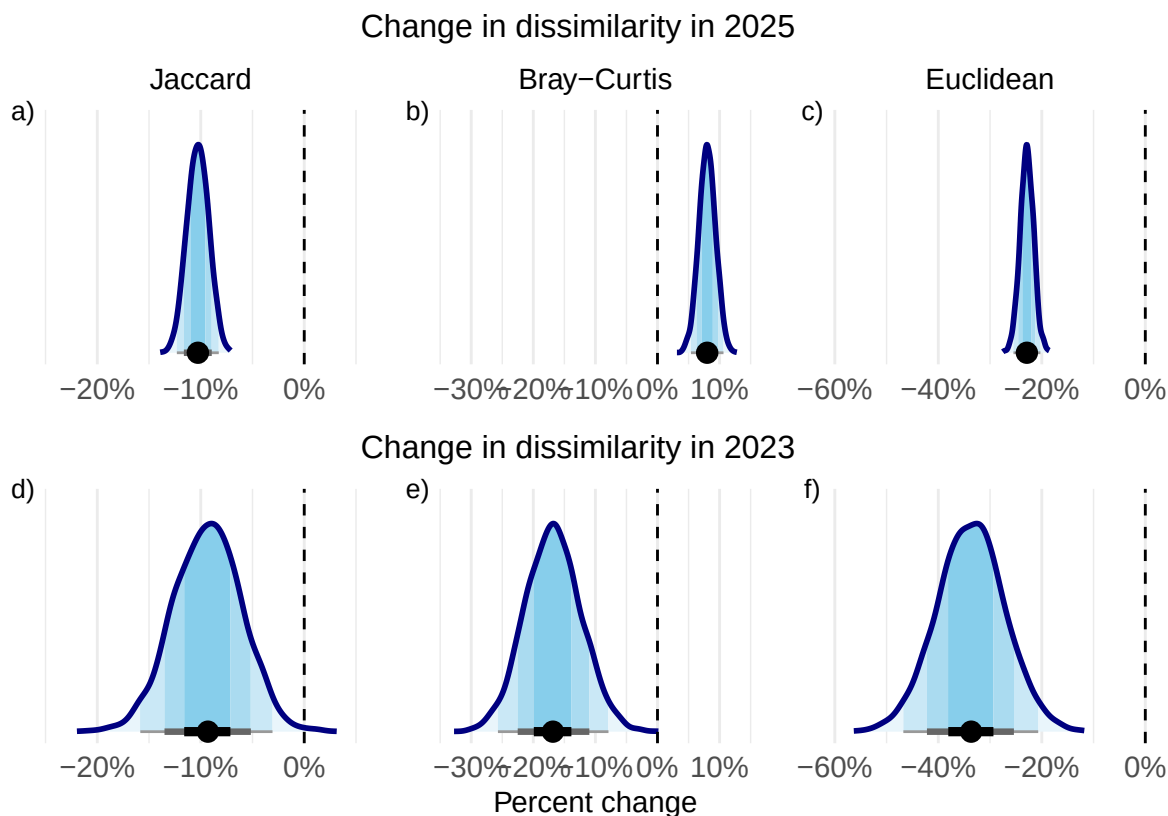
#beta lift %

r1_lift <- wrap_plots(plot_lift_jac_25, plot_lift_bray_25, plot_lift_euc_25,
  axes = 'collect') &
  theme(axis.title.x = element_blank(),
    plot.title = element_text(hjust = 0.5, size = 12),
    margins = margin_part(l = 10, r = 10))

r2_lift <- wrap_plots(plot_lift_jac_23, plot_lift_bray_23, plot_lift_euc_23,
  axes = 'collect') &
  theme(margins = margin_part(l = 10, r = 10))

wrap_plots(grid::textGrob("Change in dissimilarity in 2025", gp = grid::gpar(fontsize = 12)),
  r1_lift,
  grid::textGrob("Change in dissimilarity in 2023", gp = grid::gpar(fontsize = 12)),
  r2_lift,
  ncol = 1, heights = rep(c(1, 10), times = 2)) +
  plot_annotation(tag_levels = list(c("", paste0(letters[1:3], "")),
    "", paste0(letters[4:6], "")))) &
  theme(plot.title = element_text(size = 11),
    axis.text = element_text(size = 11),
    plot.tag = element_text(size = 10),
    plot.tag.location = 'panel',
```

```
plot.tag.position = c(-.02, 1))
```



```
ggsave('output/beta_lift.png', bg = 'white', scale = 1.1)
```

Session info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: x86_64-redhat-linux-gnu
## Running under: Nobara Linux 43 (KDE Plasma Desktop Edition)
##
## Matrix products: default
## BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP; LAPACK version 3.12.1
##
## locale:
##  [1] LC_CTYPE=it_IT.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=it_IT.utf8       LC_COLLATE=it_IT.UTF-8
##  [5] LC_MONETARY=it_IT.utf8   LC_MESSAGES=it_IT.UTF-8
##  [7] LC_PAPER=it_IT.utf8      LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
```

```

## [11] LC_MEASUREMENT=it_IT.utf8 LC_IDENTIFICATION=C
##
## time zone: Europe/Rome
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats      graphics  grDevices datasets  utils      methods    base
##
## other attached packages:
## [1] vegan_2.7-2      permute_0.9-10    ggdist_3.3.3
## [4] tidybayes_3.0.7  distributional_0.6.0 bayesplot_1.15.0
## [7] brms_2.23.0      Rcpp_1.1.1        patchwork_1.3.2
## [10] readxl_1.4.5     lubridate_1.9.5   forcats_1.0.1
## [13] stringr_1.6.0    dplyr_1.2.0       purrr_1.2.1
## [16] readr_2.2.0      tidyr_1.3.2       tibble_3.3.1
## [19] ggplot2_4.0.2    tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] svUnit_1.0.8      tidymodels_1.2.1    farver_2.1.2
## [4] loo_2.9.0         S7_0.2.1            fastmap_1.2.0
## [7] tensorA_0.36.2.1  digest_0.6.39       timechange_0.4.0
## [10] lifecycle_1.0.5   cluster_2.1.8.2     StanHeaders_2.32.10
## [13] magrittr_2.0.4    posterior_1.6.1     compiler_4.5.2
## [16] rlang_1.1.7       tools_4.5.2         utf8_1.2.6
## [19] yaml_2.3.12       knitr_1.51          labeling_0.4.3
## [22] bridgesampling_1.2-1 bit_4.6.0           pkgbuild_1.4.8
## [25] plyr_1.8.9        RColorBrewer_1.1-3  abind_1.4-8
## [28] withr_3.0.2       grid_4.5.2          stats4_4.5.2
## [31] inline_0.3.21     ggthemes_0.5.7      scales_1.4.0
## [34] MASS_7.3-65       cli_3.6.5           mvtnorm_1.3-3
## [37] rmarkdown_2.30    crayon_1.5.3        ragg_1.5.0
## [40] generics_0.1.4    otel_0.2.0          RcppParallel_5.1.11-1
## [43] rstudioapi_0.18.0 reshape2_1.4.5      tzdb_0.5.0
## [46] rstan_2.32.7      splines_4.5.2       parallel_4.5.2
## [49] cellranger_1.1.0  CoprManager_0.5.8   matrixStats_1.5.0
## [52] vctr_0.7.1        Matrix_1.7-4        hms_1.1.4
## [55] arrayhelpers_1.1-0 bit64_4.6.0-1       systemfonts_1.3.1
## [58] glue_1.8.0        codetools_0.2-20    stringi_1.8.7
## [61] gtable_0.3.6      QuickJSR_1.9.0      quadprog_1.5-8
## [64] pillar_1.11.1     htmltools_0.5.9     Brodingtonag_1.2-9
## [67] R6_2.6.1          textshaping_1.0.4   vroom_1.7.0
## [70] evaluate_1.0.5    lattice_0.22-9      backports_1.5.0
## [73] rstantools_2.6.0  gridExtra_2.3       coda_0.19-4.1
## [76] nlme_3.1-168      checkmate_2.3.4     mgcv_1.9-4
## [79] xfun_0.56         pkgconfig_2.0.3

```